

De novo identification of universal cell mechanics gene signatures

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This **important** study uses machine learning-based network analysis on transcriptomic data from different tissue cell types to identify a small set of conserved (pan-tissue) genes associated with changes in cell mechanics. The new method, which provides a new type of approach for mechanobiology, is accessible, **compelling**, and well-validated using in silico and experimental approaches. The study provides motivation for researchers to test hypotheses concerning the identified five-gene network, and the method will be strengthened over time with expanded sets of validations, such as testing genes with hitherto unknown roles and different perturbation techniques.

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Abstract

Cell mechanical properties determine many physiological functions, such as cell fate specification, migration, or circulation through vasculature. Identifying factors that govern the mechanical properties is therefore a subject of great interest. Here we present a mechanomics approach for establishing links between single-cell mechanical phenotype changes and the genes involved in driving them. We combine mechanical characterization of cells across a variety of mouse and human systems with machine learning-based discriminative network analysis of associated transcriptomic profiles to infer a conserved

network module of five genes with putative roles in cell mechanics regulation. We validate in silico that the identified gene markers are universal, trustworthy and specific to the mechanical phenotype across the studied mouse and human systems, and demonstrate experimentally that a selected target, *CAV1*, changes the mechanical phenotype of cells accordingly when silenced or overexpressed. Our data-driven approach paves the way towards engineering cell mechanical properties on demand to explore their impact on physiological and pathological cell functions.

Introduction

The extent to which cells can be deformed by external loads is determined by their mechanical properties, such as cell stiffness. Since the mechanical phenotype of cells has been shown to reflect functional cell changes, it is now well established as a sensitive label-free biophysical marker of cell state in health and disease (1, 2). Beyond being a passive property that can be correlated with cell state, cell stiffness is increasingly recognized as an important feature involved in processes such as development (3, 4) and cancer progression (5, 6). Identifying the molecular targets for on-demand tuning of mechanical properties is, thus, essential for exploring the precise impact that cell mechanics has on physiological and pathological processes in living organisms.

The mechanical properties of cells are determined by various intracellular structures and their dynamics, with cytoskeletal networks at the forefront (7). According to current knowledge, the most prominent contributor to the global mechanical phenotype is the actin cortex and its contractility regulated via Rho signaling (8, 9). Intermediate filaments, including vimentin and keratin, reside deeper inside the cell and can also contribute to measured cell stiffness, especially at high strains (10, 11). Although there is some evidence of the contribution of microtubules to cell stiffness at high strains (12), their role has been difficult to address directly, since drug-induced microtubule disassembly evokes reinforcement of actin cytoskeleton and cell contractility (13). Apart from cytoskeletal contributions, the cell mechanical phenotype can be influenced by the level of intracellular packing (14, 15) or mechanical properties of organelles occupying the cell interior, such as the cell nucleus (16). When aiming at modulating the mechanical properties of cells, it may not be practical to target cytoskeletal structures, which are central to a multitude of cellular processes, because their disruption is generally toxic to cells. It is therefore important to identify targets that enable subtle, alternative ways of intervening with cell stiffness.

Most of our knowledge about the molecular contributors to cell mechanics has been derived from drug perturbations or genetic modifications targeting structures known a priori. The challenge of identifying novel targets determining the mechanical phenotype can be addressed on a large scale by performing screens using RNA interference (17–19) or small-molecule compound libraries. Alternatively, the problem can be reverse-engineered, in that omics datasets for systems with known mechanical phenotype changes are used for prediction of genes involved in the regulation of mechanical phenotype in a mechanomics approach. Broadly speaking, mechanomics is a study of omics data within the context of mechanobiology. So far, this term has been used with regard to changes in omics profiles in response to an external mechanical stimulus such as shear flow, tensile stretch, or mechanical compression (20–22), or to collectively name all of the mechanical forces acting on or within cells (23–27). However, it can also be used to address omics changes related to changes in the mechanical properties of cells (28, 29) — a context much closer to our study.

Here we extend the concept of mechanomics to a data-driven methodology for de novo identification of genes associated with the mechanical phenotype based on omics data (Fig. 1). To demonstrate this approach, we perform a machine learning-based discriminative network

analysis termed PC-corr (28) on transcriptomics data from two unrelated biological systems with known mechanical phenotype changes (29, 30) and elucidate a conserved functional module of five candidate genes putatively involved in the regulation of cell mechanics. We then test the ability of each gene to classify cell states according to cell stiffness in silico on six further transcriptomic datasets and show that the individual genes, as well as their compression into a combinatorial marker, are universally, specifically and trustworthily associated with the mechanical phenotype across the studied mouse and human systems. Finally, we confirm experimentally that one of the candidate genes, caveolin-1 (*CAV1*), has the capacity to alter the mechanical phenotype in the predicted direction when downregulated or overexpressed. The systematic approach presented here, combining omics data with mechanical phenotypes across different systems, has the power to identify genes that ubiquitously contribute to cell mechanical phenotype in a hypothesis-free manner. Such genes can, in the future, be used as knobs for adjusting mechanical cell properties to explore their role in the homeostasis of multicellular systems or to therapeutically intervene in relevant pathologies.

Results

Cross-system identification of genes involved in cell mechanical changes

We introduce an inference approach for de novo identification of genes involved in cell mechanical changes across different systems that we refer to as mechanomics. The general workflow of this approach is presented in Fig. 1 and consists of four steps: data curation, target prediction, in silico validation and experimental validation. In the first step, mechano-transcriptomic datasets representing a broad spectrum of biological systems are collected (Fig. 1A). Each dataset encompasses two or more cell states characterized by a distinct mechanical phenotype, for which transcriptomic data is available. In the second step, a subset of the transcriptomic datasets is used to identify a conserved network module of putative target genes involved in the regulation of cell mechanical phenotype (Fig. 1B). The ability of the obtained target genes to correctly classify soft and stiff cell states is next tested in silico on the validation datasets (Fig. 1C). Finally, the best scoring targets are validated experimentally by monitoring mechanical phenotype changes upon their overexpression and downregulation in the cells of choice (Fig. 1D).

Model systems characterized by mechanical phenotype changes

To curate the mechano-transcriptomic datasets, we screened the projects ongoing in our group and identified five biological systems for which published transcriptomic data were available, and the concomitant mechanical phenotype changes were either already documented or implicated (Table 1). The mechanical phenotypes of the different cell states within each dataset were characterized primarily using real-time deformability cytometry (RT-DC), a microfluidics-based method that enables rapid analysis of thousands of cells (31) (Fig. S1) — a feature particularly useful when setting out to explore a large variety of systems and states. RT-DC relies on flowing cells through a narrow constriction of a microfluidic channel and high-speed imaging to assess the ensuing cell deformation (31) (Fig. S1, A and B). In the context of this method, the mechanical phenotype is understood as whole-cell elasticity quantified by an apparent Young's modulus, E , deduced from cell size and deformation under given experimental conditions (32) (Fig. S1, C and D). Young's modulus quantifies how much stress (force per unit area) is necessary to deform a cell to a certain extent (i.e., strain), thus higher Young's modulus values indicate that a cell is more difficult to deform, or stiffer. In two of the datasets (see Table 1), selected cell states were additionally characterized using atomic force microscopy (AFM)-based assays on adherent cells to

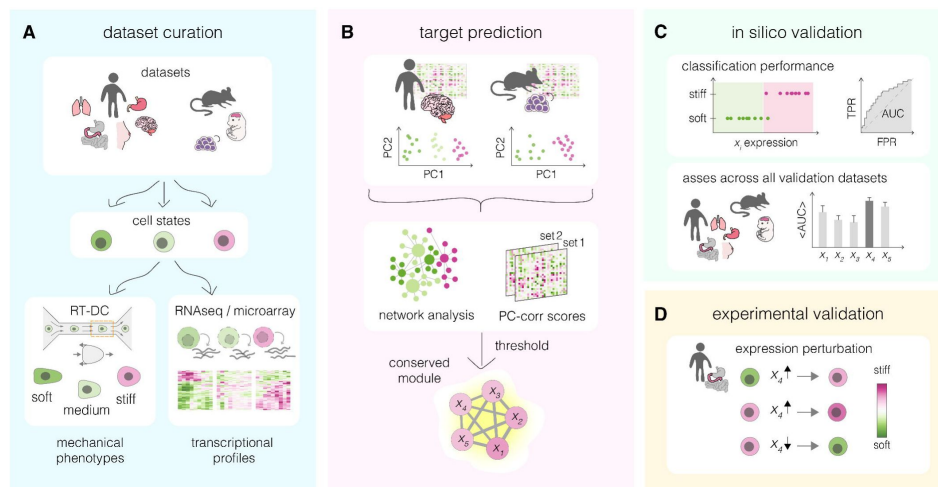


Fig. 1.

Overview of a mechanomics approach for de novo identification of genes involved in cell mechanics regulation. (A) Data curation. Datasets originating from different biological systems encompassing cell states with distinct mechanical phenotypes, as characterized by real-time deformability cytometry (RT-DC), and associated transcriptomics profiles are collected. (B) Target prediction. A subset of collected datasets is used to perform machine learning-based network analysis on transcriptomic data and identify conserved module of genes associated with cell mechanics changes. PC – principal component. (C) In silico validation. The classification performance of individual genes from module identified in (B) is evaluated in silico on remaining datasets. TPR – true positive rate, FPR – false positive rate, AUC – area under the curve. (D) Experimental validation. Targets with highest classification performance in silico are verified experimentally in perturbation experiments.

confirm the mechanical differences observed with RT-DC. The transcriptional profiles related to each system, generated by either RNA sequencing (RNA-Seq) or microarray analysis, were retrieved from entries previously deposited in online databases (**Table 1**).

We curated mechano-transcriptomic data assemblies originating from five different biological systems (**Fig. 2**) that included a total of eight transcriptomic datasets (**Table 1**). Two of the transcriptomic datasets were used for target prediction, and the remaining six for target validation. The first studied system encompassed patient-derived glioblastoma cell lines cultured in conditions supporting different levels of activation of the STAT3-Ser/Hes3 signaling axis involved in cancer growth regulation. As previously demonstrated, the higher the STAT3-Ser/Hes3 activation in the characterized states, the stiffer the measured phenotype of glioblastoma cells (29) (**Fig. 2A**).

The second system included small-cell and adenocarcinoma cell lines originating from human intestine, lung, and stomach. Consistently across tissues, small cell-carcinoma cells had a lower apparent Young's modulus compared to their adenocarcinoma counterparts (**Fig. 2B**). Small-cell carcinomas have comparatively small cell sizes, short doubling times and high metastatic potential, all connected with poor clinical prognosis in patients (40, 41). Apart from the main transcriptomic dataset for the carcinoma project, in which all mechanically characterized cell lines are represented (FANTOM5 (33)), we collected three additional transcriptomic datasets generated with different expression profiling techniques (RNA-Seq or microarray profiling), and originating from different groups: Cancer Cell Line Encyclopedia (CCLE) microarray (34), CCLE RNA-Seq (35), and Genentech (36) (see **Table 1** for overview). In the third studied system, a non-tumorigenic breast epithelium MCF10A cell line bearing single-allele oncogenic mutation H1047R in the catalytic subunit alpha of the phosphatidylinositol-4,5-bisphosphate 3-kinase (PIK3CA) (37, 42) showed increased stiffness compared to the wild type control (**Fig. 2C**). H1047R mutation causes constitutive activation of PIK3CA and an aberrant triggering of the PI3K-AKT-mTOR signaling pathway leading to growth factor-independent proliferation (43, 44). In the fourth system, the fuzzy-colony forming (F-class) state of iPSCs had a lower stiffness as compared to the bone-fide compact-colony forming (C-class) state (30) (**Fig. 2D**). C-class cells establish endogenous expression of reprogramming factors at moderate levels towards the end of reprogramming, while F-class cells depend on the ectopic expression of the pluripotency factors and are characterized by a fast proliferation rate (38). Finally, we characterized two stages of developing neurons isolated from embryonic mouse brain (39), and observed that the newborn neurons (NNs) had higher apparent Young's moduli than proliferating progenitors (PPs) (**Fig. 2E**). Cell areas and deformations used for Young's modulus extraction for all datasets are visualized in Fig. S2.

The mechano-transcriptomic datasets collected within the framework of our study (**Table 1**) represent a broad spectrum of biological systems encompassing distinct cell states associated with mechanical phenotype changes. The included systems come from two different species (human and mouse), several tissues (brain, intestine, lung, stomach, breast, as well as embryonic tissue) and are associated with processes ranging from cancerogenic transformations to cell morphogenesis. This high diversity is important for focusing the analysis on genes universally connected to the change in mechanical properties, rather than on genes specific for processes captured by individual datasets.

Discriminative network analysis on prediction datasets

After characterizing the mechanical phenotype of the cell states, we set out to use the accompanying transcriptomic data to elucidate genes associated with the mechanical phenotype changes across the different model systems. To this end, we utilized a method for inferring phenotype-associated functional network modules from omics datasets termed PC-Corr (28), that relies on combining loadings obtained from the principal component (PC) analysis and Pearson's correlation for every pair of genes. PC-Corr was performed individually on two

general information				transcriptomic data					mechanics data	
source	dataset name	used for	cell states	accession number	reference	method	unique entries	total samples used	method	reference
human	glioblastoma	Pred	FGFJ EGF serum	GEO: GSE77751	Poser <i>et al.</i> (29)	HT seq	39400	27	RT-DC	Poser <i>et al.</i> (29)
	carcinoma	Val PI	small-cell adeno	DDBJ: DRA000991 [‡]	FANTOM5 consortium (33)	CAGE	18821	12	RT-DC, AFM	this paper
		Val PII & N		GEO: GSE36139 [†]	CCLE microarray Barretina <i>et al.</i> (34)	micro array	18925	162		
		Val PII		DepMap: release 21Q4 [#]	CCLE RNA-Seq Ghandi <i>et al.</i> (35)	HT seq	51304	179		
		Val PII & N		GEO: GSE30611	Genentech Klijn <i>et al.</i> (36)	HT seq	25996	82		
	MCF10A	Val PI	wt H1047R	GEO: GSE69822	Kiselev <i>et al.</i> (37)	HT seq	38508	6	RT-DC	this paper
mouse	iPSCs	Pred	F-class C-class	GEO: GSE49940	Tonge <i>et al.</i> (38)	micro array	18118	28	RT-DC, AFM	Urbanska <i>et al.</i> (30)
	developing neurons	Val PI	PPs NNs	GEO: GSE51606	Apra <i>et al.</i> (39)	HT seq	21110	9	RT-DC	this paper

[‡]data for samples of interest was extracted using TET tool from the FANTOM5 website <https://fantom.gsc.riken.jp/5/>

[†]data was downloaded using the ArrayExpress archive <https://www.ebi.ac.uk/arrayexpress/experiments/E-MTAB-2706/>

[#]DepMap Public 21Q4 Primary Files, accessed via DepMap portal <https://depmap.org/portal/download>

Table 1.

Mechano-transcriptomic datasets used in this study. Pred – prediction, Val – validation, PI/II – positive hypothesis I/II, N – negative hypothesis, CCLE – cancer cell line encyclopedia, HT Seq – high-throughput RNA sequencing, CAGE – cap analysis gene expression, AFM – atomic force microscopy, adeno – adenocarcinoma, wt – wild type, PP – proliferating progenitors, NNs – newborn neurons.

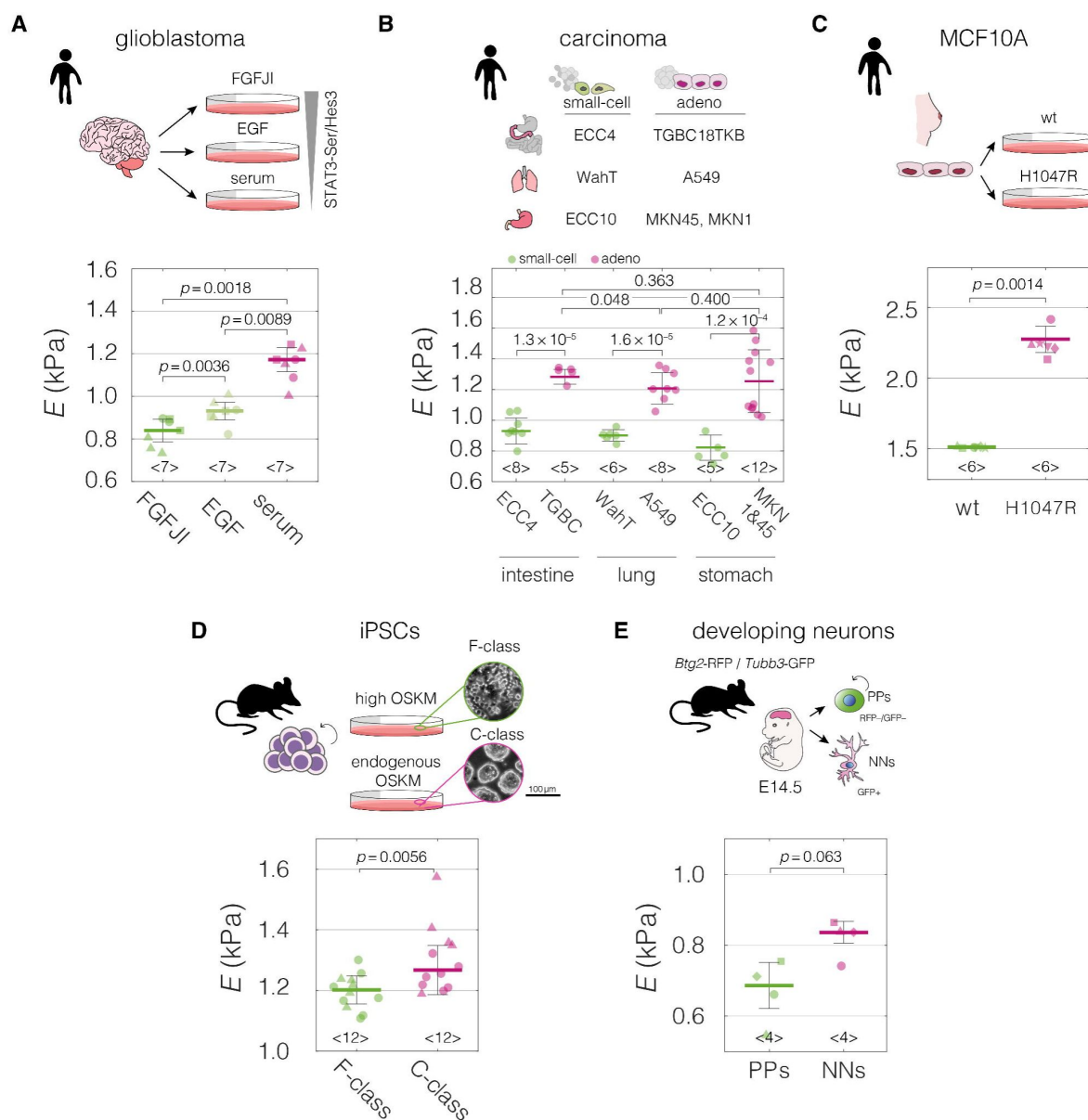


Fig. 2.

Mechanical properties of divergent cell states in five biological systems. Schematic overviews of the systems used in our study, alongside with the cell stiffness of individual cell states parametrized by Young's moduli E . (A) Human patient-derived glioblastoma cells with three distinct signaling states maintained by indicated culture conditions. (B) Human small-cell carcinoma and adenocarcinoma cell lines originating from intestine, lung, and stomach. (C) Human breast epithelium MCF10A cell line bearing single-allele H1047R mutation in the PIK3CA with parental wild type (wt) as a control. (D) Murine F- and C-class iPSCs cultured in the presence or absence of doxycycline (dox) activating ectopic expression of OSKM factors (Oct4, Sox2, Klf4, and cMyc). (E) Proliferating progenitors (PPs) and newborn neurons (NNs) isolated from brains of mouse embryos. Horizontal lines delineate medians with mean absolute deviation (MAD) as error, datapoints represent medians of the individual replicates, the number of independent biological replicates is indicated below each box. Statistical analysis was performed using generalized linear mixed effects model. The symbol shapes represent cell lines derived from three different patients (A), matched experimental replicates (C), two different reprogramming series (D), and different cell isolations (E). Data presented in (A) and (D) were previously published in ref (29) and (30), respectively.

prediction datasets, and the obtained results were overlaid to derive a conserved network module. Owing to the combination of the Pearson's correlation coefficient and the discriminative information included in the PC loadings, the PC-corr analysis does not only consider gene co-expression — as is the case for classical co-expression network analysis — but also incorporates the relative relevance of each feature for discriminating between two or more conditions; in our case, the conditions representing soft and stiff phenotypes. The overlaying of the results from two different datasets allows for a multi-view analysis (utilizing multiple sets of features) and effectively merges the information from two different biological systems.

For the network construction, we chose two datasets that originate from different species, concern unrelated biological processes, and have a high number of samples included in the transcriptional analysis: human glioblastoma and murine iPSCs (**Table 1**). PC-corr analysis was performed on these prediction datasets individually using a subset of transcripts at which the two datasets intersect (**Fig. 3A**). First, the 9,452 unique genes from the intersection were used to perform principal component analysis (PCA) (**Fig. 3B** and **C**). Next, the PC loadings for the component showing good separation between the different cell states (PC1 for both of presented datasets) were normalized and scaled (see Methods for details). The processed PC loadings, V , were then combined with Pearson's correlation coefficients, c , to obtain a $PC\text{-corr}$ value for each pair of genes i, j for every n -th dataset according to the following formula:

$$PC\text{-corr}_{i,j}^n = \text{sgn}(c_{i,j}^n) \min(|V_i^n|, |V_j^n|, |c_{i,j}^n|). \quad (1)$$

The sign of the $PC\text{-corr}$ value corresponds to the correlated (positive) or anti-correlated (negative) expression of genes i, j , and the magnitude of $PC\text{-corr}$ conveys the combined information about the strength of the expression correlation and the contribution of the individual genes to the phenotype-based separation of samples along the principal component.

To merge the PC-corr results obtained for the individual prediction datasets, a combined $PC\text{-corr}$ value, $PC\text{-corr}_{i,j}^{comb}$, was calculated either as a mean or as a minimum of the individual $PC\text{-corr}$ values. For n datasets:

$$PC\text{-corr}_{i,j}^{comb} = \begin{cases} \delta_{i,j} \frac{1}{N} \sum_{n=1}^N |PC\text{-corr}_{i,j}^n| & \text{if } \delta_{i,j} \neq 0 \\ \delta_{i,j} \min(|PC\text{-corr}_{i,j}^1|, \dots, |PC\text{-corr}_{i,j}^N|) & \text{if } \delta_{i,j} = 0 \end{cases} \quad (2)$$

where $\delta_{i,j} \in \{-1, 1\}$ defines the sign of $PC\text{-corr}_{i,j}^{comb}$, and is equal to the mode of $PC\text{-corr}_{i,j}$ signs over all individual datasets. In our implementation on two datasets, gene pairs with opposing $PC\text{-corr}$ signs were masked by setting their $PC\text{-corr}^{comb}$ values to zero.

To obtain the network of putative target genes, a cut-off was applied to the absolute $PC\text{-corr}^{comb}$ value. We explored several cut-off strategies in order to obtain a wide overview of the meaningful conserved network structures. By looking at $PC\text{-corr}^{comb}$ calculated as mean and setting the threshold for its absolute value to 0.75, we obtained a network of 29 nodes connected by 30 edges (**Fig. 3E**). The edges describe the connection between the genes in the network and their thickness is defined by the $PC\text{-corr}^{comb}$ values (Table S1). The node colors reflect the strength of the contribution of individual genes to the separation of the different classes as described by the mean of the processed PC loadings V .

The obtained network can be made more conservative by using the minimum $PC\text{-corr}^{comb}$ instead of the mean, or by changing the cut-off value. Utilizing the $PC\text{-corr}^{comb}$ calculated as minimum value (Table S2) and setting the cut-off value to 0.70, we obtained a network with 22 nodes connected by 29 edges (**Fig. 3F**). Increasing the cut-off value to 0.75 resulted in a network of 9

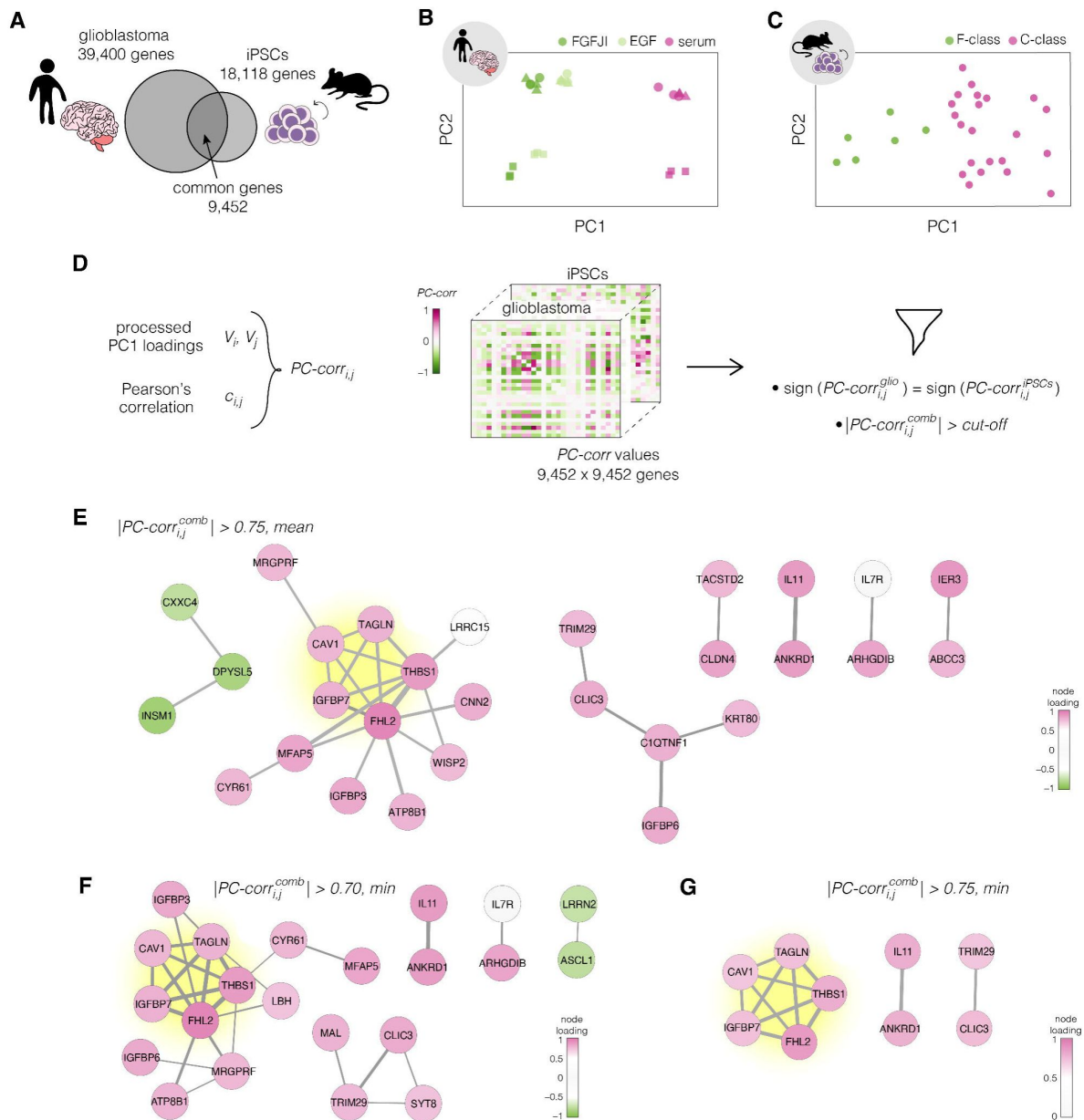


Fig. 3.

Identification of putative targets involved in cell mechanics regulation. (A) Glioblastoma and iPSC transcriptomes used for the target prediction intersect at 9,452 genes. (B, C) PCA separation along two first principal components of the mechanically distinct cell states in the glioblastoma (B) and iPSC (C) datasets. The analysis was performed using the gene expression data from the intersection presented in (A). The symbol shapes in (B) represent cell lines derived from three different patients. (D) Schematic representation of PC-corr analysis and the combination of the PC-corr results for two systems. (E to G) Gene networks based on filtering gene pairs by the combined $PC-corr$ score. The presented networks were obtained by setting the cut-off value to 0.75, when using the mean $PC-corr$ approach (E), and to 0.70 (F) and 0.75 (G), when using the minimum value approach. In (E to G) edge thicknesses represent the $|PC-corr_{i,j}^{comb}|$ (Table S1 and S2) and the colors of the nodes represent the average processed PC loadings (Table S3).

genes connected by 12 edges (**Fig. 3G**). The list of genes from the three networks presented in **Fig. 3C**, **E** to **G**, together with their full names and processed PC loading values, is presented in Table S3.

We performed gene ontology enrichment analysis for biological processes on the nodes of the network presented in **Fig. 3G**, as well as the union of all nodes presented in **Fig. 3C**, **E** to **G** (Fig. S3). The top two significantly enriched terms in the 9-gene set were the negative regulation of transcription by polymerase II (GO:000122) and negative regulation of endothelial cell proliferation (GO:0001937). In the 34-gene set, apart from a broad term of signal transduction (GO:0007165), the significantly enriched terms included negative regulation of transcription by polymerase II (GO:000122), regulation of cell growth (GO:0001558), and negative regulation of cell proliferation (GO:0008285), among others. The fact that these GO terms are not obviously related to cell mechanics might be an indicator that the association of the identified genes with cell mechanics is relative unknown, and that our mechanomics approach can identify such associations de novo. The aforementioned categories included mostly genes showing higher expression in the stiff states. Since the upregulated genes are associated with negative regulation of growth and transcription, our results point towards a targeted reduction in transcriptional activity and reduced growth/proliferation in stiff compared to soft cells.

The identified conserved functional network module comprises five genes

Regardless of the strategy chosen for the selection of the network-building gene pairs, a strongly interconnected module of 5 genes (**Table 2**) — highlighted in yellow in **Fig. 3C**, **E** to **G** — emerged. We focused on the five genes from this conserved network module as putative targets for regulating cell mechanics: *CAV1*, *FHL2*, *IGFBP7*, *TAGLN*, and *THBS1*.

Caveolin-1, *CAV1*, is a protein most prominently known for its role as a structural component of caveolae. Caveolae are small cup-shaped invaginations in the cell membrane that are involved, among other functions, in the mechanoprotective mechanism of buffering the plasma membrane tension (45, 46). Recent data suggests that *CAV1* can also confer its mechanoprotective role independently of caveolae (47). Apart from membrane organization and membrane domain scaffolding, *CAV1* plays a role in an array of regulatory functions such as metabolic regulation or Rho-signalling (46). The second identified target, four and a half LIM domains 2, *FHL2*, is a multifaceted LIM domain protein with many binding partners and a transcription factor activity (48). *FHL2* has recently been shown to remain bound to actin filaments under high tension, and be shuttled to the nucleus under low cytoskeletal tension (49, 50) — a property conserved among many LIM domain-containing proteins (50, 51). The third target, Insulin-like growth factor binding protein 7, *IGFBP7*, is a secreted protein implicated in a variety of cancers. It is involved in the regulation of processes such as cell proliferation, adhesion, and senescence (52). Transgelin, *TAGLN*, is an actin-binding protein whose expression is up-regulated by high cytoskeletal tension (53) and is also known to play a role in cancer (54). Finally, thrombospondin 1, *THBS1*, is a matricellular, calcium-binding glycoprotein that mediates cell-cell and cell-matrix adhesions and has many regulatory functions (55, 56).

Before validating the performance of the five target genes, we inspected their expression across the divergent cell states in the collected datasets. The target genes show clear differences in expression levels between the soft and stiff cell states and provide for clustering of the samples corresponding to different cell stiffnesses in both prediction and validation datasets (**Fig. 4**, Figs. S4-S6). The relation between normalized apparent Young's modulus change and fold-change in the expression of the target genes is presented in Fig. S7. The direction of changes in the expression levels between the soft and stiff cell states in the validation datasets was not always following the same direction (**Fig. 4**, **C** to **F**, Fig. S7). This suggests that the genes associated with cell mechanics may not have a monotonic relationship with cell stiffness, but rather are characterized

symbol	gene description	HGNC ID	MGI ID
<i>CAV1</i>	caveolin 1	HGNC:1527	MGI:102709
<i>FHL2</i>	four and a half LIM domains 2	HGNC:3703	MGI:1338762
<i>IGFBP7</i>	insulin like growth factor binding protein 7	HGNC:5476	MGI:1352480
<i>TAGLN</i>	transgelin	HGNC:11553	MGI:106012
<i>THBS1</i>	thrombospondin 1	HGNC:11785	MGI:98737

Table 2.

List of identified target genes comprising the conserved module.

by different expression regimes in which the expression change in opposite directions can have the same effect on cell stiffness. Additionally, in specific cases a relatively high change in Young's modulus did not correspond to marked expression changes of a given gene — see for example low CAV1 changes observed in MCF10A PIK3CA mutant (Fig. S7A), or low IGFBP7 changes in intestine and lung carcinoma samples (Fig. S7C). This indicates that the importance of specific targets for the mechanical phenotype change may vary depending on the origin of the sample.

Universality, specificity, and trustworthiness of the identified markers

Next, we validated whether the five identified genes individually, as well as their association into a unique combinatorial marker (computed as the mean of the five log-normalized genes, see Methods), are universal and specific markers of cell mechanics. To assess that, we tested three hypotheses using combinations of transcriptomic data from six validation datasets as detailed in **Table 3** [↗](#). The classification performance of each marker was assessed using the area under the curve of the receiver-operator characteristics (AUC-ROC) [\(57\)](#) [↗](#), which takes values from 0 to 1, with 1 corresponding to a perfect classifier and 0.5 to a random classifier. Importantly, for each hypothesis multiple datasets were used, and the discriminative performance was assessed in a joint multiview way by looking at the minimum value of AUC-ROC across multiple comparisons.

We first tested whether the obtained markers are universal across systems of different biological origin (positive hypothesis I) by estimating their ability to discriminate between stiff and soft cell phenotypes in three validation datasets: developing neurons (mouse), carcinoma cell lines originating from three tissues (human), and MCF10A (human) (**Table 4** [↗](#)). Particularly high minimum AUC-ROC values (≥ 0.78) were obtained for CAV1, FHL2, and TAGLN, and the combinatorial marker outperformed all of the individual genes with a minimum AUC-ROC of 0.97. The ROC curves for individual datasets are presented in Fig. S5.

Next, we tested whether the identified markers provide good sample classification across similar datasets obtained from different sources (positive hypothesis II). For this purpose, we used three carcinoma datasets that were generated by two different research group using either microarray or RNA-Seq (see **Table 1** [↗](#) and 3). Within these datasets, we looked at the discrimination between the small-cell and adenocarcinoma samples from lung. This choice was dictated by the highest number of available samples from this tissue across the datasets. Also here, the multiview AUC-ROC values were high, reaching 0.89 for CAV1, 0.88 for FHL2, and 0.86 for THBS1. The combinatorial marker had an AUC-ROC value of 0.92.

To assess whether the predicted markers are specific to the mechanical phenotype, we tested their performance in classification of the adenocarcinoma samples grouped by the tissue they were derived from (negative hypothesis). This groups did not show clear mechanical differences (**Fig. 2B** [↗](#)). For the combinatorial marker, the min AUC-ROC value was equivalent to a random classifier (0.51), and for the individual markers reached values between 0.51 and 0.65 (**Table 4** [↗](#)). Since the discriminative power of the obtained markers vanished (reached AUC-ROC close to 0.50 corresponding to a random classifier) when tested on groups that do not encompass cell mechanic phenotype difference, we can conclude that the identified markers are specific to the mechanical phenotype.

Finally, to test the trustworthiness of obtained markers, we evaluated how easy it is to generate markers with equivalent discriminative power at random. For that purpose, we devised a novel methodology called joint-view trustworthiness (JVT). JVT is a resampling technique that creates a null model distribution according to which an empirical p value is computed to evaluate the probability to sample at random a marker that offers a joint multiview discrimination equal or better to the one of the predicted markers (see Methods for details). A low JVT p value (< 0.05

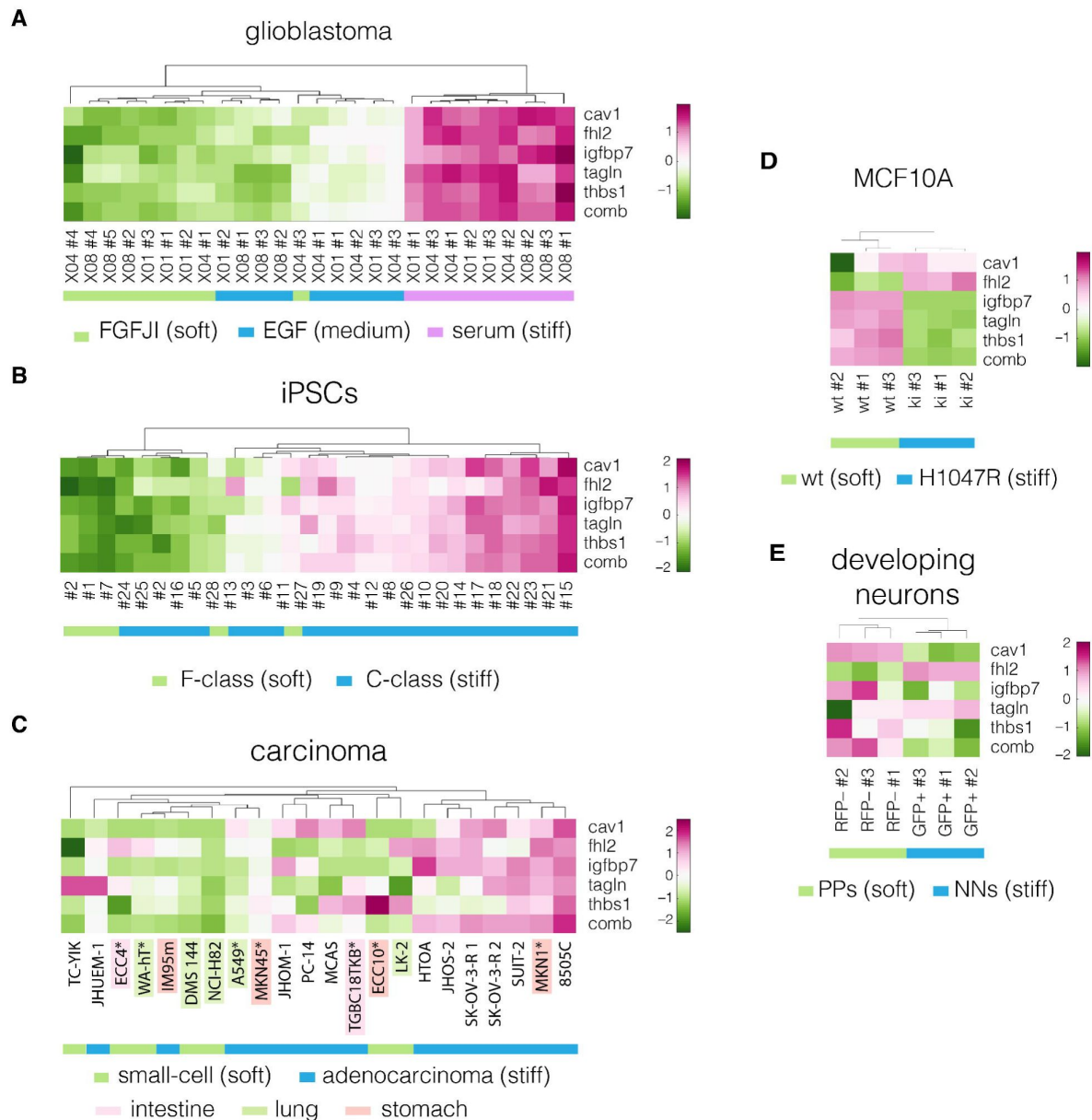


Fig. 4.

Expression of identified target genes in the prediction and validation datasets. Panels show unsupervised clustering heat maps of expression data from transcriptomic datasets corresponding to the following systems: (A) glioblastoma, (B) induced pluripotent stem cells (iPSCs), (C) carcinoma, cell lines from intestine, lung and stomach used for positive hypothesis I (see [Table 3](#)) are highlighted in pink, green, and orange, respectively; *mechanically tested cell lines (here the FANTOM5 dataset is presented as it contains all the cell lines that were tested mechanically in our study, for the remaining carcinoma datasets see Figs. S4-S6), (D) non-tumorigenic breast epithelia MCF10A, and (E) developing neurons (dev. neurons). Comb – combinatorial marker, wt – wild type, PPs – proliferating progenitors, NNs – newborn neurons. Clustering was performed using *clustergram* function in *MATLAB* (R2020a, MathWorks) on log-normalized expression data.

Table 3.

Overview of the hypotheses and datasets used for validating universality and specificity of obtained markers. Hypotheses are listed in the column headings. Under every hypothesis, sample groups used for the hypothesis testing are listed. Numbers of samples used in every group are indicated in brackets.

positive hypothesis I markers are discriminative of samples with stiff/soft mechanical phenotype independent of the studied biological system	positive hypothesis II markers are discriminative of samples with stiff/soft mechanical phenotype independent of data source	negative hypothesis markers are discriminative of samples from different tissue of origin (but with no mechanical difference)
Carcinoma FANTOM5 1. small-cell (6) vs adeno (6) (lung, intestine and stomach)	Carcinoma CCLE microarray 1. small-cell (51) vs adeno (49) (lung)	Carcinoma CCLE microarray 1. lung (49) vs stomach (19) (adeno) 2. large intestine (43) vs stomach (19) (adeno)
MCF10A 2. wt (3) vs H1047R (3)	Carcinoma CCLE RNA-Seq 2. small-cell (51) vs adeno (77) (lung)	
developing neurons 3. PPs (3) vs NNs (3)	Carcinoma Genentech (RNA-Seq) 3. small-cell (30) vs adeno (38) (lung)	Carcinoma Genentech (RNA-Seq) 3. lung (38) vs stomach (14) (adeno)

Table 4.

Validation of identified target genes and the combinatorial marker. Minimum AUC-ROC (min AUC-ROC) and JVT *p* values are reporter for the two positive hypotheses and one negative hypothesis for each target genes and the combinatorial marker (comb). The specific datasets and comparisons used for testing of each hypothesis are listed in Table 3 [above](#). The results presented in this table can be reproduced using the code and data available under the GitHub link reported in the methods section.

	measure	CAV1	FHL2	IGFBP7	TAGLN	THBS1	comb
positive hypothesis I	min AUC-ROC	0.78	0.89	0.67	0.78	0.56	0.97
	JVT <i>p</i> value	0.14	0.04	0.30	0.14	0.81	0.01
positive hypothesis II	min AUC-ROC	0.89	0.88	0.73	0.56	0.86	0.92
	JVT <i>p</i> value	0.02	0.03	0.19	0.59	0.04	0.01
negative hypothesis	min AUC-ROC	0.54	0.51	0.51	0.52	0.61	0.51
	JVT <i>p</i> value	0.40	0.76	0.90	0.61	0.06	0.91

significant level) means that it is rare to randomly generate a joint multiview marker with performance equal or better than the tested one. As summarized in [Table 4](#), the combinatorial marker had remarkably low JVT p values ($p = 0.01$) in positive hypotheses I and II, i.e., it is very unlikely to generate a similarly performing combinatorial marker at random. Conversely, in the negative hypothesis, the JVT p value of the combinatorial marker is not significant ($p = 0.91$). The performance of the tested genes individually was varied, with FHL2 showing a significant JVT p value in positive hypothesis I, and FHL2, CAV1 and THBS1 reaching significant JVT p values in positive hypothesis II. It is important to note that our implementation of JVT is conservative, as we consider the minimum discriminative performance on multiple datasets. This may lead to underestimating the performance of individual markers. In sum, the results provided in [Table 3](#) pointed towards CAV1 and FHL2 as promising markers of the mechanical phenotype.

Perturbing expression levels of CAV1 changes cells stiffness

We decided to focus our attention on CAV1 as a potential target for modulating mechanical properties of cells, as it has previously been linked to processes intertwined with cell mechanics. In the context of mechanosensing, CAV1 is known to facilitate buffering of the membrane tension ([45](#)), play a role in β 1-integrin-dependent mechanotransduction ([58](#)) and modulate the mechanotransduction in response to substrate stiffness ([59](#)). CAV1 is also intimately linked with actin cytoskeleton — it was shown to be involved in cross-talk with Rho-signaling and actin cytoskeleton regulation ([46](#), [60–62](#)), filamin A-mediated interactions with actin filaments ([63](#)), and co-localization with peripheral actin. The evidence directly relating CAV1 levels with the mechanical properties of cells ([47](#), [62](#), [65](#), [66](#)) and tissues ([66](#), [67](#)), is only beginning to emerge.

In most of the mechano-transcriptomic datasets considered in our study, the increase in apparent Young's modulus was accompanied by an increase in CAV1 levels (Fig. S7A), corroborating previous reports ([62](#), [65](#), [66](#)). Additionally, we observed that mouse embryonic fibroblasts isolated from CAV1 knock out mice (CAV1KO) are softer than the wild type cells (WT) (Fig. S6). Thus, we set out to test whether artificially decreasing the levels of CAV1 results in cell softening, and conversely, increasing the level of CAV1 in higher cell stiffness. To this end, we perturbed the levels of CAV1 in the cell lines representing two intestine carcinoma types: ECC4, the small-cell carcinoma with a comparably soft phenotype, and TGBC18TKB (TGBC), the adenocarcinoma with a comparatively stiff phenotype.

Before perturbations, we confirmed that TGBC cells have higher levels of CAV1 compared to ECC4 cells on a protein level (Fig. 5A), and that they are characterized by a stiffer phenotype, not only when measured with RT-DC (Fig. 2B, Fig. 5B), but also with atomic force microscopy (AFM) using both standard indentation experiments (Fig. 5C), as well as oscillatory measurements at different frequencies, referred to as AFM microrheology (Fig. 5D).

The three techniques for characterizing mechanical properties of cells — RT-DC, AFM indentation and AFM microrheology — differ in several aspects (summarized in Table S9), most notably in the frequency at which the force is applied to cells during the measurements, with RT-DC operating at the highest frequency (~600 Hz), AFM microrheology at a range of frequencies in-between (3–200 Hz), and AFM indentation operating at lowest frequency (5 Hz) (see Table S9 and Figure S10A). Even though the apparent Young's moduli obtained for TGBC cells were consistently higher than those for ECC4 cells across all three methods, the absolute values measured for a given cell line varied depending on the methods: RT-DC measurements yielded higher apparent Young's moduli compared to AFM indentation, while the apparent Young's moduli derived from AFM microrheology measurements were frequency-dependent and fell between the other two methods (Fig. 5B–D, Fig. S10B). The observed increase in apparent Young's modulus with probing frequency aligns with previous findings on cell stiffening with increased probing rates observed for both AFM indentation ([68](#), [69](#)) and microrheology assays ([70–72](#)).

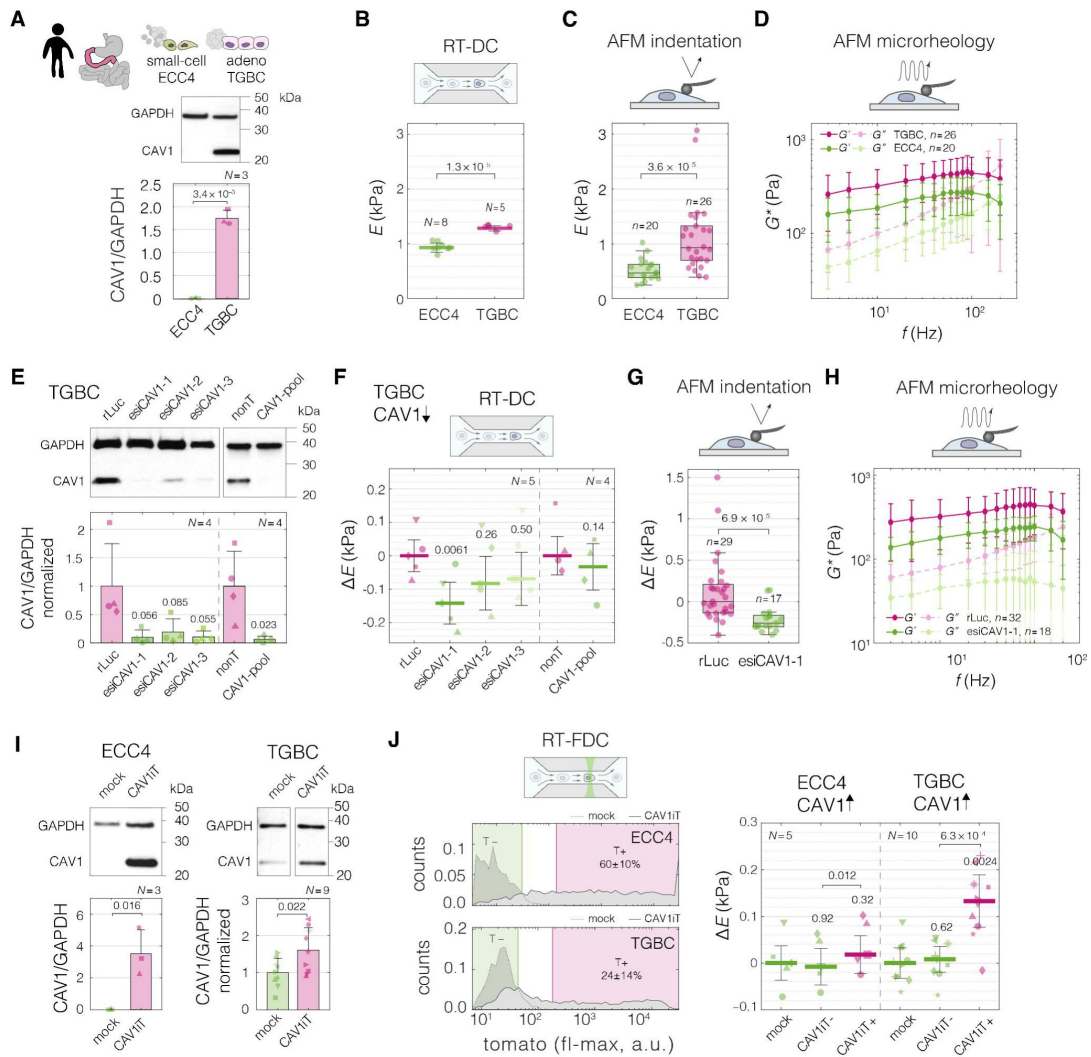


Fig. 5.

Perturbing levels of CAV1 affects the mechanical phenotype of intestine carcinoma cells. (A) CAV1 levels in small-cell (ECC4) and adenocarcinoma (TGBC) cell lines from intestine. (B to D) Mechanical phenotype of ECC4 and TGBC cells measured with RT-DC (B, as in Fig. 2B), AFM indentation (C), and AFM microrheology (D). (E) Verification of CAV1 knock-down in TGBC cells using two knock-down system: three esiRNA constructs (esiCAV1-1, esiCAV1-2, and esiCAV1-3 with rLuc as a control), and pooled siRNA mixture (CAV1-pool with non-targeting mixture (nonT) as a control). (F to H) Mechanical phenotype change of TGBC cells upon CAV1 knock-down as measured by RT-DC (F), AFM indentation (G), and AFM microrheology (H). (I) Verification of transient CAV1 overexpression in ECC4 and TGBC cells. (J) Mechanical phenotype change of ECC4 and TGBC cells upon CAV1 overexpression as measured by RT-FDC. Gating for fluorescence positive and negative cells based on dTomato expression in ECC4 (top) and TGBC (bottom) cells (left-hand side). Fluorescence positive cells correspond to cells expressing CAV1-IRES-dTomato (CAV1IT). For comparison, mock transfection sample is shown in the background (mock). Apparent Young's modulus changes of ECC4 and TGBC cells upon CAV1 overexpression (right-hand side). CAV1IT⁻ and CAV1IT⁺ are dTomato negative and positive cells, respectively. For protein quantification in (A), (E), and (I), representative Western blots (top) as well as quantification of specified replicate numbers *N* (bottom) are shown. In (B), (F) and (J), horizontal lines delineate medians with mean absolute deviation (MAD) as error, datapoints represent medians of *N* experiment replicates, statistical analysis was performed using generalized linear mixed effects model. In (C) and (G), box plots spread from 25th to 75th percentiles with a line at the median, whiskers span 1.5 × interquartile range (IQR), individual datapoints correspond to values obtained for *n* individual cells, statistical analysis was performed using two sample two-sided Wilcoxon rank sum test. In (D) and (H), datapoints correspond to means ± standard deviation of all measurements at given oscillation frequencies for *n* cells. Lines connecting datapoints serve as guides for the eye. *E* – apparent Young's modulus, *G** – complex shear modulus, Δ*E* – apparent Young's modulus change relative to respective control measurements. In (E), (F), (I), and (J), the symbol shapes represent matching experiment replicates.

To decrease the levels of CAV1 in the TGBC cells, we performed knock-down experiments using two RNA interference (RNAi) systems, endoribonuclease-prepared siRNA (esiRNA) targeting three different parts of CAV1 transcript (esiCAV1-1, esiCAV1-2, and esiCAV1-3), and a pool of conventional siRNAs (CAV1-pool) (**Fig. 5E**). All the RNAi approaches resulted in the decrease of the apparent Young's modulus of TGBC cells as measured by RT-DC (**Fig. 5F**, Fig. S7, C and D). The most prominent effect was observed using esiCAV1-1. We further confirmed that CAV1 knock-down with esiCAV1-1 resulted in decreased stiffness of TGBC cells using AFM indentation (**Fig. 5F**) and microrheology (**Fig. 5G**) (for overview of the results from all three methods see Fig. S10C).

To investigate the influence of increased CAV1 levels on cell stiffness, we performed transient overexpression experiments of CAV1 with a dTomato reporter under independent ribosomal entry site, IRES, (CAV1iT) in both ECC4 and TGBC cell lines. At 72 hours post transfection, we observed elevated levels of CAV1 in both cell lines on a protein level in bulk (**Fig. 5I**). Since in the transient overexpression experiments not all of the cells are transfected, we leveraged the possibility to monitor the fluorescence of single cells in parallel with their mechanical phenotype offered by real-time fluorescence and deformability cytometry (RT-FDC) (31) to gate for the fluorescence-positive cells (T⁺, gate marked in magenta in **Fig. 5J**). The fluorescence-positive cells in the CAV1-transfected sample, CAV1iT⁺, showed higher apparent Young's moduli as compared to fluorescence-negative cells in both control sample (mock) and CAV1-transfected sample (CAV1iT⁻, internal control) (**Fig. 5J**, Fig. S7, C and D). The effect was observed in ECC4 as well as TGBC cells. However, it was more pronounced in the TGBC cells, suggesting that the cells may be more responsive to the artificial increase in CAV1 levels when natively expressing a basal level of this protein.

Finally, we performed CAV1 perturbation experiments in a breast epithelial cell model of cancerous transformation, MCF10A-ER-Src cells, in which the Src proto-oncogene can be induced by treatment with tamoxifen (TAM). As previously shown, TAM addition triggers Src phosphorylation and cellular transformation (73), which is associated with F-actin cytoskeletal changes and, after a transient stiffening, the acquisition of a soft phenotype evident at 36 hours post induction (74). We inspected a previously published microarray dataset and determined that the expression of CAV1 diminishes over time after TAM treatment (75) (**Fig. 6A**). We then showed that the decrease of CAV1 could be observed at the protein level 72 hours post induction (**Fig. 6B**), a timepoint at which the TAM-induced MCF10A-ER-Src cells show a significant decrease in cell stiffness (ref (74) and **Fig. 6C**). We next showed that knocking down CAV1 decreased the stiffness of uninduced MCF10A-ER-Src cells, similar to the effect of TAM induction (**Fig. 6D**). Finally, we performed an inverse experiment, in which we rescued the CAV1 levels in TAM-induced MCF10A-ER-Src cells by transient overexpression. The cells with CAV1 overexpression showed a stiff phenotype, corresponding to the one of uninduced cells (**Fig. 6E**).

Taken together, the results obtained with the intestine carcinoma cell lines and MCF10A-ER-Src cells show that CAV1 not only correlates with, but also is causative of mechanical phenotype change.

Discussion

The mechanical phenotype of cells is recognized as a hallmark of many physiological and pathological processes. Understanding how to control it is a necessary next step that will facilitate exploring the impact of cell mechanics perturbations on cell and tissue function (76). The increasing availability of transcriptional profiles accompanying cell state changes has recently been complemented by the ease of screening for mechanical phenotypes of cells thanks to the advent of high-throughput microfluidic methods (77). This provides an opportunity for data-driven identification of genes associated with the mechanical cell phenotype change in a hypothesis-free manner. Here we leveraged this opportunity by performing discriminative

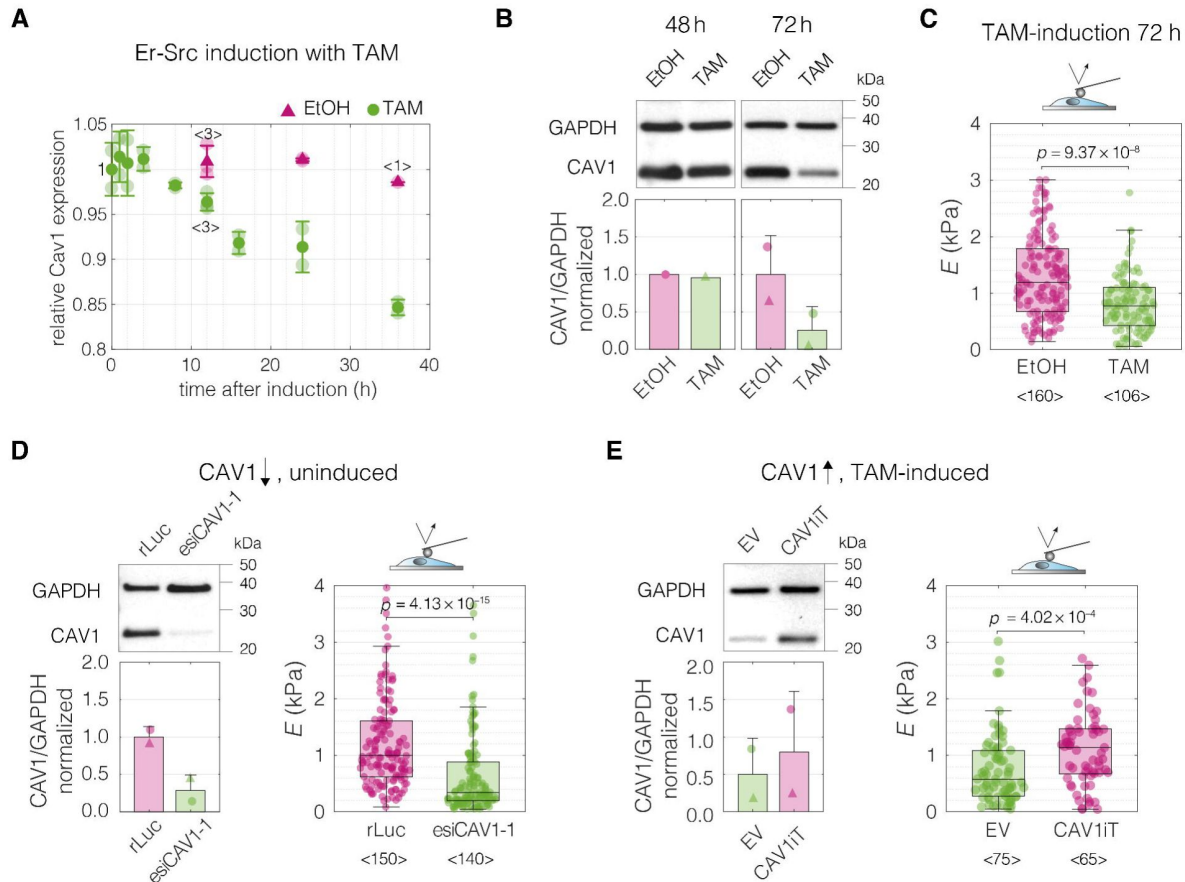


Fig. 6.

Perturbations of CAV1 levels in MCF10A-ER-Src cells result in cell stiffness changes. (A) Inducing transformation of MCF10A-ER-Src cells by tamoxifen (TAM) treatment, as opposed to vehicle control (ethanol, EtOH), causes a decrease of CAV1 expression over time, as captured by microarray analysis (GEO accession number: GSE17941, data previously published in (75)). Datapoints with error bars represent means $\pm$ standard deviation ($N = 2$, unless indicated otherwise). (B) Western blot analysis shows the decrease of CAV1 at protein level 72 h post induction. (C) MCF10A-ER-Src cells show decreased apparent Young's moduli 72 h post TAM induction. (D) CAV1 knock-down in uninduced MCF10A-ER-Src cells results in lowering of the apparent Young's modulus. (E) Overexpression of CAV1 in TAM-induced MCF10A-ER-Src cells causes increase in the apparent Young's modulus and effectively reverts the softening caused by TAM induction (compare to panel c). Box plots in (C to E) spread from 25th to 75th percentiles with a line at the median, whiskers span $1.5 \times$ interquartile range (IQR), individual datapoints correspond to values obtained for individual cells, the number of measured cells per conditions, pooled from $N = 3$ independent experiments, is indicated below each box. Statistical analysis was performed using a two-sided Wilcoxon rank sum test. In (B), (D), and (E), the symbol shapes represent experiment replicates.

network analysis on transcriptomes associated with mechanical phenotype changes to elucidate a conserved module of five genes potentially involved in cell mechanical phenotype regulation. We provided evidence that the inferred conserved functional network module contains an ensemble of five genes that, in particular when combined in a unique combinatorial marker, are universal, specific and trustworthy markers of mechanical phenotype across the studied mouse and human systems. We further demonstrated on the example of a selected marker gene, CAV1, that its experimental up- and downregulation impacts the stiffness of the measured cells. This demonstrates that the level of CAV1 not only correlates with, but also is causative of mechanical phenotype change. The mechanistic insights into how precisely the identified genes are involved in regulating mechanical properties, how they interact with each other, and whether they are universal and dominant in various contexts all remain to be established in future studies.

The workflow presented here is a blueprint for data-driven discovery of cell mechanics markers that can serve as targets for modulating cell mechanical properties. Its key features are the hypothesis-free *modus operandi* and the integration of information from different biological systems, that allows to focus on genes that play a relatively general role in cell mechanics rather than on genes specific to the individual experimental models. Noteworthy, by including the PC loadings in the scores used for thresholding, the PC-corr method implemented for network analysis in our study offers a multivariate alternative to classical co-expression analysis, that highlights not only the correlation between the genes but also their relative importance for separating samples based on their mechanical phenotype. Despite its simplicity, PC-corr offers a robust performance on different types of omics data, and has already proven its efficacy in several studies (28, 29, 78).

The mechanical phenotype of single cells is a global readout of cell's resistance to deformation that integrates contributions from all cellular components. The three techniques implemented for measuring cell mechanics in this study — RT-DC, AFM indentation and AFM microrheology — exert comparatively low deformations ($< 3 \mu\text{m}$, see Table S9), at which the actin cortex is believed to dominate the measured response. However, other cellular components, including the membrane, microtubules, intermediate filaments, nucleus, other organelles, and cytoplasmic packing, can also contribute to the measured deformations (reviewed in detail in (79)) and, for a particular system, it is hard to speculate without further investigation which parts of the cell have a dominant effect on the measured deformability.

Among the target genes elucidated in our analysis, we did not observe enrichment of gene ontology terms related to actin cytoskeleton organization, actomyosin contractility, or cell migration — processes that are typically associated with cell mechanics (Fig. S3). This can be partially explained by looking at the mRNA rather than the protein level, its supramolecular assembly, activation state or localization. Upon closer inspection of the obtained gene targets, we found some links connecting them with cell mechanics in the literature. As indicated above, CAV1 has been shown to be involved in cross-talk with Rho-signaling and actin-related processes, as well as physical interactions with actin (46, 60–64). It is thus conceivable that CAV1 is involved in cell mechanics regulation via its influence on the actin cytoskeleton and its contractility. Furthermore, CAV1 is known to modulate the activation of transcriptional cofactor yes-associated protein, YAP, in response to changes in stiffness of cell substrate (59) and in the mechanical stretch-induced mesothelial to mesenchymal transition (80). YAP is an established transducer of not only various mechanical stimuli, but also of cell shape and the changes in the actin cytoskeleton tension (81), the latter being an important determinant of cell stiffness. Conversely, YAP is an essential co-activator of CAV1 expression (82). In the extended networks (Fig. 3, E and F), we found three further genes that are identified (CYR61, ANKRD1) (83, 84) or implicated (THBS1) (81) as transcriptional targets of YAP. The next identified marker gene, transgelin, TGLN (also known as SM22 α) is an actin-binding protein, that stabilizes actin filaments and is positively correlated with cytoskeletal tension (85). Transgelin is a member of the calponin protein family, one further member of which, calponin 2, CNN2, is present in the broader

sets of genes identified in this study (**Fig. 3**, **E** and **F**, Table S3). The expression of calponin 2, likewise, stabilizes actin filaments and is increased in cells with high cytoskeletal tension (86). Finally, FHL2 is a transcriptional coactivator that is found, together with other LIM domain protein families such as zyxin and paxillin, to localize to actin filaments that are under stress (49–51). When the cytoskeletal tension is low, FHL2 translocates to the nucleus, thus serving as a nuclear transducer of actomyosin contractility (49).

To our knowledge, there are no prior studies that aim at identifying gene signatures associated with single-cell mechanical phenotype changes, in particular across different cell types. There are, however, several studies that investigated changes in expression upon exposure of specific cell types to mechanical stimuli such as compression (87, 88) or mechanical stretch (22, 80, 89), and one study that investigated difference in expression profiles between stiffer and softer cells sorted from the same population (90). Even though the studies concerned with response to mechanical stimuli answer a fundamentally different question (how gene expression changes upon exposure to external forces vs which genes are expressed in cells of different mechanical phenotype), we did observe some similarities in the identified genes. For example, in the differentially expressed genes identified in the lung epithelia exposed to compression (87), three genes from our module overlapped with the immediate response (CAV1, FHL2, TGLN) and four with the long-term one (CAV1, FHL2, TGLN, THBS1). We speculate that this substantial overlap is caused by the cells undergoing change in their stiffness during the response to compression (and concomitant unjamming transition). Another previous study explored the association between the stiffness of various tissues and their proteomes. Despite the focus on the tissue-scale rather than single-cell elasticity, the authors identified polymerase I and transcript release factor (PTRF, also known as cavin 1 and encoding for a structural component of the caveolae) as one of the proteins that scaled with tissue stiffness across samples (91).

As seen from the example of the target genes included in the conserved module, their change is correlated with cell mechanics across all datasets, but it does not always follow the same trend (**Figs. 4** and S7). This non-monotonic relationship between gene expression and the mechanical phenotype change suggests that there may be different regimes at which the expression change in the same direction has an opposite effect on the property of interest. Furthermore, the effect of expression change may be contextual and depend on the state of cells. This observation carries some parallels to the role of several of our target genes in cancer progression. For example, CAV1 has been indicated as both promoting and suppressing cancer progression in a variety of tissues. One way in which this can be reconciled is that the change in CAV1 expression may have different roles depending on the stage of cancer progression (60, 92, 93). A similar ambiguity of their role in cancer progression was indicated for THBS1 (56) and IGFBP7 (52). Of note, a non-monotonic cell stiffness response has also been described for treatments with actin-disrupting drugs. For example, treating cells with Latrunculin B makes cells progressively more deformable up to a certain concentration, beyond which the cells become less deformable again and eventually even stiffer than non-treated cells (see (77) and discussion therein for more references). Apart from characterizing the response regimes, it will be also important to consider the temporal dynamics of cell response to the change in expression of a given gene. Trying to push the cell out of its equilibrium may cause the system to respond actively to counterbalance the induced change, which, in turn, may lead to oscillations in both expression levels of manipulated protein and its effectors, as well as the mechanical properties of the cell.

Among all different types of omics data, looking at the transcriptome is advantageous and disadvantageous at the same time. Its limitation is that mRNA levels do not necessarily reflect protein content in cells. Furthermore, for many proteins it is not the absolute level that has a functional relevance, but rather the protein activation by, for example, phosphorylation or binding with co-activators, or its localization. However, identifying the players at the transcriptome level has the advantage of easy implementation in perturbation experiments with established genetic tools, such as CRISPR-Cas9 technology or RNAi. Furthermore, our analysis

framework is readily applicable to other types of omics data, including proteomic, metabolomic, lipidomic, or glycomic data, the analysis of which would complement our study and provide different insights into the regulation of cell mechanics. Lipidomic data, for example, could reveal possible contributors to cell mechanics related to the composition of the cell membrane.

For the approaches such as the one pioneered in this study to flourish, it is necessary that the mechanical datasets become routinely published and annotated in a manner similar to omics datasets. With the recent advent of high-throughput cell mechanical characterization techniques, such as deformability cytometry methods (77 [↗](#)), the establishment of a database for cell mechanics gains immediate relevance. In our group alone, within the timespan of nine years since the RT-DC method was originally published (31 [↗](#)), we have accumulated over 200,000 individual mechanical characterization experiments, comprising roughly two billion of single cells measured. Once a vast number of mechanics datasets connected to omics profiles is available, it will be straightforward to develop a next generation artificial intelligence algorithm predicting cell stiffness from given omics profiles. Apart from analyzing divergent cell states, the search for mechanical regulators could be complemented by looking into omics data of cells from unimodal populations sorted by their mechanical properties — a pursuit that with the advent of high-throughput methods for mechanics-based sorting of cells, such as sorting RT-DC (94 [↗](#), 95 [↗](#)) or passive filtration-based approaches (90 [↗](#)), becomes a realistic objective.

In conclusion, this work brings together machine learning-based discriminative network analysis and high-throughput mechanical phenotyping to establish a blueprint workflow for data-driven de novo identification of genes involved in the regulation of cell mechanics. Ultimately, identifying ways to tune the mechanical properties on demand will enable turning cell mechanics from a correlative phenomenological parameter to a controllable property. Such control will, in turn, allow us to interfere with important processes such as tissue morphogenesis, cell migration, or circulation through vasculature.

Methods

Cell culture

Glioblastoma cell lines

The glioblastoma dataset contained three primary human brain tumor cell lines (X01, X04, and X08) in three distinct signaling states. The cells were cultured and characterized within a framework of a previous study (29 [↗](#)). In brief, the three signaling states characterized by low, medium, and high activation of STAT3-Ser/Hes3 signaling axis, were maintained by growth media containing fetal bovine serum (serum), epidermal growth factor (EGF), or basic fibroblast growth factor combined with a JAK inhibitor (FGFJI), respectively. Upon thawing, cells were expanded in a serum-free DMEM/F12 medium (10-090-CV, Mediatech, Corning, NY, USA) containing N2 supplement and 20 ng ml⁻¹ EGF (R&D Systems, MN, USA) at 37°C in a 5% oxygen incubator. Each cell line was then plated into three separate flasks and cultured in the DMEM/F12 medium containing N2 supplement and additional supplementation of either serum (10%), EGF (20 ng ml⁻¹), or FGFJI (20 ng ml⁻¹, bFGF, R&D Systems; and 200 nM JAK inhibitor, Calbiochem, Merck Millipore, Germany). Cells were collected for mechanical characterization and RNA-Seq after 5-day exposure to the respective culture conditions (29 [↗](#)).

Carcinoma cell lines

Small-cell and adenocarcinoma cell lines from intestine, stomach and lung were acquired from RIKEN BioResource Research Center, Japan (see Table S4 for the list of cell lines and media). Cells were cultured in growth media supplemented with 5% (TGBC) or 10% (rest) heat-inactivated fetal

bovine serum (10270106, Gibco, ThermoFisher Scientific, MA, USA) and 100 U ml⁻¹/100 µg ml⁻¹ penicillin/streptavidin (15140122, Gibco), at 37°C and 5% CO₂. Sub-culturing was performed using trypsin (25200072, Gibco). Cells were collected for mechanical characterization at 70% confluency. The RNA-Seq data was retrieved from FANTOM5 consortium (33 [↗](#)). Additional transcriptomic datasets were retrieved from the CCLE project (microarray (34 [↗](#)) and RNA-Seq (35 [↗](#))) and from the study conducted by Genentech (36 [↗](#)) (see **Table 1** [↗](#) for overview).

MCF10A PIK3CA cell lines

MCF10A cell line with single-allele PIK3CA H1024R mutation was previously generated by homologous recombination by Horizon Discovery LTD, UK (42 [↗](#)) and was kindly provided, together with an isogenic wild type (wt) control, by L.R. Stephens (Babraham Institute, UK). Cells used for mechanical characterization were cultured in DMEM/F12 medium (31330038, Gibco) supplemented with 5% horse serum (PAA Laboratories), 10 µg ml⁻¹ insulin (I9278, Sigma-Aldrich, MO, USA), 0.2 µg ml⁻¹ hydrocortisone (H0888, Sigma-Aldrich), 0.1 µg ml⁻¹ cholera toxin (C8052, Sigma-Aldrich), and 100 U ml⁻¹/100 µg ml⁻¹ penicillin/streptomycin (15140122, Gibco). The wt cells were additionally supplemented with 10 ng ml⁻¹ EGF (E9644, Sigma-Aldrich), while mutant cell lines were maintained without EGF. Sub-confluent cells were collected for mechanical characterization using trypsin (25200056, Gibco). Mechanical data were collected from two biological replicates with three technical repetitions each. The RNA-Seq data was retrieved from a previous study (37 [↗](#)), in which cells were cultured in a reduced medium (DMEM/F12 supplemented with 1% charcoal dextran treated fetal bovine serum, 0.2 µg ml⁻¹ hydrocortisone and 0.1 µg ml⁻¹ cholera toxin).

Induced pluripotent stem cells

F- and C-class iPSCs were derived through reprogramming of murine fetal neural progenitor cells with Tet-On system for doxycycline-inducible expression of OSKM (Oct4, Sox2, Klf4, cMyc) factors in a previous study (30 [↗](#)). Both iPSCs classes were cultured on 0.1% gelatin-coated dishes in FCS/LIF medium (DMEM+Glutamax (61965059, Gibco), 15% fetal calf serum (Pansera ES, PAN-Biotech, Germany), 100 µM β-mercaptoethanol (PAN-Biotech), 2 mM L-glutamine, 1 mM sodium pyruvate, 1× nonessential amino acids, 15 ng ml⁻¹ recombinant LIF (MPI-CBG, Dresden, Germany)). The F-class iPSCs were additionally supplemented with 1 µg ml⁻¹ doxycycline, and the C-class iPSCs with a mixture of two inhibitors (2i): 1 µM MEK inhibitor (PD0325901, Calbiochem) and 3 µM GSK3 inhibitor (CH99021, Calbiochem). Cells were passaged and harvested using 0.1% trypsin solution. The mechanical characterization was performed not earlier than at the 27th day of reprogramming (30 [↗](#)). The microarray expression profiles were retrieved from a previous study, in which the F- and C-class iPSCs were derived from embryonic fibroblasts using similar doxycycline-inducible OSKM expression system (38 [↗](#)).

Developing Neurons

For isolation of neurons at different developmental stages, we used a double-reporter mouse line *Btg2^{RFP}/Tubb3^{GFP}*, in which the proliferating progenitors are double negative (RFP-/GFP-), newborn neurons are double positive (RFP+/GFP+), and the cells positive for RFP but negative for GFP are (RFP+/GFP-) are the proliferating progenitors that were not used in this study. Lateral cortices dissected from E14.5 murine embryos were dissociated using a papain-based neural dissociation kit (Miltenyi Biotech, Germany) and the cell populations of interest were separated based on the RFP/GFP expression using FACS as described in detail elsewhere (39 [↗](#)). The three types of sorted cells were then subjected to RNA sequencing (39 [↗](#)) and mechanical characterization.

Mouse embryonic fibroblasts

Previously established, immortalized WT and CAV1KO mouse embryonic fibroblasts derived from WT and CAV1KO littermate C57BL/9 mice (96 [DOI](#)) were used in this study. Cells were cultured in DMEM medium (11960044, Gibco), supplemented with 10% fetal bovine serum (10270106, Gibco), 2 mM glutamine (25030081, Gibco), 100 U ml⁻¹/100 µg ml⁻¹ penicillin/streptomycin (15070063, Gibco), at 37°C and 5% CO₂. Sub-confluent cells were collected for mechanical measurements by trypsinization (25200056, Gibco).

MCF10A ER-Src cell line

The MCF10A ER-Src cells were a kind gift from K. Struhl (Harvard Medical School, MA, USA). ER-Src is a fusion of the v-Src (viral non-receptor tyrosine kinase) with the ligand-binding domain of the estrogen receptor, that can be induced by cell treatment with tamoxifen (TAM) (73 [DOI](#)). Cells were grown at 37°C under 5% CO₂ in DMEM/F12 medium (11039047, Gibco), supplemented with 5% charcoal (C6241, Sigma-Aldrich)-stripped horse serum (16050122, Gibco), 20 ng ml⁻¹ EGF (AF-100-15, Peprotech), 10 mg ml⁻¹ insulin (I9278, Sigma-Aldrich), 0.5 mg ml⁻¹ hydrocortisone (H0888, Sigma-Aldrich), 100 ng ml⁻¹ cholera toxin (C8052, Sigma-Aldrich), and 100 U ml⁻¹/100 µg ml⁻¹ penicillin/streptomycin (15070063, Gibco). To induce the Src expression cells were plated at 50% confluency, and after allowing to adhere for 24 h, treated with 1 µM 4OH-TAM (H7904, Sigma-Aldrich) or with identical volume of ethanol as a control. Cells were characterized in adherent state using AFM at timepoints specified in the text.

Mechanical measurements

Mechanical characterization of cells using RT-DC

RT-DC measurements for mechanical characterization of cells were performed at room temperature according to previously established procedures (97 [DOI](#)). In brief, cells were harvested by trypsinization (adherent cells) and/or centrifugation at 400 g for 3–5 min, and suspended in a measurement buffer (MB). MB (osmolality 310–315 mOsm kg⁻¹, pH 7.4) was based on phosphate buffered saline without Mg²⁺ and Ca²⁺ and contained 0.5% or 0.6% (w/w) methylcellulose (4000 cPs, Alfa Aesar, Germany) for increased viscosity. Cells were introduced into a microfluidic chip using a syringe pump (NemeSys, Cetoni, Germany), and focused into a 300-µm long channel constriction (with a square cross-section of 20 × 20 or 30 × 30 µm) by sheath flow infused at a flow rate three times as high as that of the cell suspension. The imaging was performed at the end of the channel constriction (Fig. S1B) at 2,000 frames s⁻¹. The cell area and deformation were derived from the fitted cell contours in real-time by the acquisition software (*ShapeIn2*; Zellmechanik Dresden, Germany). Apparent Young's modulus values were assigned to each cell based on its area and deformation under given experimental conditions (flow rate, channel size, viscosity of the medium, temperature) using a look-up table obtained through numerical simulations of an elastic solid (32 [DOI](#)) with the aid of *ShapeOut* (*ShapeOut 1.0.10*; <https://github.com/ZELLMECHANIK-DRESDEN/ShapeOut> [DOI](#); Zellmechanik Dresden). The events were filtered for area ratio (the ratio between the area enclosed by the convex hull of the cell contour and the raw area enclosed by the contour) to discard incomplete contours or cells with rough surface, and for cell area and aspect ratio to discard derbies and doublets. Experimental details (channel sizes, flow rates, measurement buffers) and gates used for filtration in respective datasets are listed in Table S5.

Mechanical characterization of cells using AFM

For AFM measurements, cells were seeded on glass bottom dishes (FluoroDish; FD35100, WPI, FL, USA) at least one day in advance. Mechanical characterization was performed on adherent cells in a sub-confluent culture in CO₂-independent medium (18045054, Gibco) at 37°C (temperature was maintained by a petri dish heater, JPK Instruments, Germany). AFM measurements on TGBC and

ECC4 cell lines were conducted on a *Nanowizard 4* (JPK Instruments/Bruker). Tip-less cantilevers (PNP-TR-TL, nominal spring constant $k = 0.08 \text{ N m}^{-1}$, Nanoworld, Switzerland) decorated a polystyrene bead of 5- μm diameter (PS-R-5.0, microParticles, Germany) each were used as the indenters. The cantilever spring constants were measured prior to each experiment using the thermal noise method implemented in the *JPK SPM* software (JPK Instruments). For each cell three indentation curves were recorded with a piezo extension speed of $5 \mu\text{m s}^{-1}$ to a maximum set force of 2 nN. For the microrheology analysis, the cantilever was lowered using a piezo extension speed of $5 \mu\text{m s}^{-1}$ until a force set point of 1 nN was reached, corresponding to an approximate indentation depth δ_0 of 1 μm . The lowered cantilever was then oscillated by a sinusoidal motion of the piezo elements at an amplitude of 10 nm for a period of 10 cycles. The oscillations were performed sequentially at different frequencies in the range of 3–200 Hz. Indentation experiments on MCF10A ER-Src cells were conducted as described above, except different tip-less cantilevers (Arrow TL1, nominal spring constant $k = 0.035\text{--}0.045 \text{ N m}^{-1}$, Nanoworld) with a 5- μm bead glued at the end were used as the indenter.

AFM indentation data analysis

Recorded force-distance curves were converted into force-indentation curves and fitted in JPK data processing software (*JPK DP*, JPK Instruments/Bruker) using Sneddon's modification of the Hertz model for a spherical indenter (98 [C](#)):

$$F = \frac{E}{1-\nu^2} \left(\frac{a^2+r^2}{2} \ln \frac{r+a}{r-a} - ar \right), \quad (3)$$

With

$$\delta = \frac{a}{2} \ln \frac{r+a}{r-a}, \quad (4)$$

where F denotes the indentation force, E the elastic modulus, ν the Poisson's ratio, a the radius of the projected contact area formed between the sample and the indenter, r the radius of the indenter, and δ the indentation depth. Poisson ratio was set to 0.5. The curves were fitted to a maximal indentation of 1.5 μm .

AFM microrheology data analysis

The force and indentation signals from oscillatory measurements were fitted using a sinusoidal function to extract the amplitude and phase angle of each signal. Data were analyzed analogously to the procedure described by Alcaraz et al. (70 [C](#)) but for a spherical not a pyramidal indenter. Briefly, the method relies on the linearization of the Hertz model for a spherical indenter due to small oscillations by using the first term of the Taylor expansion and subsequent transformation to the frequency domain:

$$F(\omega) = 2 \frac{E^*(\omega)}{(1-\nu^2)} \sqrt{R\delta_0} \delta(\omega), \quad (5)$$

where $F(\omega)$ and $\delta(\omega)$ are the force and indentation signals in the frequency domain, respectively, $E^*(\omega)$ is the complex Young's modulus, ν is the Poisson's ratio assumed to be 0.5, R is the radius of the indenter and ω is the angular frequency. The complex shear modulus $G^*(\omega)$ can be written using $G^*(\omega) = \frac{E^*(\omega)}{2(1+\nu)}$:

$$G^*(\omega) = G'(\omega) + iG''(\omega) = \frac{(1-\nu)}{4\sqrt{R\delta_0}} \frac{F(\omega)}{\delta(\omega)}, \quad (6)$$

where $G'(\omega)$ is the storage modulus and $G''(\omega)$ is the loss modulus. The ratio of the force $F(\omega)$ and indentation $\delta(\omega)$ is calculated from the measured amplitudes $A^F(\omega)$ and $A^\delta(\omega)$ and the phase shifts $\theta^F(\omega)$ and $\theta^\delta(\omega)$ of the oscillatory signals (99 [↗](#)):

$$\frac{F(\omega)}{\delta(\omega)} = \frac{A^F(\omega)}{A^\delta(\omega)} e^{i(\theta^F(\omega) - \theta^\delta(\omega))}, \quad (7)$$

where the difference of the phase shifts ($\theta^F(\omega) - \theta^\delta(\omega)$) is in the range of 0° (elastic solid) and 90° (viscous fluid). Furthermore, the hydrodynamic drag contribution to the cantilever oscillation was estimated and subtracted from the complex shear modulus as previously described (100 [↗](#)):

$$G^*(\omega) = \frac{(1-\nu)}{4\sqrt{R\delta_0}} \left[\frac{F(\omega)}{\delta(\omega)} - i\omega b(0) \right], \quad (8)$$

where $b(h)$ is the hydrodynamic drag coefficient function measured from non-contact oscillations of the cantilever at different distances h from the sample, and $b(0)$ is the extrapolation to distance 0 from the sample. For PNP-TR-TL cantilevers, the hydrodynamic drag coefficient was estimated to be $b(0) = 5.28 \mu\text{N s m}^{-1}$.

Perturbation experiments

CAV1 knock-down

For RNAi experiments, cells were transfected using RNAiMax reagent (13778030, Thermo Fisher Scientific) and a reverse transfection protocol. Per transfection, 200 ng of esiRNA (Eupheria Biotech, Germany) or 300 ng of ON-TARGETplus siRNA (Dharmacon, CO, USA) and 2 μl RNAiMax were prepared in OptiMEM (31985062, Gibco) according to the manufacturer's instructions and pipetted onto 12-well plates (see Table S6 for full list of siRNAs used). Cells in 1 ml of culture medium were plated on top of the transfection mix at a density allowing for sub-confluent growth within the experimental timeframe. 72 h post transfection, cells were collected for the mechanical characterization and Western blot analysis.

Plasmid for CAV1 overexpression

The cDNA of *CAV1* was amplified by PCR, introducing NheI and XhoI restriction sites in the flanking regions. The PCR product was then cloned into the pCGIT destination vector (a kind gift from P. Serup, University of Copenhagen, Denmark) under the CAG promoter and with dTomato fluorescent marker under internal ribosomal entry site (IRES) downstream of *CAV1*.

Transient CAV1 overexpression in ECC4 and TGBC cells

ECC4 and TGBC cells were transiently transfected with the *CAV1* overexpression plasmid by electroporation (Neon Transfection System, MPK5000, Thermo Fisher Scientific). Per transfection 0.3×10^6 ECC4 cells, or 0.2×10^6 TGBC cells were mixed with 1 μg of plasmid DNA in PBS. Electroporation was conducted using 10 μl Neon tips (MPK1096, Thermo Fisher Scientific) and a program of two pulses of 1050 V and 30 ms duration each. Electroporated cells were transferred to 500 μl of fresh culture medium in a 24-well plate. The cells were collected for mechanical characterization and Western blot analysis 72 h post transfection. To identify fluorescent cells during mechanical characterization, the combined real-time fluorescence and deformability cytometry (RT-FDC) (19 [↗](#)) setup was used, and the maximum intensity of the fluorescence signal from channel 2 (excitation 561 nm, 10% laser power; collection 700/75) was utilized for gating.

Transient CAV1 overexpression in MCF10A-ER-src cells

MCF10A-ER-src cells were transiently transfected with the CAV1 overexpressing plasmid using Effectene transfection reagent (301425, Qiagen). One day before transfection, cells were seeded on glass bottom 35-mm dishes (FluoroDish; FD35100, WPI, FL, USA) at a density of 20,000 cells per well. Transfection was performed according to the manufacturer's instruction using 75 μ l EC buffer, 0.6 μ g plasmid DNA, 4.8 μ l Enhancer and 6 μ l Effectene reagent per well. 24 h post transfection cells were induced with 1 μ M TAM. Mechanical analysis was performed after additional 72 h of culture.

Western blotting

For Western blot analysis of carcinoma and MCF10A-ER-Src cell lines, cell pellets were collected in parallel with mechanical measurements and lysed using ice-cold RIPA buffer (89900, ThermoFisher Scientific) supplemented with protease/phosphatase inhibitor cocktail (78441, ThermoFisher Scientific) and benzonase (E1014, Sigma-Aldrich). The lysates were cleared at 4°C by 10-minute sonication followed by 10-minute centrifugation at 16,900 g. Obtained supernatants were mixed with Laemmli buffer (final concentration: 62.5 mM Tris-HCl (pH 6.8), 2% SDS, 10% glycerol, 5% β -mercaptoethanol, 0.01% bromophenol blue), boiled (5 min at 95°C), and separated by SDS-PAGE electrophoresis on 4–20% gradient gels (Mini-PROTEAN TGX Precast Gels; 4561093, Biorad, CA, USA) in MOPS SDS Running buffer (B0001, ThermoFisher Scientific). After transferring the proteins onto a PVDF membrane (Merck Millipore), the membranes were blocked in TBS-T (20 mM Tris, 137 mM NaCl, 0.1% Tween) containing 5% w/v skimmed milk powder (T145.1, Carl Roth, Germany) for 40 minutes. Next, membranes were incubated with the primary anti-Cav1 (1:1000; D46G3; #3267, Cell Signaling Technology, MA, USA) and anti-GAPDH (1:5000; ab9485, Abcam, UK) antibodies at 4°C overnight in 5% milk/TBS-T, washed, and incubated with anti-rabbit HRP-conjugated secondary antibody (1:4000; ab97069, Abcam). Chemiluminescence detection was performed using Pierce Enhanced Chemi-Luminescence (ECL) substrate (32109, ThermoFisher Scientific) and ECL films (GE28-9068-37, Merck Millipore). Films were developed in an OptiMax X-ray film processor (KODAK, NY, USA). Quantitative analysis was performed on scanned films using the gel analysis tool in *Jimage* version 2.0.0-rc-69/1.52p (<https://imagej.nih.gov/>). For western blot analysis of MEFs the same anti-Cav1 antibody (1:1000; D46G3; #3267, Cell Signaling) was used, and anti-tubulin antibody (1:2000; DM1A; #3873, Cell Signaling) was used as a loading control. Goat anti-mouse 680 and goat anti-rabbit 800 (1:2000; A28183 and A32735, ThermoFisher Scientific) antibodies were used for secondary detection. Membranes were scanned with the Odyssey imaging system (LI-COR Biosciences, NE, USA).

Computational analysis

Transcriptomic datasets

Transcriptomic datasets were retrieved from online databases (Gene Expression Omnibus, GEO and DNA Data Bank of Japan, DDBJ) with accession numbers listed in **Table 1**. Overview of experimental detail for RNA profiling procedures and data analysis in individual datasets is presented in Table S7. The IDs of samples used in respective categories in each dataset are listed in Table S8. In case of multiple entries for the same gene in a given transcriptomic dataset, the expression values were averaged, so that only one entry per gene and sample was available.

PC-corr analysis

Before performing the PC-corr analysis, the glioblastoma and iPSC datasets were intersected and normalized by taking the log₁₀ (glioblastoma dataset) or z-score (iPSC dataset) of the subset of 9,452 overlapping genes. The PC-corr analysis was conducted on individual datasets as described in detail elsewhere (28). In brief, PCA was performed using *svd* function in *MATLAB* (R2020a,

MathWorks, MA, USA) on normalized datasets. The original PC loadings from the component providing good separation of sample categories (PC1 for both analyzed datasets) were processed in a two-step procedure including the normalization and scaling. The processing of the PC loadings is performed to adjust the distribution of the loadings to the range of Pearson's correlation values $[-1, 1]$, so that they are comparable when computing the *PC-corr* value. The normalization was performed using a custom function developed previously (28) of the following formula:

$$V_i^* = \text{sgn}(V_i^0) \log_{10} \left(1 + \frac{|V_i^0|}{\langle |V^0| \rangle} \right), \quad (9)$$

where V_i^* denotes the normalized loading corresponding to the i -th feature, V_i^0 the original loading corresponding to the i -th feature, and $\langle |V^0| \rangle$ the average of all absolute loadings of the vector V^0 .

The normalized loadings were then scaled to fall on the interval $[-1, 1]$ using a previously developed custom function (28):

$$V_i = \text{sgn}(V_i^*) \frac{|V_i^*| - \min(|V^*|)}{\max(|V^*|) - \min(|V^*|)}, \quad (10)$$

where V_i denotes the processed loading corresponding to the i -th feature, and $|V^*|$ the vector containing absolute values of all normalized loadings.

The *PC-corr* values for each pair of features were computed according to Equation 1. The *PC-corr* results of the glioblastoma and iPSC datasets were combined as described in the results section. Gene pairs showing different *PC-corr* signs were masked by setting the *PC-corr*^{comb} to zero. The genes and edges comprising the network were obtained via thresholding strategies described in the main text. The network was visualized using *cytoscape* (*cytoscape* 3.8.0; <https://cytoscape.org/>) (101).

Combinatorial marker

To compute a combinatorial marker associated with a gene functional network module composed of n genes, we use the following three-step procedure:

(1) Dataset normalization

To scale the features to a comparable range and reduce the dominant influence of highly expressed genes, each dataset is normalized. Possible normalization approaches include logarithm normalization ($x = \log(x + 1)$) and z-score normalization. Since both normalization approaches lead to comparable results, we decided to proceed with the logarithm normalization because it is one of the most widely adopted in computational genomics for combinatorial markers (102).

(2) Direction alignment

The direction of the gene expression change between different samples is analyzed and, if necessary, aligned. The pairwise Pearson's correlation of the n genes from the dataset used for inference of the combinatorial marker is computed. If all pairs of genes are positively correlated between each other there is no need of direction alignment — this was the specific case in our study. Otherwise, the directions of genes whose correlation with the reference gene is negative need to be aligned before the compression. The reference gene for the direction alignment is the gene with the highest average pairwise Pearson's correlation with the other $n-1$ genes in the functional module. The alignment is performed by subtracting the mean value of the normalized

expression across samples, $\bar{g}$, from the normalized expression of the given gene, g , inverting its trend using the multiplication by -1 , and finally adding again the mean value to regain the original expression level:

$$\text{align}(g) = -(g - \bar{g}) + \bar{g} = 2\bar{g} - g. \quad (11)$$

Once defined, the aligned values should be used for any further validation analysis, including the computation of the JVT. The alignment step is necessary to make sure that the information contained in the anticorrelated genes does not annihilate each other during the compression into the combinatorial marker. Below, an example is provided to illustrate this issue.

(3) Compression

To perform compression and obtain the combinatorial marker g_{comb} we employ one of the most employed compression operators in computational genomics (102 [↗](#)), the mean operator:

$$g_{comb} = \frac{1}{n} \sum_{i=1}^n g_i, \quad (12)$$

where g_i indicates the normalized and aligned expression value of the i -th gene of the functional module from which the combinatorial marker is derived.

To illustrate the importance of the alignment, let us consider a simple example of two anticorrelated genes in four samples: $g_1 = [1 \ 1 \ 3 \ 3]$ and $g_2 = [3 \ 3 \ 1 \ 1]$. When the compression is performed without alignment, following values of the combinatorial marker are obtained: $g_{comb} = \frac{g_1 + g_2}{2} = [2 \ 2 \ 2 \ 2]$. The so-obtained combinatorial marker is non-discriminative, even though the individual genes are. On the contrary, if the alignment function is applied prior to compression: $g_{comb} = \frac{g_1 + \text{align}(g_2)}{2} = \frac{[1 \ 1 \ 3 \ 3] + [4 - [3 \ 3 \ 1 \ 1]]}{2} = \frac{[1 \ 1 \ 3 \ 3] + [1 \ 1 \ 3 \ 3]}{2} = [1 \ 1 \ 3 \ 3]$, original discriminative information is conserved in the combined marker.

Joint-view trustworthiness (JVT)

The single-view trustworthiness measure was recently introduced by us in studies on pattern recognition to assess the extent to which the geometrical discrimination of samples of a dataset might emerge at random along a dimension of embedding in a geometrical space (78 [↗](#), 103 [↗](#)). In brief, the single-view trustworthiness measure is an empirical p value computed from a null model distribution obtained by a resampling technique, which randomly shuffles the labels of the samples and computes what is the probability to generate at random a matching between sample labels and sample geometrical location that offers a discrimination that is equal or larger than the one tested. The obtained p value assesses whether the visualized and measured sample discrimination along a dimension of a geometrical space is significant (because rare to appear at random) or no significant (because frequent to appear at random). This is particularly useful to assess the trustworthiness of a discriminative result when the number of samples for each class is small or when it is unbalanced, as is the case for some datasets in our study. To assess the trustworthiness of a marker's discrimination performance jointly on many datasets, we introduce a joint-view extension to this method which we refer to as the joint-view trustworthiness (JVT).

To ensure that the proposed markers have a joint multiview discrimination that is rare to obtain by chance, JVT samples markers at random from the data and compares their performance to the one of predicted targets according to the following procedure:

(1) Data preparation

Collect datasets that support (positive hypothesis: for instance, discriminative presence of a cell mechanic phenotype: soft/stiff) or not support (negative hypothesis: for instance, discriminative absence of a cell mechanic phenotype: soft/stiff) your hypothesis, and make sure that you consider

for all of them only the features (genes) that are common to each dataset in your study.

(2) Data normalization

Perform only when computing combinatorial marker (see point (1) in the combinatorial marker section above).

(3) Null model distribution sampling and p-value estimation

- *Single marker test:* Sample at random a gene and extract its expression from each dataset, compute its joint multiview discrimination performance as the minimum performance measure (we adopted the AUC-ROC because it is one of the most used in classification assessment, but any classification performance measure can be employed) across the datasets. Repeat this procedure sampling at random for T times (in our study we used $T = 10,000$) a gene from the datasets and computing its minimum classification performance measure across the datasets. The ensemble of the T minimum classification performance measures can be used to draw an empirical distribution that forms the null model. The p value of the tested marker is computed counting the proportion of genes that within the T samplings have a minimum classification performance that is equal or larger than that for the tested marker. Please note that here we compute the joint multiview discrimination performance using the minimum performance across the datasets because we pursue a conservative estimation. Other operators, such as mean, median, or mode can be employed instead of the minimum operator to make the JVT estimation less conservative.
- *Combinatorial marker test:* Given a combinatorial marker of m genes, sample at random m genes and extract their expressions from each dataset, compute the combinatorial marker (apply the same compression formula of the tested combinatorial marker) joint multiview discrimination performance as the minimum performance measure (we adopted the AUC-ROC, see point (3.1) above for details) across the datasets. Repeat this procedure T times (in our study we used $T = 10,000$). The p value is computed as for the single marker test (see point (3.1) above).

The JVT pseudocode and time complexity analysis are provided in Supplementary Material. In brief, the overall complexity of JVT considering a scenario like in our study is $O(Z+T)$, i.e., JVT is linear in Z (number of common genes in the datasets) and T (number of samplings). The JVT code (in MATLAB, R and Python) and datasets to replicate the results in **Table 4** of this study are available at: <https://github.com/biomedical-cybernetics/Joint-View-trustworthiness-JVT>.

Statistical analysis

The RT-DC datasets were compared using generalized linear mixed effects models with the aid of *ShapeOut* (*ShapeOut 1.0.1*; <https://github.com/ZELLMECHANIK-DRESDEN/ShapeOut>; Zellmechanik Dresden) as described in detail elsewhere (104). AFM datasets were compared using two-sided Wilcoxon rank sum test in *MATLAB (R2020a, MathWorks)*. Western blot results were compared using a two-sided two-sample t-test in *MATLAB (R2020a, MathWorks)*.

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Author contributions

JG and CVC conceived the project. YG under supervision of CVC performed the computational analysis on transcriptomics data. CVC invented the joint-view trustworthiness (JVT) theory and algorithm. SSA, under the supervision of CVC, wrote the JVT pseudo-code and the time complexity analysis. CVC coded the MATLAB function and SSA coded the R and Python functions. All codes were benchmarked by SSA, YG and CVC. MU, MW, MH, MK, and NT performed the measurements of mechanical states. MU and MW performed the genetic manipulation experiments. SA and AT provided methodological support with AFM measurements and data analysis. JD and OF were involved in MCF10A PIK3CA experiments. MD under supervision of FC isolated the developing neurons. FL under supervision of MADP performed MEF experiments. MU and MW analyzed the experimental data. MU visualized the data and prepared figures. MU, with the support of JG, CVC, and YG prepared the initial version of the manuscript. All authors revised and edited the manuscript. JG, CVC, MADP and AT acquired funding.

Competing interests

S.A., M.K., and J.G. are co-founders and shareholders of the company Rivercyte GmbH that is commercializing deformability cytometry technology. The remaining authors declare no competing interests.

Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the manuscript or available on public repositories as detailed below. The transcriptomic data used in this study were obtained from public repositories, their accession numbers are listed in **Table 1** [↗](#). The mechanical characterization data are available as a collection on figshare: <https://doi.org/10.6084/m9.figshare.c>

.5399826 [DOI](#). The MATLAB code for performing the PC-corr analysis was based on the code deposited alongside a previous publication (28 [DOI](#)), accessible on GitHub: https://github.com/biomedical-cybernetics/PC-corr_net [DOI](#). The JVT code (in MATLAB, R and Python) and datasets for replicating the results presented in **Table 4** [DOI](#) are available on GitHub: <https://github.com/biomedical-cybernetics/Joint-View-trustworthiness-JVT> [DOI](#) and figshare: <https://doi.org/10.6084/m9.figshare.20123159> [DOI](#).

References

1. Guck J., Chilver E. R. (2013) **Mechanics meets medicine** *Science Translational Medicine* **5**:3–6
2. Nematbakhsh Y., Lim C. T. (2015) **Cell biomechanics and its applications in human disease diagnosis** *Acta Mechanica Sinica* **31**:268–273
3. Lecuit T., Lenne P. F. (2007) **Cell surface mechanics and the control of cell shape, tissue patterns and morphogenesis** *Nature Reviews Molecular Cell Biology* **8**:633–644
4. Hannezo E., Heisenberg C. P. (2019) **Mechanochemical Feedback Loops in Development and Disease** *Cell* **178**:12–25
5. Suresh S. (2007) **Biomechanics and biophysics of cancer cells** *Acta Materialia* **55**:3989–4014
6. Gensbittel V., Kräter M., Harlepp S., Busnelli I., Guck J., Goetz J. G. (2021) **Mechanical Adaptability of Tumor Cells in Metastasis** *Developmental Cell* **56**:164–179
7. Fletcher D. A., Mullins R. D. (2010) **Cell mechanics and the cytoskeleton** *Nature* **463**:485–492
8. Chugh P., Paluch E. K. (2018) **The actin cortex at a glance** *Journal of Cell Science* **131**:1–9
9. Kelkar M., Bohec P., Charras G. (2020) **Mechanics of the cellular actin cortex: From signalling to shape change** *Current Opinion in Cell Biology* **66**:69–78
10. Seltmann K., Fritsch A. W., Käs J. A., Magin T. M. (2013) **Keratins significantly contribute to cell stiffness and impact invasive behavior** *Proceedings of the National Academy of Sciences of the United States of America* **110**:18507–18512
11. Patteson A. E., Carroll R. J., Iwamoto D. V., Janmey P. A. (2020) **The vimentin cytoskeleton: When polymer physics meets cell biology** *Physical Biology* **18**
12. Kubitschke H., Schnauss J., Nnetu K. D., Warmt E., Stange R., Kaes J. (2017) **Actin and microtubule networks contribute differently to cell response for small and large strains** *New Journal of Physics* **19**
13. Chang Y. C., Nalbant P., Birkenfeld J., Chang Z. F., Bokoch G. M. (2008) **GEF-H1 couples nocodazole-induced microtubule disassembly to cell contractility via RhoA** *Molecular Biology of the Cell* **19**:2147–2153
14. Zhou E. H., Treppe X., Park C. Y., Lenormand G., Oliver M. N., Mijailovich S. M., Hardin C., Weitz D. A., Butler J. P., Fredberg J. J. (2009) **Universal behavior of the osmotically compressed cell and its analogy to the colloidal glass transition** *Proceedings of the National Academy of Sciences* **106**:10632–10637
15. Guo M. *et al.* (2017) **Cell volume change through water efflux impacts cell stiffness and stem cell fate** *Proceedings of the National Academy of Sciences*
16. Caille N., Thoumine O., Tardy Y., Meister J. J. (2002) **Contribution of the nucleus to the mechanical properties of endothelial cells** *Journal of Biomechanics* **35**:177–187

17. Chugh P., Clark A. G., Smith M. B., Cassani D. A. D., Dierkes K., Ragab A., Roux P. P., Charras G., Salbreux G., Paluch E. K. (2017) **Actin cortex architecture regulates cell surface tension** *Nature Cell Biology* **19**:689–697
18. Toyoda Y., Cattin C. J., Stewart M. P., Poser I., Theis M., Kurzchalia T. V., Buchholz F., Hyman A. A., Müller D. J. (2017) **Genome-scale single-cell mechanical phenotyping reveals disease-related genes involved in mitotic rounding** *Nature Communications* **8**:1–10
19. Rosendahl P. *et al.* (2018) **Real-time fluorescence and deformability cytometry** *Nat Methods* **15**:355–358
20. Wang J., Lü D., Mao D., Long M. (2014) **Mechanomics: An emerging field between biology and biomechanics** *Protein and Cell* **5**:518–531
21. Putra V. D. L., Song M. J., McBride-Gagyi S., Chang H., Poole K., Whan R., Dean D., Sansalone V., Knothe Tate M. L. (2020) **Mechanomics Approaches to Understand Cell Behavior in Context of Tissue Neogenesis, During Prenatal Development and Postnatal Healing** *Frontiers in Cell and Developmental Biology* **7**:1–8
22. Zhang F., Wang J., Lü D., Zheng L., Shangguan B., Gao Y., Wu Y., Long M. (2020) **Mechanomics analysis of hESCs under combined mechanical shear, stretch, and compression** *Biomechanics and Modeling in Mechanobiology* <https://doi.org/10.1007/s10237-020-01378-5>
23. Lang M. (2008) **Lighting up the mechanome** *Frontiers of engineering: Reports on leading-edge engineering from the 2007 symposium*
24. Van Loon J. J. W. A. (2009) **Mechanomics and physicomics in gravisensing** *Microgravity Science and Technology* **21**:159–167
25. Song M. J., Brady-Kalnay S. M., McBride S. H., Phillips-Mason P., Dean D., Tate M. L. K. (2012) **Mapping the mechanome of live stem cells using a novel method to measure local strain fields in situ at the fluid-cell interface** *PLoS ONE* **7**:1–12
26. Song M. J., Dean D., Knothe Tate M. L. (2013) **Mechanical modulation of nascent stem cell lineage commitment in tissue engineering scaffolds** *Biomaterials* **34**:5766–5775
27. Wang H. *et al.* (2021) **Mechanomics Biomarker for Cancer Cells Unidenti fi able through Morphology and Elastic Modulus** *Nano Letters* **21**:1538–1545 <https://doi.org/10.1021/acs.nanolett.1c00003>
28. Ciucci S. *et al.* (2017) **Enlightening discriminative network functional modules behind principal component analysis separation in differential-omic science studies** *Scientific Reports* **7**:1–24
29. Poser S. W. *et al.* (2019) **Controlling distinct signaling states in cultured cancer cells provides a new platform for drug discovery** *The FASEB Journal* **33**:9235–9249
30. Urbanska M., Winzi M., Neumann K., Abuhattum S., Rosendahl P., Müller P., Taubenberger A., Anastassiadis K., Guck J. (2017) **Single-cell mechanical phenotype is an intrinsic marker of reprogramming and differentiation along the mouse neural lineage** *Development* :4313–4321
31. Otto O. *et al.* (2015) **Real-time deformability cytometry: on-the-fly cell mechanical phenotyping** *Nature Methods* **12**:199–202

32. Mokbel M., Mokbel D., Mietke A., Träber N., Salvatore G., Otto O., Guck J., Aland S. (2017) **Numerical Simulation of Real-Time Deformability Cytometry To Extract Cell Mechanical Properties** *ACS Biomaterials Science & Engineering* **3**:2962–2973
33. Forrest A. R. R. *et al.* (2014) **A promoter-level mammalian expression atlas** *Nature* **507**:462–470
34. Barretina J. *et al.* (2012) **The Cancer Cell Line Encyclopedia enables predictive modelling of anticancer drug sensitivity** *Nature* **483**:603–607
35. Ghandi M. *et al.* (2019) **Next-generation characterization of the Cancer Cell Line Encyclopedia** *Nature* **569**:503–508
36. Klijn C. *et al.* (2015) **A comprehensive transcriptional portrait of human cancer cell lines** *Nature Biotechnology* **33**:306–312
37. Kiselev V. Y., Juvin V., Malek M., Luscombe N., Hawkins P., Novère N. L., Stephens L. (2015) **Perturbations of PIP3 signalling trigger a global remodelling of mRNA landscape and reveal a transcriptional feedback loop** *Nucleic Acids Research* **43**:9663–9679
38. Tonge P. D. *et al.* (2014) **Divergent reprogramming routes lead to alternative stem-cell states** *Nature* **516**:192–197
39. Aprea J. *et al.* (2013) **Transcriptome sequencing during mouse brain development identifies long non-coding RNAs functionally involved in neurogenic commitment** *EMBO Journal* **32**:3145–3160
40. Brenner B., Tang L. H., Klimstra D. S., Kelsen D. P. (2004) **Small-cell carcinomas of the gastrointestinal tract: A review** *Journal of Clinical Oncology* **22**:2730–2739
41. Kalemkerian G. P. *et al.* (2013) **Small cell lung cancer: Clinical practice guidelines in oncology** *JNCCN Journal of the National Comprehensive Cancer Network* **11**:78–98
42. Juvin V., Malek M., Anderson K. E., Dion C., Chessa T., Lecureuil C., Ferguson G. J., Cosulich S., Hawkins P. T., Stephens L. R. (2013) **Signaling via Class IA Phosphoinositide 3-Kinases (PI3K) in Human, Breast-Derived Cell Lines** *PLoS ONE* **8**
43. Bader A. G., Kang S., Vogt P. K. (2006) **Cancer-specific mutations in PIK3CA are oncogenic in vivo** *Proceedings of the National Academy of Sciences of the United States of America* **103**:1475–1479
44. Kang S., Bader A. G., Vogt P. K. (2005) **Phosphatidylinositol 3-kinase mutations identified in human cancer are oncogenic** *Proceedings of the National Academy of Sciences of the United States of America* **102**:802–807
45. Sinha B. *et al.* (2011) **Cells respond to mechanical stress by rapid disassembly of caveolae** *Cell* **144**:402–413
46. Parton R. G., Del Pozo M. A. (2013) **Caveolae as plasma membrane sensors, protectors and organizers** *Nature Reviews Molecular Cell Biology* **14**:98–112
47. Lolo F.-N. *et al.* (2022) **Caveolin-1 dolines form a distinct and rapid caveolae-independent mechanoadaptation system** *Nat Cell Biol* :1–14

48. Johannessen M., Møler S., Hansen T., Moens U., Van Ghelue M. (2006) **The multifunctional roles of the four-and-a-half-LIM only protein FHL2** *Cellular and Molecular Life Sciences* **63**:268–284
49. Nakazawa N., Sathe A. R., Shivashankar G. V., Sheetz M. P. (2016) **Matrix mechanics controls FHL2 movement to the nucleus to activate p21 expression** *Proceedings of the National Academy of Sciences of the United States of America* **113**:E6813–E6822
50. Sun X. *et al.* (2020) **Mechanosensing through Direct Binding of Tensed F-Actin by LIM Domains** *Developmental Cell* <https://doi.org/10.1016/j.devcel.2020.09.022>
51. Winkelman J. D., Anderson C. A., Suarez C., Kovar D. R., Gardel M. L. (2020) **Evolutionarily diverse LIM domain-containing proteins bind stressed actin filaments through a conserved mechanism** *Proceedings of the National Academy of Sciences of the United States of America* **117**:25532–25542
52. Jin L., Shen F., Weinfeld M., Sergi C. (2020) **Insulin Growth Factor Binding Protein 7 (IGFBP7)-Related Cancer and IGFBP3 and IGFBP7 Crosstalk** *Frontiers in Oncology* **10**:1–14
53. Liu R., Hossain M. M., Chen X., Jin J. P. (2017) **Mechanoregulation of SM22 α /Transgelin** *Biochemistry* **56**:5526–5538
54. Dvorakova M., Nenutil R., Bouchal P. (2014) **Transgelins, cytoskeletal proteins implicated in different aspects of cancer development** *Expert Review of Proteomics* **11**:149–165
55. Adams J. C., Lawler J. (2011) **The thrombospondins** *Cold Spring Harbor Perspectives in Biology* **3**:1–29
56. Huang T., Sun L., Yuan X., Qiu H. (2017) **Thrombospondin-1 is a multifaceted player in tumor progression** *Oncotarget* **8**:84546–84558
57. Hanley J. A., McNeil B. J. (1982) **The meaning and use of the area under a receiver operating characteristic (ROC) curve** *Radiology* **143**:29–36
58. del Pozo M. A., Balasubramanian N., Alderson N. B., Kiosses W. B., Grande-García A., Anderson R. G. W., Schwartz M. A. (2005) **Phospho-caveolin-1 mediates integrin-regulated membrane domain internalization** *Nature Cell Biology* **7**:901–908
59. Moreno-Vicente R. *et al.* (2018) **Caveolin-1 Modulates Mechanotransduction Responses to Substrate Stiffness through Actin-Dependent Control of YAP** *Cell Reports* **25**:1622–1635
60. Raudenska M., Gumulec J., Balvan J., Masarik M. (2020) **Caveolin-1 in oncogenic metabolic symbiosis** *International Journal of Cancer* **147**:1793–1807
61. Pol A., Morales-Paytuví F., Bosch M., Parton R. G. (2020) **Non-caveolar caveolins - Duties outside the caves** *Journal of Cell Science* **133**
62. Lin H.-H. *et al.* (2015) **Mechanical phenotype of cancer cells: cell softening and loss of stiffness sensing** *Oncotarget* **6**
63. Muriel O., Echarri A., Hellriegel C., Pavón D. M., Beccari L., del Pozo M. A. (2011) **Phosphorylated filamin A regulates actin-linked caveolae dynamics** *Journal of Cell Science* **124**:2763–2776

64. Sun R. J., Muller S., Zhuang F. Y., Stoltz J. F., Wang X. (2002) **Caveolin-1 redistribution in human endothelial cells induced by laminar flow and cytokine** *Biorheology* **40**:31–39
65. Hsu C. K. *et al.* (2018) **Caveolin-1 Controls Hyperresponsiveness to Mechanical Stimuli and Fibrogenesis-Associated RUNX2 Activation in Keloid Fibroblasts** *Journal of Investigative Dermatology* **138**:208–218
66. Le Master E., Paul A., Lazarko D., Aguilar V., Ahn S. J., Lee J. C., Minshall R. D., Levitan I. (2022) **Caveolin-1 is a primary determinant of endothelial stiffening associated with dyslipidemia, disturbed flow, and ageing** *Scientific Reports* **12**:1–15
67. Grivas D., González-Rajal Á., Guerrero Rodríguez C., Garcia R., de la Pompa J. L. (2020) **Loss of Caveolin-1 and caveolae leads to increased cardiac cell stiffness and functional decline of the adult zebrafish heart** *Sci Rep* **10**
68. Li Q. S., Lee G. Y. H., Ong C. N., Lim C. T. (2008) **AFM indentation study of breast cancer cells** *Biochemical and Biophysical Research Communications* **374**:609–613
69. Zhou Z. L., Ngan A. H. W., Tang B., Wang A. X. (2012) **Reliable measurement of elastic modulus of cells by nanoindentation in an atomic force microscope** *Journal of the Mechanical Behavior of Biomedical Materials* **8**:134–142
70. Alcaraz J., Buscemi L., Grabulosa M., Trepát X., Fabry B., Farré R., Navajas D. (2003) **Microrheology of human lung epithelial cells measured by atomic force microscopy** *Biophysical Journal* **84**:2071–2079
71. Massiera G., Van Citters K. M., Biancaniello P. L., Crocker J. C. (2007) **Mechanics of single cells: Rheology, time dependence, and fluctuations** *Biophysical Journal* **93**:3703–3713
72. Rigato A., Miyagi A., Scheuring S., Rico F. (2017) **High-frequency microrheology reveals cytoskeleton dynamics in living cells** *Nature Physics* **13**:771–775
73. Hirsch H. A., Iliopoulos D., Tschlis P. N., Struhl K. (2009) **Metformin selectively targets cancer stem cells, and acts together with chemotherapy to block tumor growth and prolong remission** *Cancer Research* **69**:7507–7511
74. Tavares S. *et al.* (2017) **Actin stress fiber organization promotes cell stiffening and proliferation of pre-invasive breast cancer cells** *Nature Communications* **8**:1–18
75. Hirsch H. A., Iliopoulos D., Joshi A., Zhang Y., Jaeger S. A., Bulyk M., Tschlis P. N., Shirley Liu X., Struhl K. (2010) **A Transcriptional Signature and Common Gene Networks Link Cancer with Lipid Metabolism and Diverse Human Diseases** *Cancer Cell* **17**:348–361
76. Guck J. (2019) **Some thoughts on the future of cell mechanics** *Biophysical Reviews* **11**:667–670
77. Urbanska M., Muñoz H. E., Shaw Bagnall J., Otto O., Manalis S. R., Di Carlo D., Guck J. (2020) **A comparison of microfluidic methods for high-throughput cell deformability measurements** *Nat Methods* **17**:587–593
78. Durán C. *et al.* (2021) **Nonlinear machine learning pattern recognition and bacteria-metabolite multilayer network analysis of perturbed gastric microbiome** *Nature communications* **12**

79. Urbanska M., Guck J. (2024) **Single-Cell Mechanics: Structural Determinants and Functional Relevance** *Annual Review of Biophysics* **53**:367–395
80. Strippoli R. *et al.* (2020) **Caveolin1 and YAP drive mechanically induced mesothelial to mesenchymal transition and fibrosis** *Cell Death and Disease* **11**:1–19
81. Dupont S. *et al.* (2011) **Role of YAP/TAZ in mechanotransduction** *Nature* **474**:179–184
82. Rausch V., Bostrom J. R., Park J., Bravo I. R., Feng Y., Hay D. C., Link B. A., Hansen C. G. (2019) **The Hippo Pathway Regulates Caveolae Expression and Mediates Flow Response via Caveolae** *Current Biology* **29**:242–255
83. Stein C. *et al.* (2015) **YAP1 Exerts Its Transcriptional Control via TEAD-Mediated Activation of Enhancers** *PLoS Genetics* **11**:1–28
84. Zhao B. *et al.* (2008) **TEAD mediates YAP-dependent gene induction and growth control** *Genes and Development* **22**:1962–1971
85. Jiang W. R., Cady G., Hossain M. M., Huang Q. Q., Wang X., Jin J. P. (2014) **Mechanoregulation of h2-calponin gene expression and the role of notch signaling** *Journal of Biological Chemistry* **289**:1617–1628
86. Hossain M. M., Crish J. F., Eckert R. L., Lin J. J. C., Jin J. P. (2005) **H2-Calponin Is Regulated By Mechanical Tension and Modifies the Function of Actin Cytoskeleton** *Journal of Biological Chemistry* **280**:42442–42453
87. De Marzio M. *et al.* (2021) **Genomic signatures of the unjamming transition in compressed human bronchial epithelial cells** *Science Advances* **7**:1–16
88. Kılıç A. *et al.* (2020) **Mechanical forces induce an asthma gene signature in healthy airway epithelial cells** *Scientific Reports* **10**:1–12
89. Rysä J., Tokola H., Ruskoaho H. (2018) **Mechanical stretch induced transcriptomic profiles in cardiac myocytes** *Scientific Reports* **8**:1–14
90. Lv J. *et al.* (2020) **Cell softness regulates tumorigenicity and stemness of cancer cells** *EMBO J.* **40**
91. Swift J. *et al.* (2013) **Nuclear lamin-A scales with tissue stiffness and enhances matrix-directed differentiation** *Science* **341**
92. Goetz J. G., Lajoie P., Wiseman S. M., Nabi I. R. (2008) **Caveolin-1 in tumor progression: The good, the bad and the ugly** *Cancer and Metastasis Reviews* **27**:715–735
93. Wang Z. *et al.* (2015) **Caveolin-1, a stress-related oncotarget, in drug resistance** *Oncotarget* **6**:37135–37150
94. Nawaz A. A. *et al.* (2020) **Intelligent image-based deformation-assisted cell sorting with molecular specificity** *Nature Methods* **17**:595–599
95. Nawaz A. A., Soteriou D., Xu C. K., Goswami R., Herbig M., Guck J., Girardo S. (2023) **Image-based cell sorting using focused travelling surface acoustic waves** *Lab Chip* **23**:372–387

96. Razani B. *et al.* (2001) **Caveolin-1 Null Mice Are Viable but Show Evidence of Hyperproliferative and Vascular Abnormalities** *Journal of Biological Chemistry* **276**:38121–38138
97. Urbanska M., Rosendahl P., Kräter M., Guck J. (2018) **High-throughput single-cell mechanical phenotyping with real-time deformability cytometry** *Methods in Cell Biology* **147**
98. Sneddon I. N. (1965) **The relation between load and penetration in the axisymmetric boussinesq problem for a punch of arbitrary profile** *International Journal of Engineering Science* **3**:47–57
99. Rother J., Nöding H., Mey I., Janshoff A. (2014) **Atomic force microscopy-based microrheology reveals significant differences in the viscoelastic response between malign and benign cell lines** *Open Biology* <https://doi.org/10.1098/rsob.140046>
100. Alcaraz J., Buscemi L., Puig-De-Morales M., Colchero J., Baró A., Navajas D. (2002) **Correction of microrheological measurements of soft samples with atomic force microscopy for the hydrodynamic drag on the cantilever** *Langmuir* <https://doi.org/10.1021/la0110850>
101. Shannon P., Markiel A., Ozier O., Baliga N. S., Wang J. T., Ramage D., Amin N., Schwikowski B., Ideker T. (2003) **Cytoscape : A Software Environment for Integrated Models of Biomolecular Interaction Networks** *Genome Res.* **13**:2498–2504
102. Danaher P., Warren S., Dennis L., D’Amico L., White A., Disis M. L., Geller M. A., Odunsi K., Beechem J., Fling S. P. (2017) **Gene expression markers of Tumor Infiltrating Leukocytes** *Journal for ImmunoTherapy of Cancer* **5**:1–15
103. Acevedo A., Duran C., Kuo M. J., Ciucci S., Schroeder M., Cannistraci C. V. (2022) **Measuring Group Separability in Geometrical Space for Evaluation of Pattern Recognition and Dimension Reduction Algorithms** *IEEE Access* **10**:22441–22471
104. Herbig M., Mietke A., Müller P., Otto O. (2018) **Statistics for real-time deformability cytometry: Clustering, dimensionality reduction, and significance testing** *Biomechanics* **12**
105. Itoh M. *et al.* (2012) **Automated workflow for preparation of cDNA for cap analysis of gene expression on a single molecule sequencer** *PLoS ONE* **7**
106. (2017) C. Herold, <http://arxiv.org/abs/1704.00572> (2017).

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Reviewer #1 (Public review):

In this work, Urbanska and colleagues use a machine-learning based crossing of mechanical characterisations of various cells in different states and their transcriptional profiles. Using this approach, they identify a core set of five genes that systematically vary together with the mechanical state of the cells, although not always in the same direction depending on the conditions. They show that the combined transcriptional changes in this gene set is strongly predictive of a change in the cell mechanical properties, in systems that were not used to identify the genes (a validation set). Finally, they experimentally alter the expression level of one of these genes, CAV1, that codes for the caveolin 1 protein, and show that, in a variety of cellular systems and contexts, perturbations in the expression level of CAV1 also induce changes in cell mechanics, cells with lower CAV1 expression being generally softer.

Overall the approach seems accessible, sound and is well described. My personal expertise is not suited to judge its validity, novelty or relevance, so I do not make comments on that. The results it provides seem to have been thoroughly tested by the authors (using different types of mechanical characterisations of the cells) and to be robust in their predictive value. The authors also show convincingly that one of the genes they identified, CAV1, is not only correlated with the mechanical properties of cells, but also that changing its expression level affects cell mechanics. At this stage, the study appears mostly focused on the description and validation of the methodological approach, and it is hard to really understand what the results obtain really mean, the importance of the biological finding - what is this set of 5 genes doing in the context of cell mechanics? Is it really central, or is it just one of the set of knobs on which the cell plays - and it is identified by this method because it is systematically modulated but maybe, for any given context, it is not the dominant player - all these fundamental questions remain unanswered at this stage. On one hand, it means that the study might have identified an important novel module of genes in cell mechanics, but on the other hand, it also reveals that it is not yet easy to interpret the results provided by this type of novel approach.

Comments on revisions:

In their point-by-point answer, the authors did a great effort to provide pedagogical answers that clarified most of the points I had raised. They also did more analysis, some of which are included as supplementary data, and added a few sentences to the main text and discussion. As far as I am concerned, I see no particular issue with the revised article. I think it will be interesting both as a new type of approach in mechanobiology, and also as a motivation for more experimentally oriented labs to test the hypothesis proposed in the article and the 'module' they found.

<https://doi.org/10.7554/eLife.87930.2.sa1>

Author response:

The following is the authors' response to the original reviews.

In summary, the changes made in the revision process include:

An addition of a paragraph in the result section that discusses the absolute values of measured Young's moduli in the light of probing frequencies, accompanied by a new supplementary figure and a supplementary table that support that discussion

- Fig. S10. Absolute Young's modulus values across the frequencies characteristic for the three measurement methods.

- Table S9. Operation parameters of the three methods used for characterizing the mechanical properties of cells.

Three new supplementary figures that display the expression matrices for the genes from the identified modules in carcinoma datasets used for validation:

- Fig. S4. Expression of identified target genes in the CCLE microarray dataset used for validation.

- Fig. S5. Expression of identified target genes in the CCLE RNA-Seq dataset used for validation.

- Fig. S6. Expression of identified target genes in the Genentech dataset used for validation.

An addition of a paragraph in the discussion section that discusses the intracellular origins of resistance to deformation and the dominance of actin cortex at low deformations.

- Refinement of the manuscript text and figures based on the specific feedback from the Reviewers.

Please see below for detailed responses to the Reviewers' comments.

Reviewer #1 (Public Review)

In this work, Urbanska and colleagues use a machine-learning based crossing of mechanical characterisations of various cells in different states and their transcriptional profiles. Using this approach, they identify a core set of five genes that systematically vary together with the mechanical state of the cells, although not always in the same direction depending on the conditions. They show that the combined transcriptional changes in this gene set is strongly predictive of a change in the cell mechanical properties, in systems that were not used to identify the genes (a validation set). Finally, they experimentally alter the expression level of one of these genes, CAV1, that codes for the caveolin 1 protein, and show that, in a variety of cellular systems and contexts, perturbations in the expression level of CAV1 also induce changes in cell mechanics, cells with lower CAV1 expression being generally softer.

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We thank the Reviewer #1 for the thoughtful evaluation of our manuscript. The primary goal of the manuscript was to present a demonstration of an unbiased approach for the identification of genes involved in the regulations of cell mechanics. The manuscript further provides a comprehensive computational validation of all genes from the identified network, and experimental validation of a selected gene, CAV1.

We agree that at the current stage, far-reaching conclusions about the biological meaning of the identified network cannot be made. We are, however, convinced that the identification of an apparently central player such as CAV1 across various cellular systems is per se meaningful, in particular since CAV1 modulation shows clear effects on the cell mechanical state in several cell types.

We anticipate that our findings will encourage more mechanistic studies in the future, investigating how these identified genes regulate mechanical properties and interact with each other. Notwithstanding, the identified genes (after testing in specific system of interest) can be readily used as genetic targets for modulating mechanical properties of cells. Access to such modifications is of huge relevance not only for performing further research on the functional consequence of cell mechanics changes (in particular in in-vivo systems where using chemical perturbations is not always possible), but also for the potential future implementation in modulating mechanical properties of the cells to prevent disease (for example to inhibit cancer metastasis or increase efficacy of cancer cell killing by cytotoxic T cells).

We have now added a following sentence in the first paragraph of discussion to acknowledge the open ends of our study:

“(…). Here we leveraged this opportunity by performing discriminative network analysis on transcriptomes associated with mechanical phenotype changes to elucidate a conserved module of five genes potentially involved in cell mechanical phenotype regulation. We provided evidence that the inferred conserved functional network module contains an ensemble of five genes that, in particular when combined in a unique combinatorial marker, are universal, specific and trustworthy markers of mechanical phenotype across the studied mouse and human systems. We further demonstrated on the example of a selected marker gene, CAV1, that its experimental up- and downregulation impacts the stiffness of the measured cells. This demonstrates that the level of CAV1 not only correlates with, but also is causative of mechanical phenotype change. The mechanistic insights into how precisely the identified genes are involved in regulating mechanical properties, how they interact with each other, and whether they are universal and dominant in various contexts all remain to be established in

future studies.”

Reviewer #2 (Public Review)

A key strength is the quantitative approaches all add rigor to what is being attempted. The approach with very different cell culture lines will in principle help identify constitutive genes that vary in a particular and predictable way. To my knowledge, one other study that should be cited posed a similar pan-tissue question using mass spectrometry proteomics instead of gene expression, and also identified a caveolae component (cavin-1, PTRF) that exhibited a trend with stiffness across all sampled tissues. The study focused instead on a nuclear lamina protein that was also perturbed in vitro and shown to follow the expected mechanical trend (Swift et al 2013).

We thank the Reviewer #2 for the positive evaluation of the breadth of the results and for pointing us to the relevant reference for the proteomic analysis related to tissue stiffness

(Swift et al., 2013). This study, which focused primarily on the tissue-level mechanical properties, identifying PTRF, a caveolar component, which links to our observation of another caveolar component, CAV1, at the single-cell level.

We have now included the citation in the following paragraph of the discussion:

“To our knowledge, there are no prior studies that aim at identifying gene signatures associated with single-cell mechanical phenotype changes, in particular across different cell types. There are, however, several studies that investigated changes in expression upon exposure of specific cell types to mechanical stimuli such as compression (87, 88) or mechanical stretch (22, 80, 89), and one study that investigated difference in expression profiles between stiffer and softer cells sorted from the same population (90). Even though the studies concerned with response to mechanical stimuli answer a fundamentally different question (how gene expression changes upon exposure to external forces vs which genes are expressed in cells of different mechanical phenotype), we did observe some similarities in the identified genes. For example, in the differentially expressed genes identified in the lung epithelia exposed to compression (87), three genes from our module overlapped with the immediate response (CAV1, FHL2, TGLN) and four with the long-term one (CAV1, FHL2, TGLN, THBS1). We speculate that this substantial overlap is caused by the cells undergoing change in their stiffness during the response to compression (and concomitant unjamming transition). Another previous study explored the association between the stiffness of various tissues and their proteomes. Despite the focus on the tissue-scale rather than single-cell elasticity, the authors identified polymerase I and transcript release factor (PTRF, also known as cavin 1 and encoding for a structural component of the caveolae) as one of the proteins that scaled with tissue stiffness across samples (91).”

Reviewer #3 (Public Review)

In this work, Urbanska et al. link the mechanical phenotypes of human glioblastoma cell lines and murine iPSCs to their transcriptome, and using machine learning-based network analysis identify genes with putative roles in cell mechanics regulation. The authors identify 5 target genes whose transcription creates a combinatorial marker which can predict cell stiffness in human carcinoma and breast epithelium cell lines as well as in developing mouse neurons. For one of the target genes, caveolin1 (CAV1), the authors perform knockout, knockdown, overexpression and rescue experiments in human carcinoma and breast epithelium cell lines. They determine the cell stiffness via RT-DC, AFM indentation and AFM rheology and confirm that high CAV1 expression levels correlate with increased stiffness in those model systems. This work brings forward an interesting approach to identify novel genes in an unbiased manner, but surprisingly the authors validate caveolin 1, a target gene with known roles in cell mechanics regulation.

I have two main concerns with the current version of this work:

(1) The authors identify a network of 5 genes that can predict mechanics. What is the relationship between the 5 genes? If the authors aim to highlight the power of their approach by knockdown, knockout or over-expression of a single gene why choose CAV1 (which has an individual p-value of 0.16 in Fig S4)? To justify their choice, the authors claim that there is limited data supporting the direct impact of CAV1 on mechanical properties of cells but several studies have previously shown its role in for example zebrafish heart stiffness, where a knockout leads to higher stiffness (Grivas et al., Scientific Reports 2020), in cancer cells, where a knockdown leads to cell softening (Lin et al., Oncotarget 2015), or in endothelial cell, where a knockout leads to cell softening (Le Master et al., Scientific Reports 2022).

We thank the reviewer for their comments. First, we do acknowledge that studying the relationship between the five identified genes is an intriguing question and would be a

natural extension of the currently presented work. It is, however, beyond the scope of presented manuscript, in which our primary goal was to introduce a general pipeline for de novo identification of genes related to cell mechanics. We did add a following statement in the discussion (yellow highlight) to acknowledge the open ends of our study:

“The mechanical phenotype of cells is recognized as a hallmark of many physiological and pathological processes. Understanding how to control it is a necessary next step that will facilitate exploring the impact of cell mechanics perturbations on cell and tissue function (76).

The increasing availability of transcriptional profiles accompanying cell state changes has recently been complemented by the ease of screening for mechanical phenotypes of cells thanks to the advent of high-throughput microfluidic methods (77). This provides an opportunity for data-driven identification of genes associated with the mechanical cell phenotype change in a hypothesis-free manner. Here we leveraged this opportunity by performing discriminative network analysis on transcriptomes associated with mechanical phenotype changes to elucidate a conserved module of five genes potentially involved in cell mechanical phenotype regulation. We provided evidence that the inferred conserved functional network module contains an ensemble of five genes that, in particular when combined in a unique combinatorial marker, are universal, specific and trustworthy markers of mechanical phenotype across the studied mouse and human systems. We further demonstrated on the example of a selected marker gene, CAV1, that its experimental up- and downregulation impacts the stiffness of the measured cells. This demonstrates that the level of CAV1 not only correlates with, but also is causative of mechanical phenotype change. The mechanistic insights into how precisely the identified genes are involved in regulating mechanical properties, how they interact with each other, and whether they are universal and dominant in various contexts all remain to be established in future studies.”

Regarding the selection of CAV1 as the gene that we used for validation experiment; as mentioned in the introductory paragraph of the result section “Perturbing expression levels of CAV1 changes cells stiffness” (copied below), we were encouraged by the previous data already linking CAV1 with cell mechanics when selecting it as our first target. The relationship between CAV1 and cell mechanics regulation, however, is not very well established (of note, two of the latest manuscripts came out after the initial findings of our study).

Regarding the citations suggested by the reviewer: two are already included in the original manuscript (Lin et al., Oncotarget 2015 – Ref (63), Le Master –2022 Ref (67)), along with an additional one (Hsu et al 2018 (66)), and the third one (Grivas et al, 2020 (68)) is now also added to the manuscript. Though, we would like to highlight that even though Grivas et al state that the CAV1 KO cells are stiffer, the AFM indentation measurements were performed on the cardiac tissue, with a spherical tip of 30 μm radius and likely reflect primarily supracellular, tissue-scale properties, as opposed to cell-scale measurements performed in our study (we used cultured cells which mostly lack the extracellular tissue structures, deformability cytometry was performed on dissociated cells and picks up on cell properties exclusively, and in case of AFM measurements a spherical tip with 5 μm radius was used).

“We decided to focus our attention on CAV1 as a potential target for modulating mechanical properties of cells, as it has previously been linked to processes intertwined with cell mechanics. In the context of mechanosensing, CAV1 is known to facilitate buffering of the membrane tension (45), play a role in β 1-integrin-dependent mechanotransduction (58) and modulate the mechanotransduction in response to substrate stiffness (59). CAV1 is also intimately linked with actin cytoskeleton — it was shown to be involved in cross-talk with Rho-signaling and actin cytoskeleton regulation (46, 60–62), filamin A-mediated interactions with actin filaments (63), and co-localization with peripheral actin (64). The evidence directly

relating CAV1 levels with the mechanical properties of cells (47, 62, 65, 66) and tissues (66, 67), is only beginning to emerge.”

Regarding the cited p-value of 0.16, we would like to clarify that it is the p-value associated with the coefficient of the crude linear regression model fitted to the data for illustrative purposes in Fig S4. This value only says that from the linear fit we cannot conclude much about the correlation of the level of Cav1 with the Young's modulus change. Much more relevant parameters to look at are the AUC-ROC values and associated p-values reported in the Table 4 in the main text (see below), which show good performance of CAV1 in separating soft and stiff cell states.

The positive hypothesis I assumes that markers are discriminative of samples with stiff/soft mechanical phenotype regardless of the studied biological system, and CAV1 has a clear trend with the minimum AUC-ROC on 3 datasets of 0.78, even though the p-value is below the significance level. The positive hypothesis II assumes that markers are discriminative of samples with stiff/soft mechanical phenotype in carcinoma regardless of data source, and CAV1 has a clear significance because the minimum AUC-ROC on 3 datasets is 0.89 and the p-value is 0.02.

(2) The authors do not show how much does PC-Corr outperforms classical co-expression network analysis or an alternative gold standard. It is worth noting that PC-Corr was previously published by the same authors to infer phenotype-associated functional network modules from omics datasets (Ciucci et al., Scientific Reports 2017).

As pointed out by the Reviewer, PC-corr has been introduced and characterized in detail in a previous publication (Ciucci et al, 2017, Sci. Rep.), where it was compared against standard co-expression analysis (below reported as: p-value network) on molecules selected using univariate statistical analysis.

See the following fragment of Discussion in Ciucci et al, 2017:

“The PC-corr networks were always compared to P-value networks. The first strategical difference lies in the way features are selected: while the PC-corr adopts a multivariate approach, i.e. it uses a combination of features that are responsible for the sample discrimination, in the P-value network the discriminating features are singly selected (one by one) with each Mann-Whitney test (followed by Benjamini-Hochberg procedure). The second strategical difference lies in the generation of the correlation weights in the network. PC-corr combines in parallel and at the same time in a unique formula the discrimination power of the PC-loadings and the association power of the Pearson correlation, directly providing in output discriminative omic associations. These are generated using a robust (because we use as merging factor the minimum operator, which is a very penalizing operator) mathematical trade-off between two important factors: multivariate discriminative significance and correlation association. In addition, as mentioned above, the minimum operator works as an AND logical gate in a digital circuit, therefore in order to have a high link weight in the PCcorr network, both the discrimination (the PC-loadings) and the association (the Pearson correlations) of the nodes adjacent to the link should be simultaneously high. Instead, the Pvalue procedure begins with the pre-selection of the significant omic features and, only in a second separated step, computes the associations between these features. Therefore, in P-value networks, the interaction weights are the result neither of multivariate discriminative significance, nor of a discrimination/association interplay.”

Here we implement PC-corr for a particular application and do not see it as central to the message of the present manuscript to compare it with other available methods. We considered it much more relevant to focus on an in-silico validation on dataset not used during the PCcorr analysis (see Table 3 and 4 for details).

Altogether, the authors provide an interesting approach to identify novel genes associated with cell mechanics changes, but the current version does not fulfill such potential by focusing on a single gene with known roles in cell mechanics.

Our manuscript presents a demonstration of an overall approach for the identification of genes involved in the regulation of cell mechanics, and the perturbations performed on CAV1 have a demonstrative role (please also refer to the explanations of why we decided to perform the verification focused on CAV1 above). The fact that we identify CAV1, which has been implicated in regulating cell mechanics in a handful of studies, de novo and in an unbiased way speaks to the power of our approach. We do agree that investigation into the effect of manipulating the expression of the remaining genes from the identified network module, as well as into the mutual relationships between those genes and their covariance in perturbation experiments, constitutes a desirable follow-up on the presented results. It is, however, beyond the scope of the current manuscript. Regardless, the other genes identified can be readily tested in systems of interest and used as potential knobs for tuning mechanical properties on demand.

Reviewer #1 (Recommendations For Authors)

I am not a specialist of the bio-informatics methods used in this study, so I will not make any specific technical comments on them.

In terms of mechanical characterisation of cells, the authors use well established methods and the fact that they systematically validate their findings with at least two independent methods (RT-DC and AFM for example) makes them very robust. So I have no concerns with this part. The experiments of perturbations of CAV 1 are also performed to the best standards and the results are clear, no concern on that.

My main concerns are rather questions I was asking myself and could not answer when reading the article. Maybe the authors could find ways to clarify them - the discussion of their article is already very long and maybe it should not be lengthened to much. In my opinion, some of the points discussed are not really essential and rather redundant with other parts of the paper. This could be improved to give some space to clarify some of the points below:

We thank the Reviewer #1 for an overall positive evaluation of the manuscript as well as the points of criticism which we addressed in a point-by-point manner below.

(1) This might be a misunderstanding of the method on my side, but I was wondering whether it is possible to proceed through the same steps but choose other pairs of training datasets amongst the 5 systems available (there are 10 such pairs if I am not mistaken) and ask whether they always give the same set of 5 genes. And if not, are the other sets also then predictive, robust, etc. Or is it that there are 'better' pairs than others in this respect. Or the set of 5 genes is the only one that could be found amongst these 5 datasets - and then could it imply that it is the only group 'universal' group of predictive genes for cell mechanics (when applied to any other dataset comprising similar mechanical measures and expression profiles, for other cells, other conditions)?

I apologize in case this question is just the result of a basic misunderstanding of the method on my side. But I could not answer the question myself based on what is in the article and it seems to be important to understand the significance of the finding and the robustness of the method.

We thank the Reviewer for this question. To clarify: while in general it is possible to proceed through the same analysis steps choosing a different pair of datasets (see below for examples), we have purposefully chosen those two and not any other datasets because they encompassed the highest number of samples per condition in the RNAseq data (see Fig 4 and Table R1 below), originated from two different species and concerned least related tissues (the other option for mouse would be neural progenitors which in combination with the glioblastoma would likely result in focusing on genes expressed in neural tissues). This is briefly explained in the following fragment of the manuscript on Page 10:

“For the network construction, we chose two datasets that originate from different species, concern unrelated biological processes, and have a high number of samples included in the transcriptional analysis: human glioblastoma and murine iPSCs (Table 1).”

To further address the comment of the reviewer: there is indeed a total of 10 possible two-set combinations of datasets, 6 of those pairs are human-mouse combinations (highlighted in orange in Author response Table 1), 3 are human-human combinations (highlighted in blue), and 1 is mouse-mouse (marked in green).

Author response table 1.

Possible two-set combinations of datasets. For each combination, the number of common genes is indicated. The number on the diagonal represents total number of transcripts in the individual datasets, n corresponds to the number of samples in the respective datasets. * include non-coding genes.

		human			mouse	
		A. glioblastoma	B. carcinoma	C. MCF10A	D. iPSCs	E. dev neurons
human	A. glioblastoma	39400* (n = 27)	16260	31688	9452	12894
	B. carcinoma		18821 (n = 22)	16267	9410	12393
	C. MCF10A			38508* (n = 6)	9598	12951
mouse	D. iPSC				18118 (n = 28)	10338
	E. dev neurons					21110 (n = 6)

POSSIBLE 2-SET COMBINATIONS (x10)

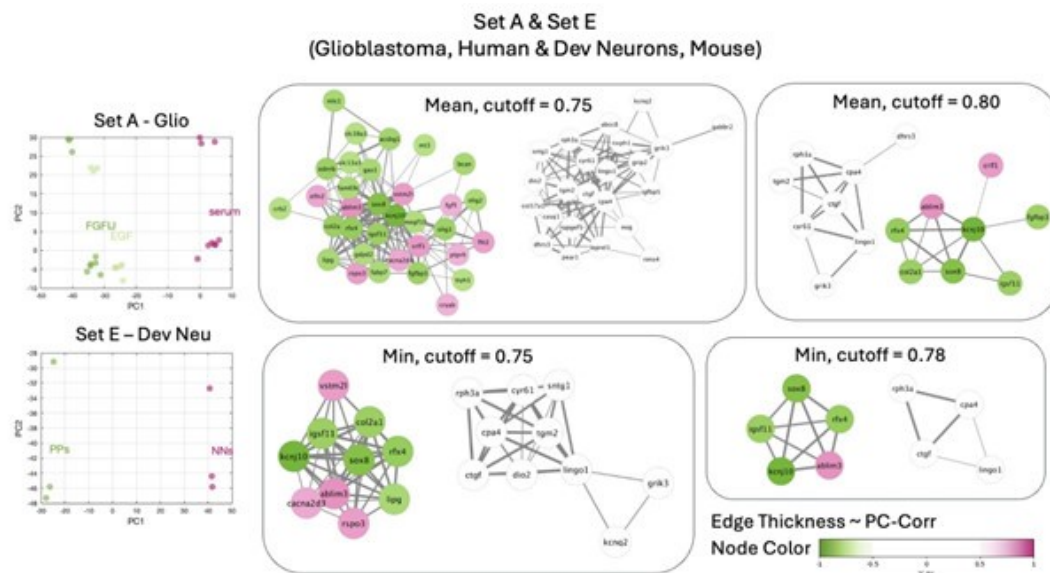
- human-mouse (x6, desired)
- human-human (x3)
- mouse-mouse (x1)

To reiterate, we have chosen the combination of set A (glioblastoma) and set D (iPSCs) to choose datasets from different species and with highest sample number.

As for the other combinations of human-mouse datasets:

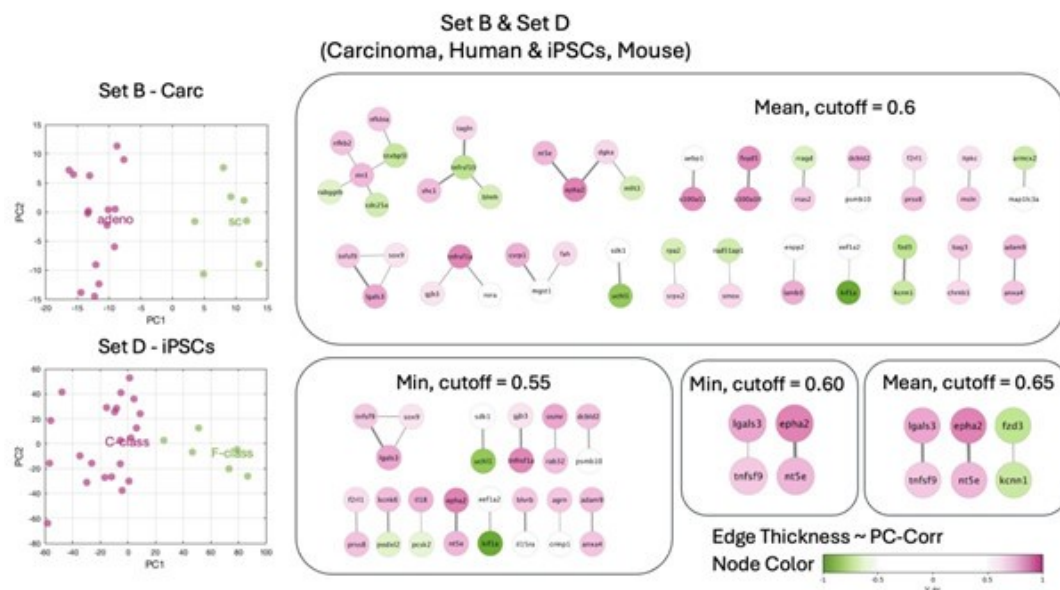
- set A & E lead to derivation of a conserved module, however as expected this module includes genes specific for neuronal tissues (such as brain & testis specific immunoglobulin IGSF11, or genes involved in neuronal development such as RFX4, SOX8)

Author response image 1.

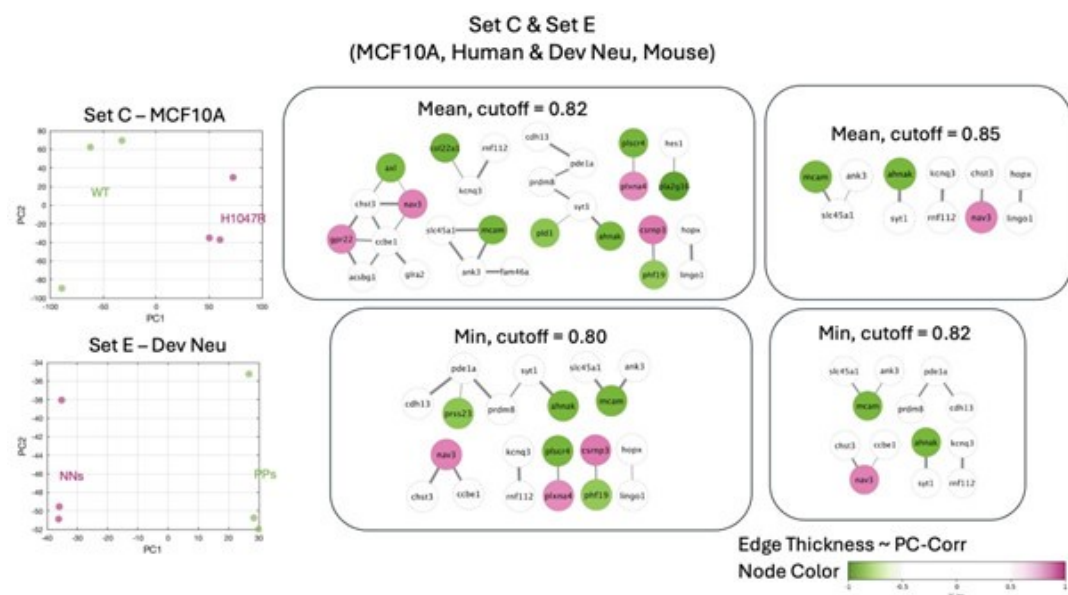


- the remaining combinations (set B&D, B&E, C&D and C&E) do not lead to a derivation of a highly interconnected module

Author response image 2.



Author response image 5.



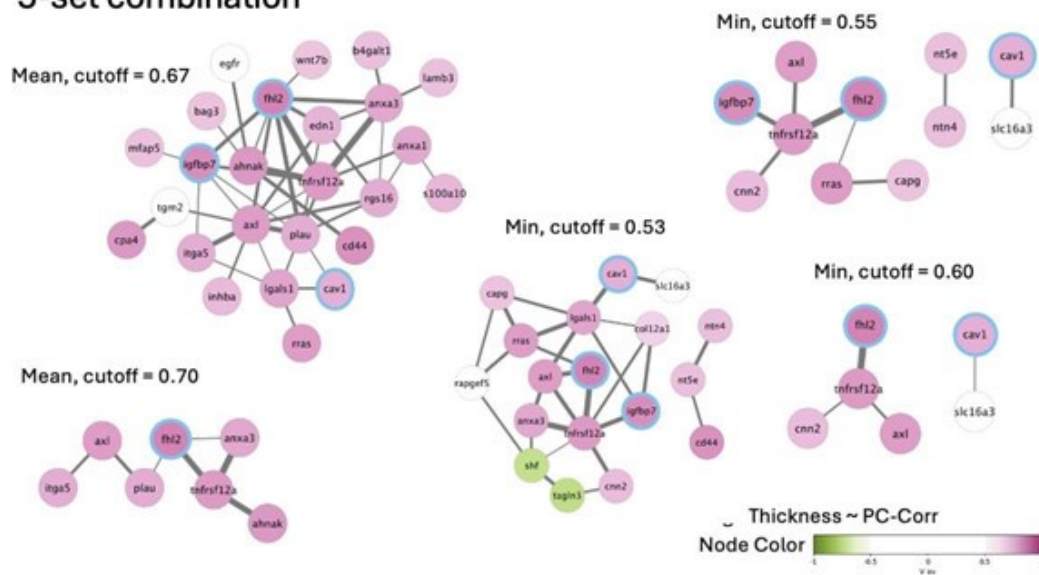
Finally, it would have also been possible to perform the combined PC-corr procedure on all 5 datasets. However, this would prevent us from doing validation using unknown datasets.

Hence, we decided to proceed with the 2 discovery and 4 validation datasets.

For the sake of completeness, we present below some of the networks obtained from the analysis performed on all 5 datasets (which intersect at 8059 genes).

Author response image 6.

5-set combination



The above network was created by calculating mean/minimum PC-corr among all five datasets and applying the threshold. The thresholding can be additionally restricted in that

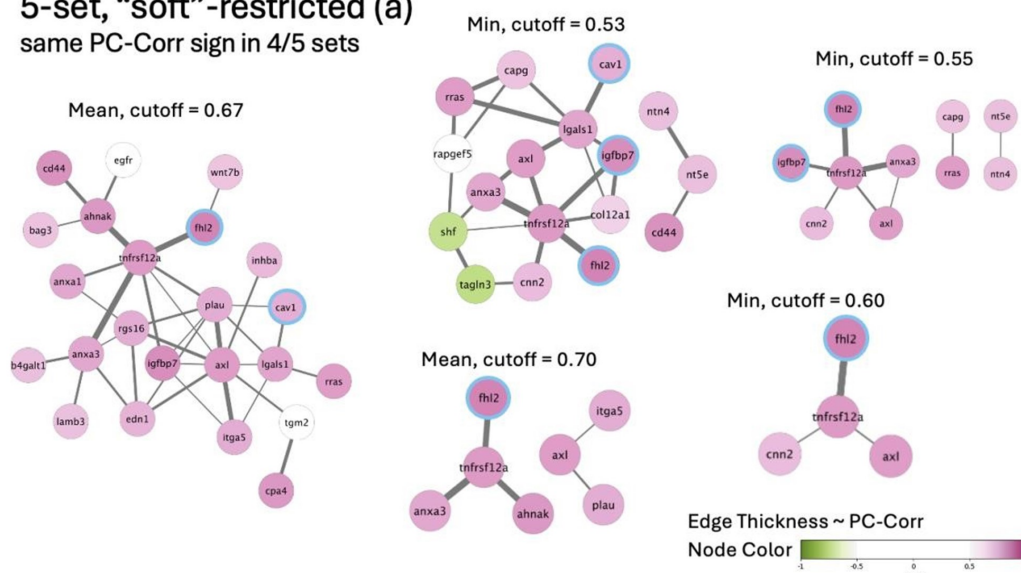
we:

- constrain the directionality of the correlation between the genes ($sgn(c)$) to be the same among all or at least n datasets
- constrain the directionality of the correlation between the cell stiffness and gene expression level ($sgn(V)$) for individual genes.

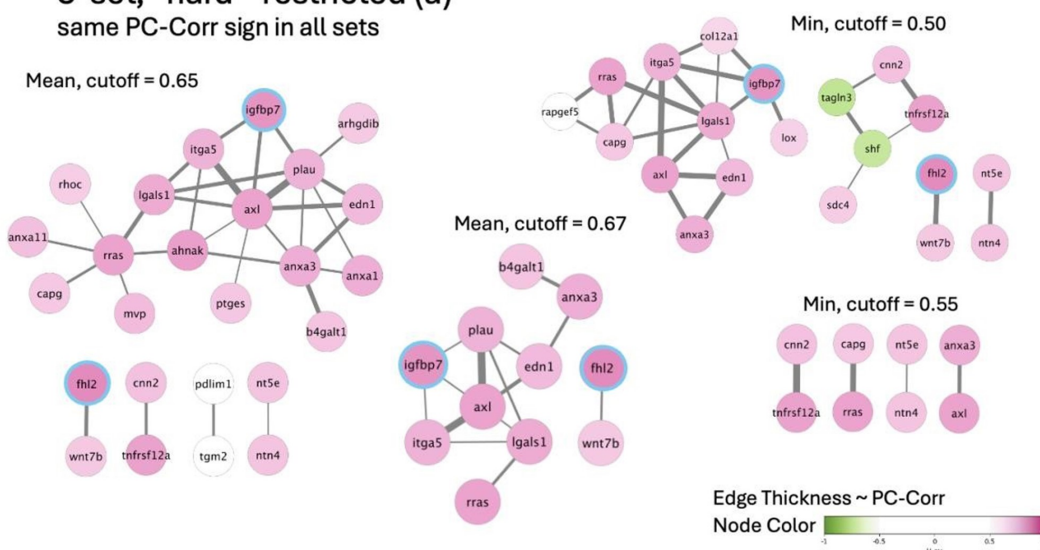
Some of the resulting networks for such restrictions are presented below.

Author response image 7.

5-set, “soft”-restricted (a) same PC-Corr sign in 4/5 sets



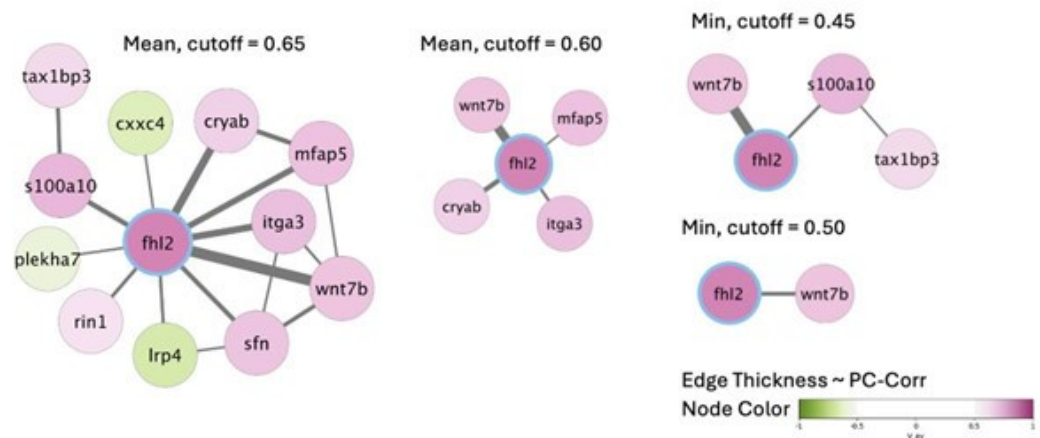
5-set, “hard”-restricted (a) same PC-Corr sign in all sets



Author response image 8.

5-set, “hard”-restricted (a) & (b)

same PC-Corr sign 5 sets, same node directionality in 5 sets



Of note, some of the nodes from the original network presented in the paper (CAV1, FHL2, and IGFBP7) are preserved in the 5-set network (and highlighted with blue rims),

(2) The authors already use several types of mechanical characterisation of the cells, but there are even more of them, in particular, some that might not directly correspond to global cell stiffness but to other aspects, like traction forces, or cell cortex rheology, or cell volume or passage time through constrictions (active or passive) - they might all be in a way or another related, but they are a priori independent measures. Would the authors anticipate finding very different 'universal modules' for these other mechanical properties, or again the same one? Is there a way to get at least a hint based on some published characterisations for the cells used in the study? Basically, the question is whether the gene set identified is specific for a precise type of mechanical property of the cell, or is more generally related to cell mechanics modulation - maybe, as suggested by the authors because it is a set of molecular knobs acting upstream of general mechanics effectors like YAP/TAZ or acto-myosin?

We thank the Reviewer for this comment. We would like to first note that in our study, we focused on single-cell mechanical phenotype understood as a response of the cells to deformation at a global (RT-DC) or semi-local (AFM indentation with 5- μ m bead) level and comparatively low deformations (1-3 μ m, see Table S9). There is of course a variety of other methods for measuring cell mechanics and mechanics-related features, such as traction force microscopy mentioned by the reviewer. Though, traction force microscopy probes how the cells apply forces and interact with their environment rather than the inherent mechanical properties of the cells themselves which were the main interest of our study.

Nevertheless, as mentioned in the discussion, we found some overlap with the genes identified in other mechanical contexts, for example in the context of mechanical stretching of cells:

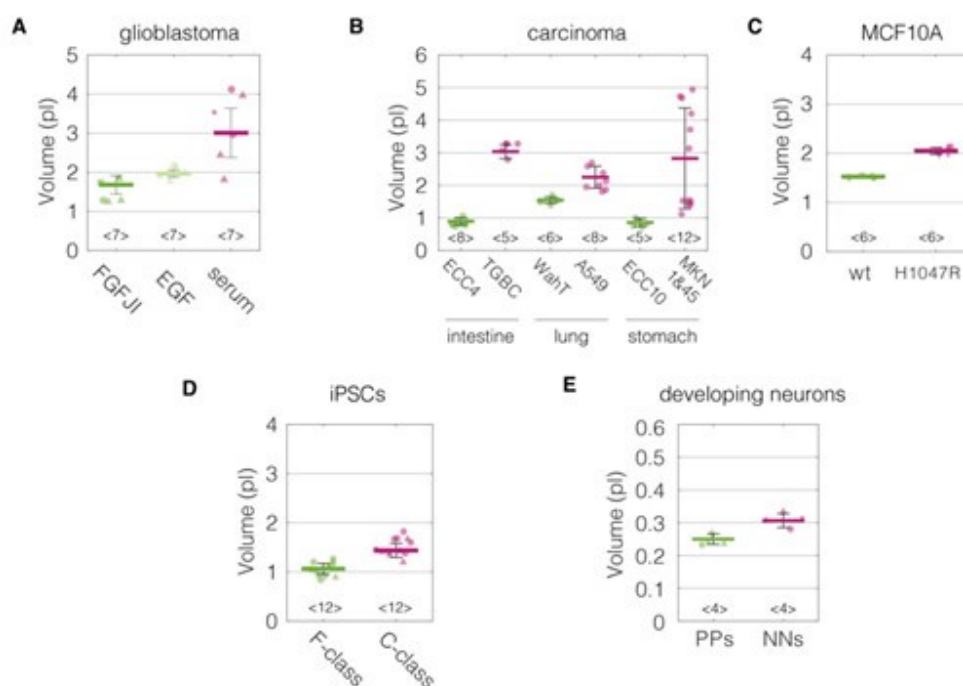
“Furthermore, CAV1 is known to modulate the activation of transcriptional cofactor yes-associated protein, YAP, in response to changes in stiffness of cell substrate (60) and in the mechanical stretch-induced mesothelial to mesenchymal transition (74).”

Which suggests that the genes identified here may be more broadly related to mechanical aspects of cells.

Of note, we do have some insights connected to the changes of cell volume — one of the biophysical properties mentioned by the reviewer — from our experiments. For all measurements performed with RT-DC, we can also calculate cell volumes from 2D cell contours (see Author response images 9, 10, and 11). For most of the cases (all apart from MEF CAV1KO), the stiffer phenotype of the cells, associated with higher levels of CAV1, shows a higher volume.

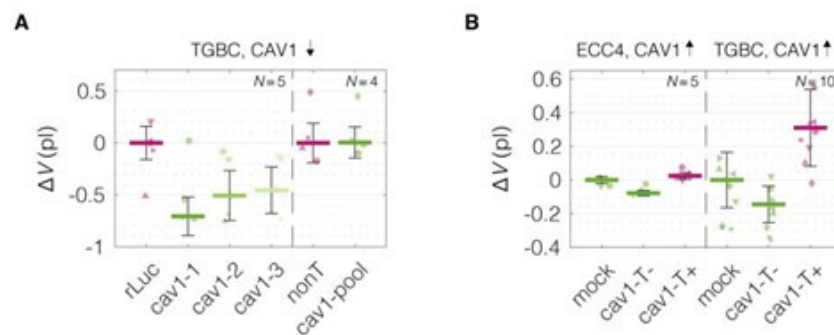
Author response image 9.

Cell volumes for the divergent cell states in the five characterized biological systems. (A) Glioblastoma. (B) Carcinoma, (C) MCF10A, (D) iPSCs, (E) Developing neurons. Data corresponds to Figure 2. Cell volumes were estimated using Shape-Out 1.0.10 by rotation of the cell contours.



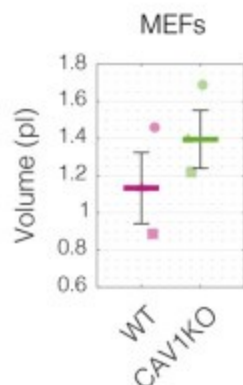
Author response image 10.

Cell volumes for CAV1 perturbation experiments. (A) CAV1 knock down performed in TGBC cells. (B) CAV1 overexpression in ECC4 and TGBC cells. Data corresponds to Figure 5. Cell volumes were estimated using Shape-Out 1.0.10 by rotation of the cell contours.



Author response image 11.

Cell volumes for WT and CAV1KO MEFs. Data corresponds to Figure S9. Cell volumes were estimated using Shape-Out 1.0.10 by rotation of the cell contours.



(3) The authors have already tested a large number of conditions in which perturbations of the level of expression of CAV1 correlates with changes in cell mechanics, but I was wondering whether it also has some direct explanatory value for the initial datasets used - for example for the glioblastoma cells from Figure 2, in the different media, would a knock-down of CAV1 prevent the increase in stiffness observed upon addition of serum, or for the carcinoma cells from different tissues treated with different compounds - if I understand well, the authors have tested a subset of these (ECC4 versus TGBC in figure 5) - how did they choose these and how general is it that the mechanical phenotype changes reported in Figure 2 are all mostly dependant on CAV1 expression level? I must say that the way the text is written and the results shown, it is hard to tell whether CAV1 is really having a dominant effect on cell mechanics in most of these contexts or only a partial effect. I hope I am being clear in my question - I am not questioning the conclusions of Figures 5 and 6, but asking whether the level of expression of CAV1, in the datasets reported in Figure 2, is the dominant explanatory feature for the differences in cell mechanics.

We thank reviewer for this comment and appreciate the value of the question about the generality and dominance of CAV1 in influencing cell mechanics.

On the computational side, we have addressed these issues by looking at the performance of CAV1 (among other identified genes) in classifying soft and stiff phenotypes across biological

systems (positive hypothesis I), as well as across data of different type (sequencing vs microarray data) and origin (different research institutions) (positive hypothesis II). CAV1 showed strong classification performance (Table 4), suggesting it is a general marker of stiffness changes.

On the experimental side, we conducted the perturbation experiments in two systems of choice: two intestinal carcinoma cell lines (ECC4 and TGBC) and the MCF10A breast epithelial cell line. These choices were driven by ease of handling, accessibility, as well as (for MCF10A) connection with a former study (Taveres et al, 2017). While we observed correlations between CAV1 expression and cell mechanics in wide range of datasets, the precise role of CAV1 in each system may vary, and further perturbation experiments in specific systems could be performed to solidify the direct/dominant role of CAV1 in cell mechanics. We hypothesize that the suggested knockdown of CAV1 upon serum addition in glioblastoma cells could reduce or prevent the increase in stiffness observed, though this experiment has not been performed.

In conclusion, while the computational analysis gives us confidence that CAV1 is a good indicator of cell stiffness, we predict that it acts in concert with other genes and in specific context could be replaced by other changes. We suggest that the suitability of CAV1 for manipulation of the mechanical properties should be tested in each system of interest before use.

To highlight the fact that the relevance of CAV1 for modulating cell mechanics in specific systems of interest should be tested and the mechanistic insights into how CAV1 regulates cell mechanics are still missing, we have added the following sentence in the discussion:

“The mechanical phenotype of cells is recognized as a hallmark of many physiological and pathological processes. Understanding how to control it is a necessary next step that will facilitate exploring the impact of cell mechanics perturbations on cell and tissue function (76). The increasing availability of transcriptional profiles accompanying cell state changes has recently been complemented by the ease of screening for mechanical phenotypes of cells thanks to the advent of high-throughput microfluidic methods (77). This provides an opportunity for data-driven identification of genes associated with the mechanical cell phenotype change in a hypothesis-free manner. Here we leveraged this opportunity by performing discriminative network analysis on transcriptomes associated with mechanical phenotype changes to elucidate a conserved module of five genes potentially involved in cell mechanical phenotype regulation. We provided evidence that the inferred conserved functional network module contains an ensemble of five genes that, in particular when combined in a unique combinatorial marker, are universal, specific and trustworthy markers of mechanical phenotype across the studied mouse and human systems. We further demonstrated on the example of a selected marker gene, CAV1, that its experimental up- and downregulation impacts the stiffness of the measured cells. This demonstrates that the level of CAV1 not only correlates with, but also is causative of mechanical phenotype change. The mechanistic insights into how precisely the identified genes are involved in regulating mechanical properties, how they interact with each other, and whether they are universal and dominant in various contexts all remain to be established in future studies.”

(4) It would be nice that the authors try to more directly address, in their discussion, what is the biological meaning of the set of 5 genes that they found - is it really mostly a product of the methodology used, useful but with little specific relevance to any biology, or does it have a deeper meaning? Either at a system level, or at an evolutionary level.

We would like to highlight that our manuscript is focused on the method that we introduce to identify sets of genes involved in the regulation of cell mechanics. The first implementation included here is only the beginning of this line of work which, in the future, will include

looking in detail at the biological meaning and the interconnectivity of the genes identified. Most likely, there is a deeper meaning of the identified module which could be revealed with a lot of dedicated future work. As it is a mere speculation at this point, we would like to refrain from going into more detail about it in the current manuscript. We provide below a few words of extended explanation and additional analysis that can shed light on the current limited knowledge of the connections between the genes and evolutionary preservation of the genes.

While it is difficult to prove at present, we do believe that the identified node of genes may have an actual biological meaning and is not a mere product of the used methodology. The PC-corr score used for applying the threshold and obtaining the gene network is high only if the Pearson's correlation between the two genes is high, meaning that the high connected module of genes identified show correlated expression and is likely co-regulated. Additionally, we performed the GO Term analysis using DAVID to assess the connections between the genes (Figure S3). We have now performed an additional analysis using two orthogonal tools the functional protein association tool STRING and KEGG Mapper.

With STRING, we found a moderate connectivity using the five network nodes identified in our study, and many of the obtained connections were based on text mining and co-expression, rather than direct experimental evidence (Author response image 12A). A more connected network can be obtained by allowing STRING to introduce further nodes (Author response image 12B). Interestingly, some of the nodes included by STRING in the extended network are nodes identified with milder PCcorr thresholds in our study (such as CNN2 or IGFBP3, see Table S3).

With KEGG Mapper, we did not find an obvious pathway-based clustering of the genes from the module either. A maximum of two genes were assigned to one pathway and those included:

- focal adhesions (pathway hsa04510): CAV1 and THBS1
- cytoskeleton in muscle cells (pathway hsa04820): FHL2 and THBS1
- proteoglycans in cancer (pathway hsa05205): CAV1 and THBS1.

As for the BRITE hierarchy, following classification was found:

- membrane trafficking(hsa04131): CAV1, IGFBP7, TAGLN, THBS, with following subcategories:

- endocytosis / lipid raft mediated endocytosis/caveolin-mediated endocytosis:

CAV1

- endocytosis / phagocytosis / opsonins: THBS1

- endocytosis / others/ insulin-like growth factor-binding proteins: IGFBP7 o others / actin-binding proteins/others: TAGLN.

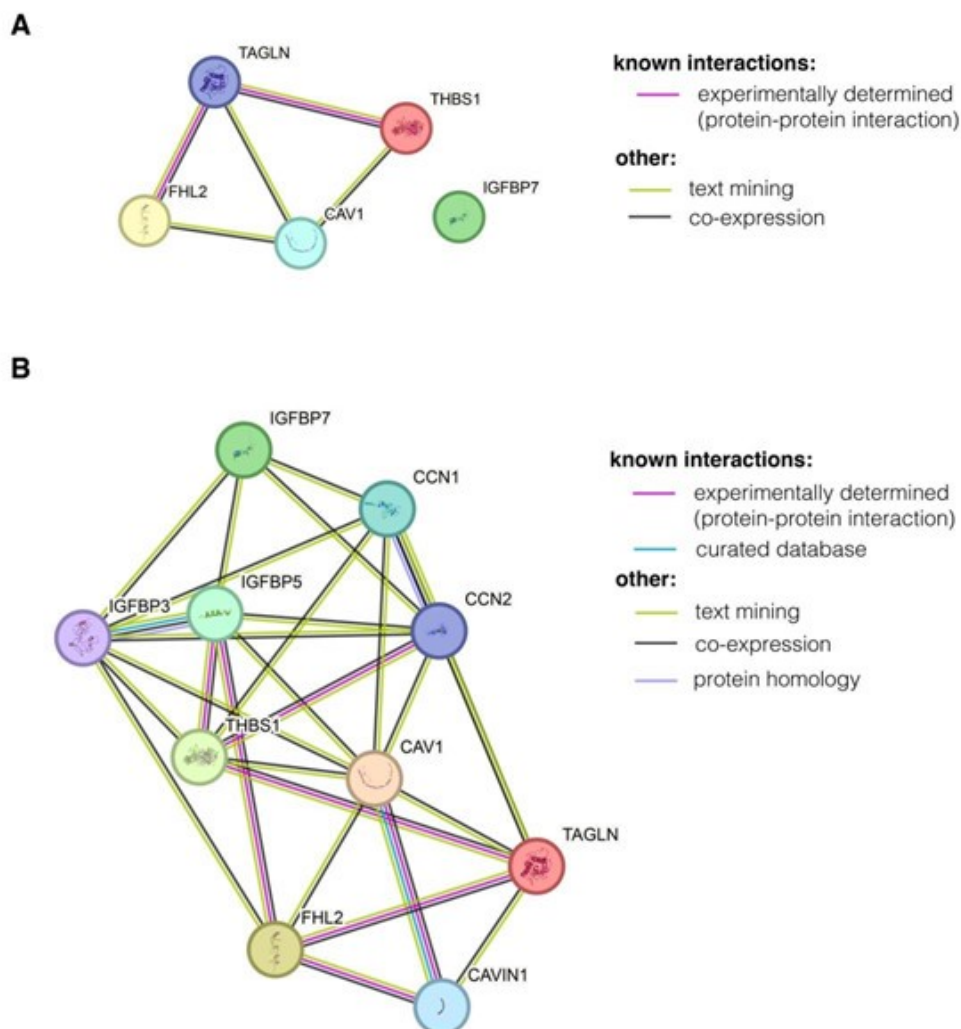
Taken together, all that analyses (DAVID, STRING, KEGG) show that at present no direct relationship/single pathway can be found that integrates all the genes from the identified modules. Future experiments, including investigations of how other module nodes are affected when one of the genes is manipulated, will help to establish actual physical or regulatory interactions between the genes from our module.

To touch upon the evolutionary perspective, we provide an overview of occurrence of the genes from the identified module across the evolutionary tree. This overview shows that the

five identified genes are preserved in phylum Chordata with quite high sequence similarity, and even more so within mammals (Author response image 13).

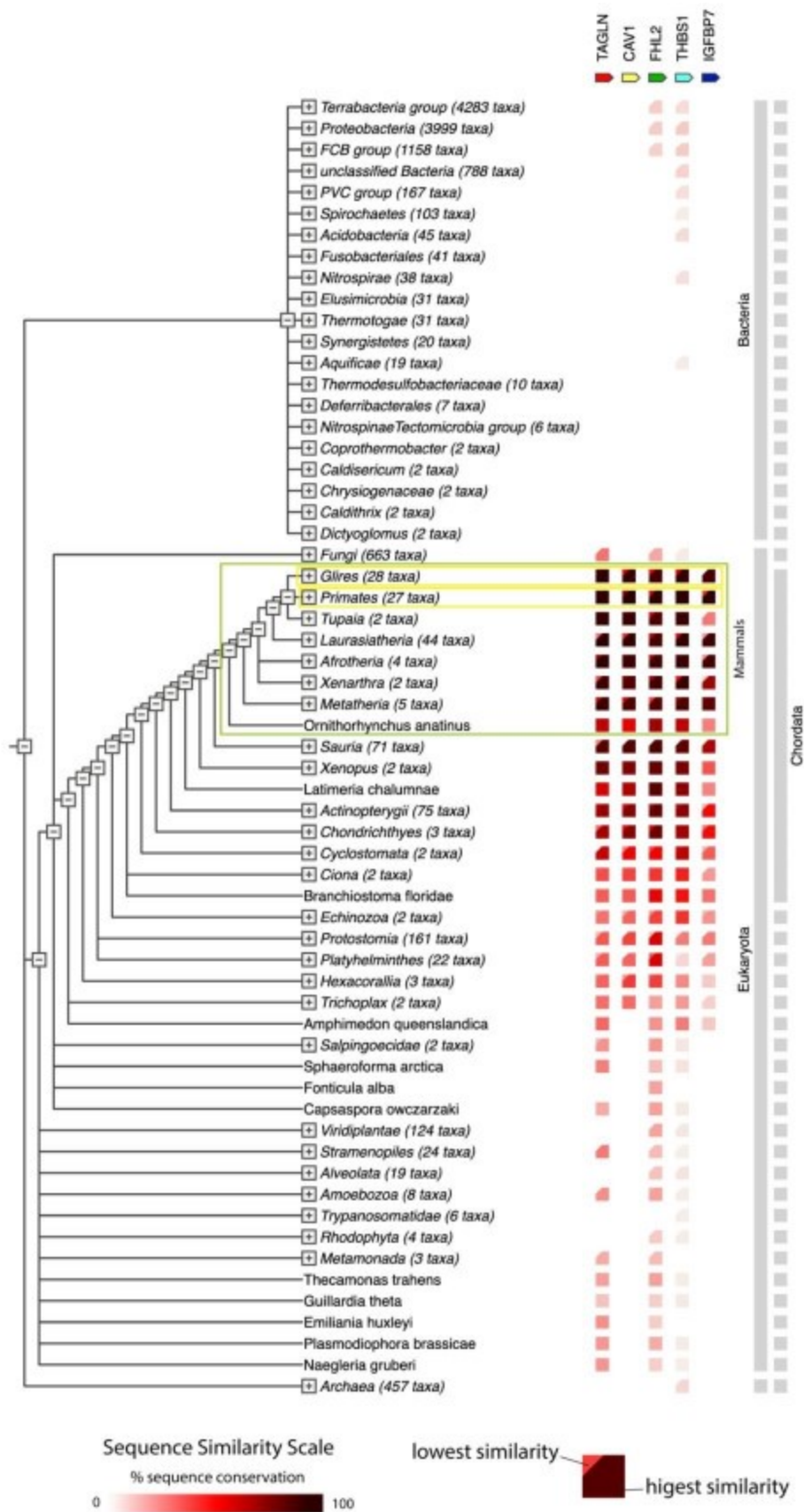
Author response image 12.

Visualisation of interactions between the nodes in the identified module using functional protein association networks tool STRING. (A) Connections obtained using multiple proteins search and entering the five network nodes. (B) Extended network that includes further genes to increase indirect connectivity. The genes are added automatically by STRING. Online version of STRING v12.0 was used with Homo sapiens as species of interest.



Author response image 13.

Co-occurrence of genes from the network module across the evolutionary tree. Mammals are indicated with the green frame, glires (include mouse), as well as primates (include human) are indicated with yellow frames. The view was generated using online version of STRING 12.0.



Reviewer #2 (Recommendations For Authors)

(1) The authors need to discuss the level of sensitivity of their mechanical measurements with RT-DC for changes to the membrane compared to changes in microtubules, nucleus, etc. The limited AFM measurements also seem membrane/cortex focused. For these and further reasons below, "universal" doesn't seem appropriate in the title or abstract, and should be deleted.

We thank the reviewer for this comment. Indeed, RT-DC is a technique that deforms the entire cell to a relatively low degree (inducing ca 17% mean strain, i.e. a deformation of approximately 2.5 μm on a cell with a 15 μm diameter, see Table S9 and Urbanska et al., Nat Methods 2020). Similarly, the AFM indentation experiments performed in this study (using a 5- μm diameter colloidal probe and 1 μm indentation) induce low strains, at which, according to current knowledge, the actin cortex dominates the measured deformations. However, other cellular components, including the membrane, microtubules, intermediate filaments, nucleus, other organelles, and cytoplasmic packing, can also contribute. We have reviewed these contributions in detail in a recent publication (Urbanska and Guck, 2024, Ann Rev Biophys., PMID 38382116). For a particular system, it is hard to speculate without further investigation which parts of the cell have a dominant effect on the measured deformability. We have added now a following paragraph in the discussion to include this information:

“The mechanical phenotype of single cells is a global readout of cell’s resistance to deformation that integrates contributions from all cellular components. The two techniques implemented for measuring cell mechanical in this study — RT-DC and AFM indentation using a spherical indenter with 5 μm radius — exert comparatively low strain on cells (< 3 μm , see Table S9), at which the actin cortex is believed to dominate the measured response. However, other cellular components, including the membrane, microtubules, intermediate filaments, nucleus, other organelles, and cytoplasmic packing, also contribute to the measured deformations (reviewed in detail in (79)) and, for a particular system, it is hard to speculate without further investigation which parts of the cell have a dominant effect on the measured deformability.”

The key strength of measuring the global mechanics is that such measurements are agnostic of the specific origin of the resistance to shape change. As such, the term “universal” could be seen as rather appropriate, as we are not testing specific contributions to cell mechanics, and we see the two methods used (RT-DC and AFM indentation) as representative when it comes to measuring global cell mechanics. And we highlighted many times throughout the text that we are measuring global single-cell mechanical phenotype.

Most importantly, however, we have used the term “universal” to capture that the genes are preserved across different systems and species, not in relation to the type of mechanical measurements performed and as such we would like to retain the term in the title.

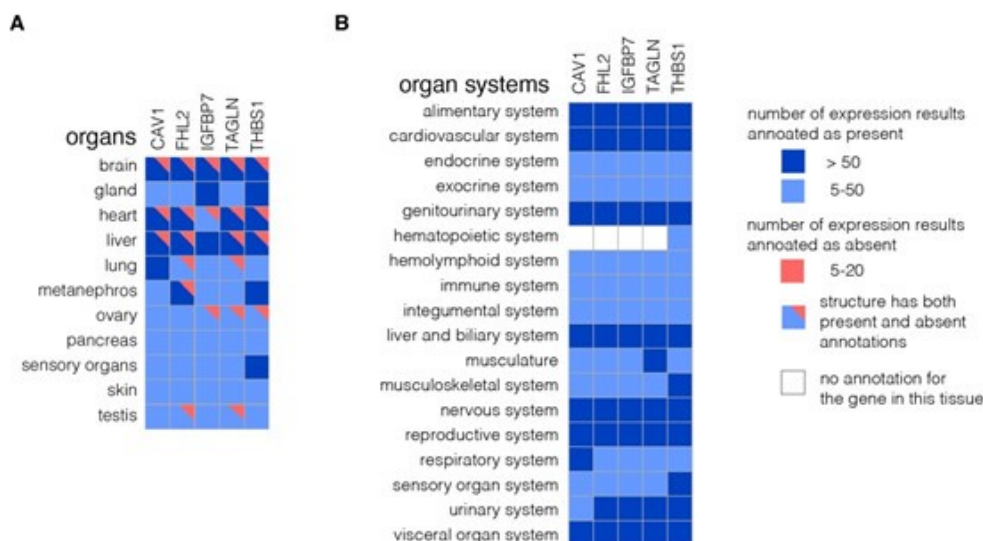
(2) Fig.2 cartoons of tissues is a good idea to quickly illustrate the range of cell culture lines studied. However, it obligates the authors to examine the relevant primary cell types in singlecell RNAseq of human and/or mouse tissues (e.g. Tabula Muris). They need to show CAV1 is expressed in glioblastoma, iPSCs, etc and not a cell culture artifact. CAV1 and the other genes also need to be plotted with literature values of tissue stiffness.

We thank the reviewer for this the comment; however, we do believe that the cartoons in Figure 2 should assist the reader to readily understand whether cultured cells derived from the respective tissues were used (see cartoons representing dishes), or the cells directly isolated from the tissue were measured (this is the case for the developing neurons dataset).

We did, however, follow the suggestion of the reviewer to use available resources and checked the expression of genes from the identified network module across various tissues in mouse and human. We first used the Mouse Genome Informatics (MGI; <https://www.informatics.jax.org/>) to visualize the expression of the genes across organs and organ systems (Author response image 14) as well as across more specific tissue structures (Author response image 15). These two figures show that the five identified genes are expressed quite broadly in mouse. We next looked at the expression of the five genes in the scRNASeq dataset from Tabula Muris (Author response image 16). Here, the expression of respective genes seemed more restricted to specific cell clusters. Finally, we also collected the cross-tissue expression of the genes from our module in human tissues from Human Protein Atlas v23 at both mRNA (Author response image 17) and protein (Author response image 18) levels. CAV1, IGFBP7, and THBS1 showed low tissue specificity at mRNA level, FHL2 was enriched in heart muscle and ovary (the heart enrichment is also visible in Author response image 15 for mouse) and TAGLN in endometrium and intestine. Interestingly, the expression at the protein level (Author response image 18) did not seem to follow faithfully the mRNA levels (Author response image 17). Overall, we conclude that the identified genes are expressed quite broadly across mouse and human tissues.

Author response image 14.

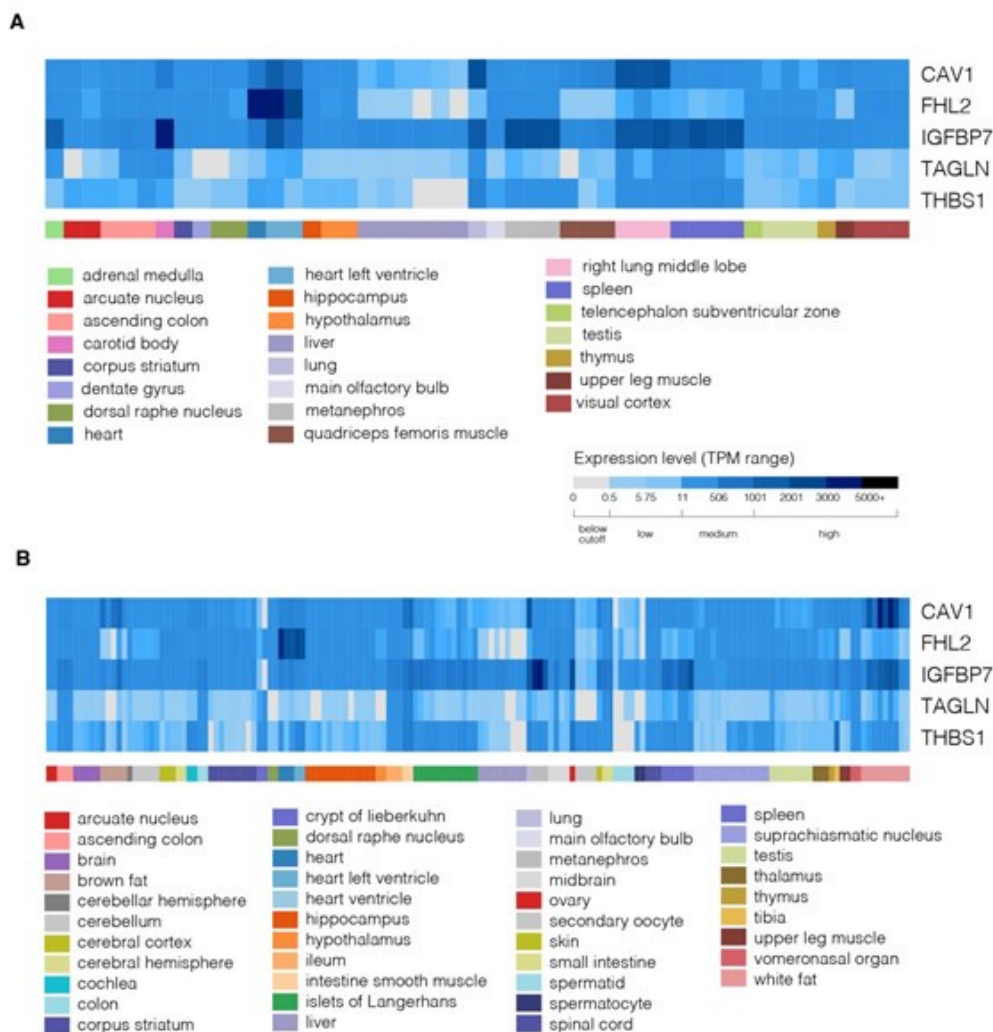
Expression of genes from the identified module across various organ and organ systems in mouse. The expression matrices for organs (A) and organ systems (B) were generated using Tissue x Gene Matrix tool of Gene eXpression Database (<https://www.informatics.jax.org/gxd/>, accessed on 22nd September 2024). No pre-selection of stage (age) and assay type (includes RNA and protein-based assays) was applied. The colors in the grid (blues for expression detected and reds for expression not detected) get progressively darker when there are more supporting annotations. The darker colors do not denote higher or lower levels of expression, just more evidence.



Author response image 15.

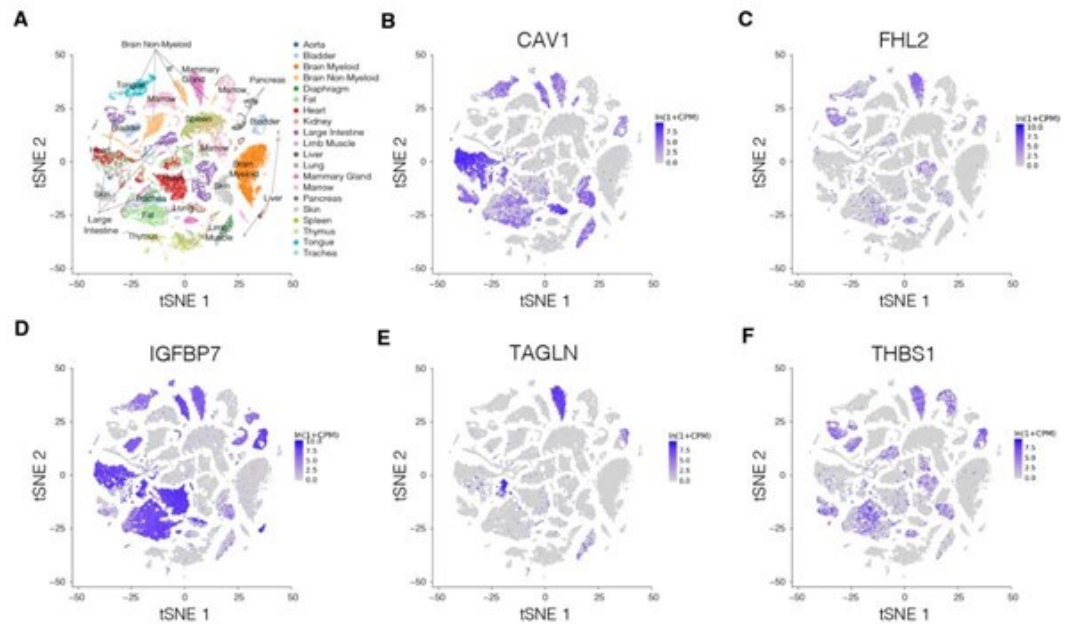
Expression of genes from the identified module across various mouse tissue structures. The expression matrices for age-selected mouse marked as adult (A) or young individuals (collected ages labelled P42-84 / P w6-w12 / P m1.5-3.0) (B) are presented and were generated

using RNASeq Heatmap tool of Gene eXpression Database (<https://www.informatics.jax.org/gxd/>, accessed on 2nd October 2024).



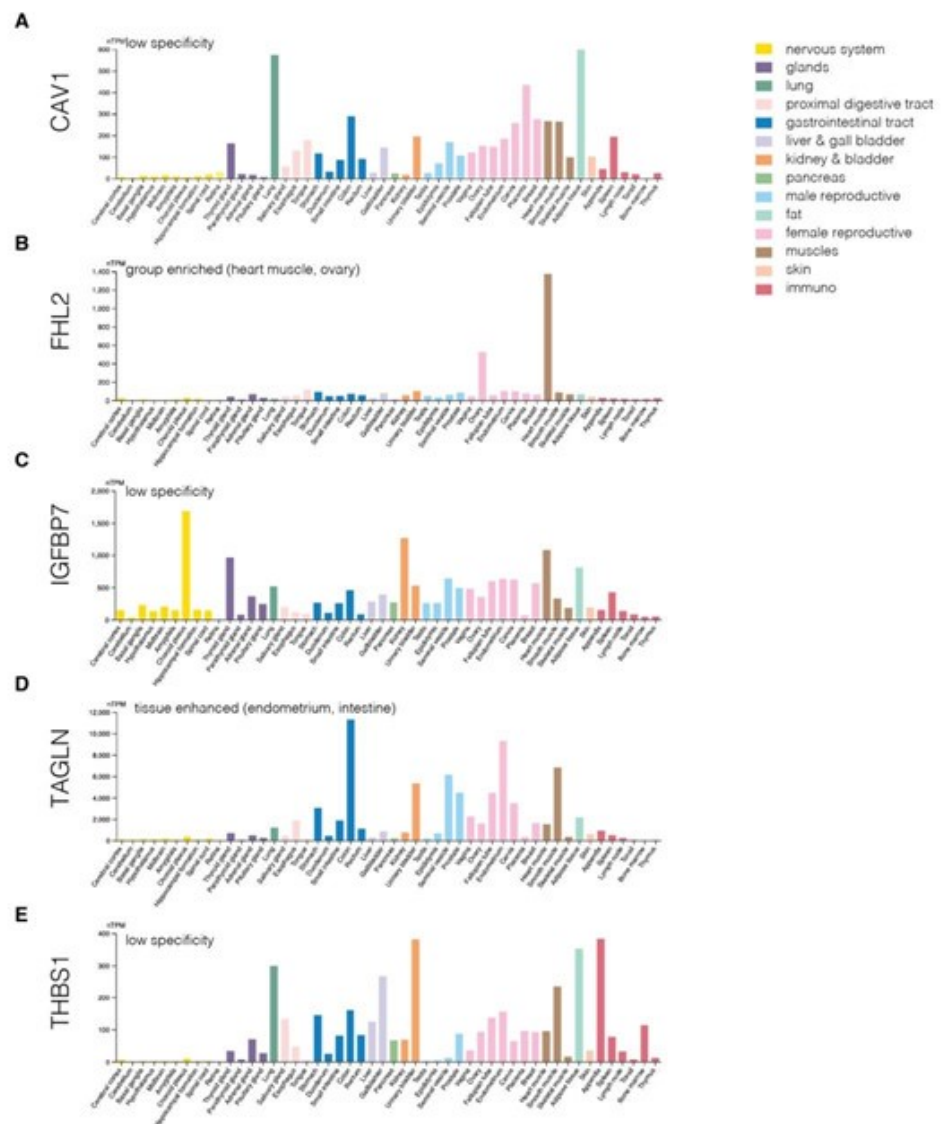
Author response image 16.

Expression of genes from the identified module across various cell types and organs in t-SNE embedding of *Tabula Muris* dataset. (A) t-SNE clustering color-coded by organ. (B-F) t-SNE clustering color-coded for expression of CAV1 (B), IGFBP7 (C), FHL2 (D), TAGLN (E), and THBS1 (F). The plots were generated using FACS-collected cells data through the visualisation tool available at <https://tabulamuris.sf.czbiohub.org/> (accessed on 22nd September 2024).



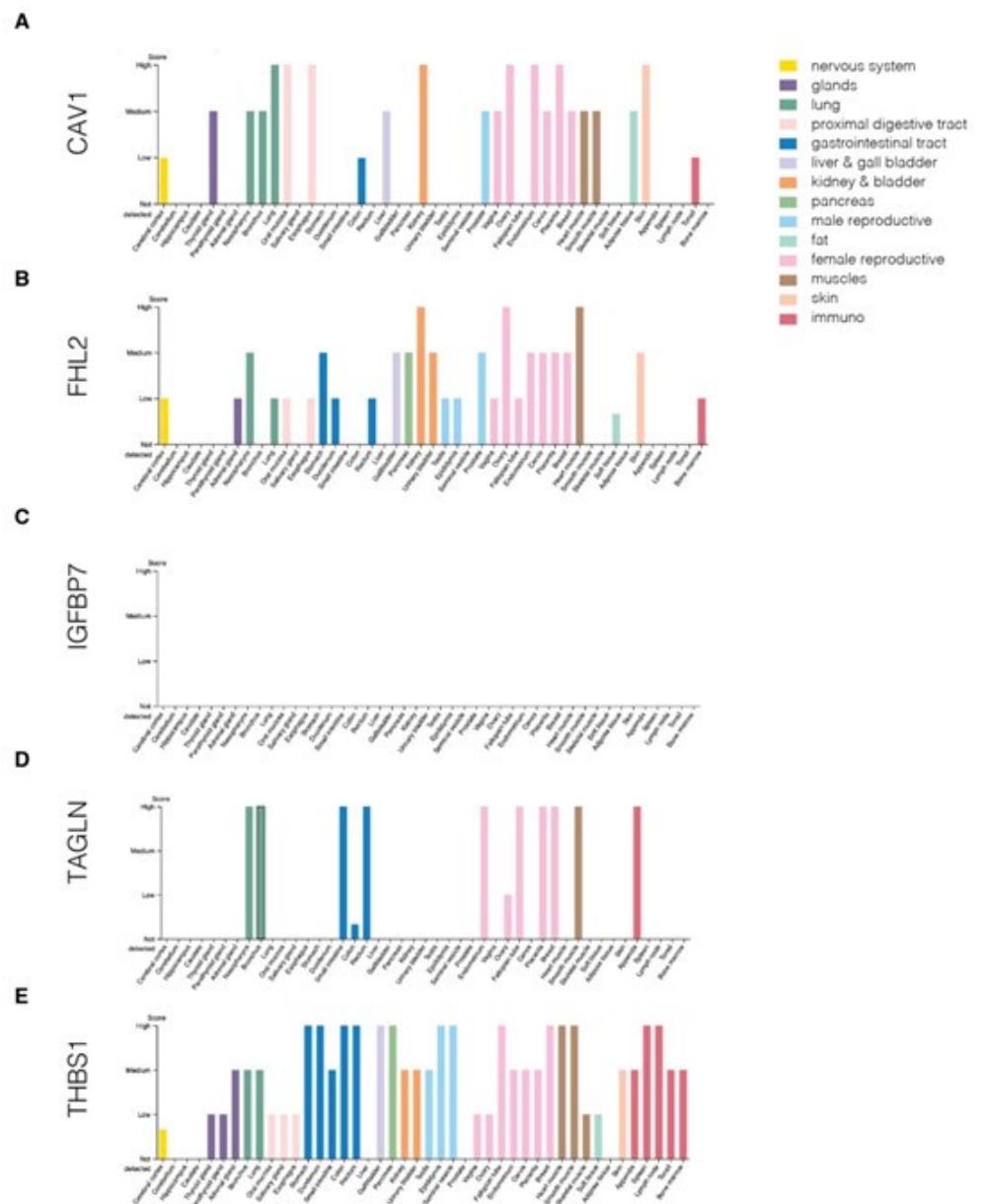
Author response image 17.

Expression of genes from the identified module at the mRNA level across various human tissues. (A-E) Expression levels of CAV1 (A), IGFBP7 (B), FHL2 (C), TAGLN (D), and THBS1 (E). The plots were generated using consensus dataset from Human Protein Atlas v23 <https://www.proteinatlas.org/> (accessed on 22nd September 2024).



Author response image 18.

Protein levels of genes from the identified module across various human tissues. (A-E) Protein levels of CAV1 (A), IGFBP7 (B), FHL2 (C), TAGLN (D), and THBS1 (E). The plots were generated using Human Protein Atlas v23 <https://www.proteinatlas.org/> (accessed on 22nd September 2024).



Regarding literature values and tissue stiffness, we would like to argue that cell stiffness is not equivalent to tissue stiffness, and we are interested in the former. Tissue stiffness is governed by a combination of cell mechanical properties, cell adhesions, packing and the extracellular matrix. There can be, in fact, mechanically distinct cell types (for example characterized by different metabolic state, malignancy level etc) within one tissue of given stiffness. Hence, we consider that testing for the correlation between tissue stiffness and expression of identified genes is not immediately relevant.

(3) Fig.5D,H show important time-dependent mechanics that need to be used to provide explanations of the differences in RT-DC (5B,F) and in standard AFM indentation expts (5C,G). In particular, it looks to me that RT-DC is a high-f/short-time measurement compared to the AFM indentation, and an additional Main or Supp Fig needs to somehow combine all of this data to clarify this issue.

We thank the reviewer for this comment. It is indeed the case, that cells typically display higher stiffness when probed at higher rates. We have now expanded on this aspect of the results and added a supplementary figure (Fig. S10) that illustrates the frequencies used in different methods and summarizes the apparent Young's moduli values into one plot in a frequencyordered manner. Of note, we typically acquire RT-DC measurements at up to three flowrates, and the increase in measurement flow rates accompanying increase in flow rate also results in higher extracted apparent Young's moduli (see Fig. S10 B,D). We have further added Table S9 that summarizes operating parameters of all three methods used for probing cell mechanics in this manuscript:

“The three techniques for characterizing mechanical properties of cells — RT-DC, AFM indentation and AFM microrheology — differ in several aspects (summarized in Table S9), most notably in the frequency at which the force is applied to cells during the measurements, with RT-DC operating at the highest frequency (~600 Hz), AFM microrheology at a range of frequencies in-between (3–200 Hz), and AFM indentation operating at lowest frequency (5 Hz) (see Table S9 and Figure S10A). Even though the apparent Young's moduli obtained for TGBCS cells were consistently higher than those for ECC4 cells across all three methods, the absolute values measured for a given cell line varied depending on the methods: RT-DC measurements yielded higher apparent Young's moduli compared to AFM indentation, while the apparent Young's moduli derived from AFM microrheology measurements were frequency-dependent and fell between the other two methods (Fig. 5B–D, Fig. S10B). The observed increase in apparent Young's modulus with probing frequency aligns with previous findings on cell stiffening with increased probing rates observed for both AFM indentation (68, 69) and microrheology assays (70–72).”

(4) The plots in Fig.S4 are important as main Figs, particularly given the cartoons of different tissues in Fig.1,2. However, positive correlations for a few genes (CAV1, IGFBP7, TAGLN) are most clear for the multiple lineages that are the same (stomach) or similar (gli, neural & pluri). The authors need to add green lines and pink lines in all plots to indicate the 'lineagespecific' correlations, and provide measures where possible. Some genes clearly don't show the same trends and should be discussed.

We thank reviewer for this comment. It is indeed an interesting observation (and worth highlighting by adding the fits to lineage-restricted data) that the relationship between relative change in Young's modulus and the selected gene expression becomes steeper for samples from similar tissue contexts.

For the sake of keeping the main manuscript compact, we decided to keep Fig. S7 (formerly Fig. S4) in the supplement, however, we did add the linear fit to the glioblastoma dataset (pink line) and a fit to the related neural/embryonic datasets (gli, neural & pluri – purple line) as advised — see below.

We did not pool the stomach data since it is represented by a single point in the figure, aligning with how the data is presented in the main text—stomach adenocarcinoma cell lines (MKN1 and MKN45) are pooled in Fig. 1B (see below).

We have also amended the respective results section to emphasize that, in certain instances, the correlation between changes in mechanical phenotype and alterations in the expression of analysed genes may be less pronounced:

“The relation between normalized apparent Young's modulus change and fold-change in the expression of the target genes is presented in Fig. S7. The direction of changes in the expression levels between the soft and stiff cell states in the validation datasets was not always following the same direction (Fig. 4, C to F, Fig. S7). This suggests that the genes associated with cell mechanics may not have a monotonic relationship with cell stiffness, but

rather are characterized by different expression regimes in which the expression change in opposite directions can have the same effect on cell stiffness. Additionally, in specific cases a relatively high change in Young's modulus did not correspond to marked expression changes of a given gene — see for example low CAV1 changes observed in MCF10A PIK3CA mutant (Fig. S7A), or low IGFBP7 changes in intestine and lung carcinoma samples (Fig. S7C). This indicates that the importance of specific targets for the mechanical phenotype change may vary depending on the origin of the sample.”

(5) Table-1 neuro: Perhaps I missed the use of the AFM measurements, but these need to be included more clearly in the Results somewhere.

To clarify: there were no AFM measurements performed for the developing neurons (neuro) dataset, and it is not marked as such in Table 1. There are previously published AFM measurements for the iPSCs dataset (maybe that caused the confusion?), and we referred to them as such in the table by citing the source (Urbanska et al (30)) as opposed to the statement “this paper” (see the last column of Table 1). We did not consider it necessary to include these previously published data. We have added additional horizontal lines to the table that will hopefully help in the table readability.

Reviewer #3 (For Authors)

Major

- I strongly encourage the authors to validate their approach with a gene for which mechanical data does not exist yet, or explore how the combination of the 5 identified genes is the novel regulator of cell mechanics.

We appreciate the reviewer's insightful comment and agree that it would be highly interesting to validate further targets and perform combinatorial perturbations. However, it is not feasible at this point to expand the experimental data beyond the one already provided. We hope that in the future, the collective effort of the cell mechanics community will establish more genes that can be used for tuning of mechanical properties of cells.

- If this paper aims at highlighting the power of PC-Corr as a novel inference approach, the authors should compare its predictive power to that of classical co-expression network analysis or an alternative gold standard.

We thank the reviewer for the suggestion to compare the predictive power of PC-Corr with classical co-expression network analysis or an alternative gold standard. PC-corr has been introduced and characterized in detail in a previous publication (Ciucci et al, 2017, Sci. Rep.), where it was compared against standard co-expression analysis methods. Here we implement PC-corr for a particular application. Thus, we do not see it as central to the message of the present manuscript to compare it with other available methods again.

- The authors call their 5 identified genes "universal, trustworthy and specific". While they provide a great amount of data all is derived from human and mouse cell lines. I suggest toning this down.

We thank the reviewers for this comment. To clarify, the terms universal, trustworthy and specific are based on the specific hypotheses tested in the validation part of the manuscript, but we understand that it may cause confusion. We have now toned that the statement by adding “universal, trustworthy and specific across the studied mouse and human systems” in the following text fragments:

(1) Abstract

“(…) We validate in silico that the identified gene markers are universal, trustworthy and specific to the mechanical phenotype across the studied mouse and human systems, and demonstrate experimentally that a selected target, CAV1, changes the mechanical phenotype of cells accordingly when silenced or overexpressed. (…)”

(2) Last paragraph of the introduction

“(…) We then test the ability of each gene to classify cell states according to cell stiffness in silico on six further transcriptomic datasets and show that the individual genes, as well as their compression into a combinatorial marker, are universally, specifically and trustworthily associated with the mechanical phenotype across the studied mouse and human systems. (…)”

(3) First paragraph of the discussion

“We provided strong evidence that the inferred conserved functional network module contains an ensemble of five genes that, in particular when combined in a unique combinatorial marker, are universal, specific and trustworthy markers of mechanical phenotype across the studied mouse and human systems.”

Minor suggestions

- The authors point out how genes that regulate mechanics often display non-monotonic relations with their mechanical outcome. Indeed, in Fig.4 developing neurons have lower CAV1 in the stiff group. Perturbing CAV1 expression in that model could show the nonmonotonic relation and strengthen their claim.

We thank reviewer for highlighting this important point. It would indeed be interesting to explore the changes in cell stiffness upon perturbation of CAV1 in a system that has a potential to show an opposing behavior. Unfortunately, we are unable to expand the experimental part of the manuscript at this time. We do hope that this point can be addressed in future research, either by our team or other researchers in the field.

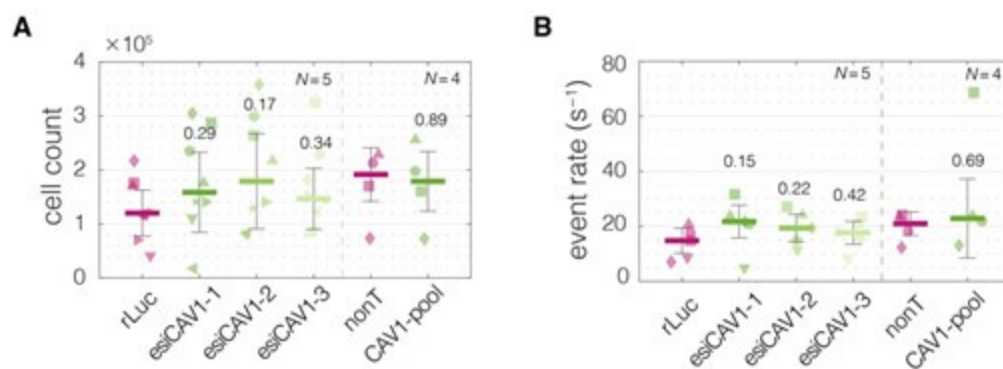
- In their gene ontology enrichment assay, the authors claim that their results point towards reduced transcriptional activity and reduced growth/proliferation in stiff compared to soft cells. Proving this with a simple proliferation assay would be a nice addition to the paper.

This is a valuable suggestion that should be followed up on in detail in the future. To give a preliminary insight into this line of investigation, we have had a look at the cell count data for the CAV1 knock down experiments in TGBC cells. Since CAV1 is associated with the GO Term “negative regulation of proliferation/transcription” (high CAV1 – low proliferation), we would expect that lowering the levels of CAV1 results in increased proliferation and higher cell counts at the end of experiment (3 days post transfection). As illustrated in Author response image 19 below, the cell counts were higher for the samples treated with CAV1 siRNAs, though, not in a statistically significant way. Interestingly, the magnitude of the effect partially mirrored the trends observed for the cell stiffness (Figure 5F).

Author response image 19.

The impact of CAV1 knock down on cell counts in TGBC cells. (A) Absolute cell counts per condition in a 6-well format. Cell counts were performed when harvesting for RT-DC measurements using an automated cell counter (Countess II, Thermo Fisher Scientific). (B) The event rates observed during the RT-DC measurements. The harvested cells are resuspended in a specific volume of measuring buffer standardized per experiment (50-100 µl); thus, the event rates reflect the absolute cell numbers in the respective samples.

Horizontal lines delineate medians with mean absolute deviation (MAD) as error, datapoints represent individual measurement replicates, with symbols corresponding to matching measurement days. Statistical analysis was performed using two sample two-sided Wilcoxon rank sum test.



Methods

- The AFM indentation experiments are performed with a very soft cantilever at very high speeds. Why? Also, please mention whether the complete AFM curve was fitted with the Hertz/Sneddon model or only a certain area around the contact point.

We thank the reviewer for this comment. However, we believe that the spring constants and indentation speeds used in our study are typical for measurements of cells and not a cause of concern.

For the indentation experiments, we used Arrow-TL1 cantilevers (nominal spring constant $k = 0.035\text{--}0.045 \text{ N m}^{-1}$, Nanoworld, Switzerland) which are used routinely for cell indentation (with over 200 search results on Google Scholar using the term: "Arrow-TL1"+"cell", and several former publications from our group, including Munder et al 2016, Tavares et al 2017, Urbanska et al 2017, Taubenberger et al 2019, Abuhattum et al 2022, among others). Additionally, cantilevers with the spring constants as low as 0.01 N m^{-1} can be used for cell measurements (Radmacher 2002, Thomas et al, 2013).

The indentation speed of $5 \mu\text{m s}^{-1}$ is not unusually high and does not result in significant hydrodynamic drag.

For the microrheology experiments, we used slightly stiffer and shorter (100/200 μm compared to 500 μm for Arrow-TL1) cantilevers: PNP-TR-TL (nominal spring constant $k = 0.08 \text{ N m}^{-1}$, Nanoworld, Switzerland). The measurement frequencies of 3-200 Hz correspond to movements slightly faster than $5 \mu\text{m s}^{-1}$, but cells were indented only to 100 nm, and the data were corrected for the hydrodynamic drag (see equation (8) in Methods section).

Author response image 20.

Exemplary indentation curve obtained using arrow-TL1 decorated with a 5- μm sphere on a ECC4 cell. The shown plot is exported directly from JPK Data Processing software. The area shaded in grey is the area used for fitting the Sneddon model.

In the indentation experiments, the curves were fitted to a maximal indentation of 1.5 μm (rarely exceeded, see Author response image 20). We have now added this information to the methods section:

- Could the authors include the dataset wt #1 in Fig 4D? Does it display the same trend?

We thank the reviewer for this comment. To clarify: in the MCF10A dataset (GEO: GSE69822) there are exactly three replicates of each wt (wild type) and ki (knock-in, referring to the H1047R mutation in the PIK3CA) samples. The numbering wt#2, wt#3, wt#4 originated from the short names that were used in the working files containing non-averaged RPKM (possibly to three different measurement replicates that may have not been exactly paired with the ki samples). We have now renamed the samples as wt#1, wt#2 and wt#3 to avoid the confusion. This naming also reflects better the sample description as deposited in the GSE69822 dataset (see Author response table 2).

Author response table 2.

Short Name	Sample No	Sample Description	Displayed Name (old)	Displayed Name (new)
wt_2_0	GSM1709515	MCF10a WT_t=0_replicate1 condition: no EGF + DMSO	wt #2	wt #1
wt_3b_0	GSM1709516	MCF10a WT_t=0_replicate2 condition: no EGF + DMSO	wt #3	wt #2
wt_4_0	GSM1709517	MCF10a WT_t=0_replicate3 condition: no EGF + DMSO	wt #4	wt #3
ki_1_0	GSM1709554	MCF10a PIK3CA H1047R_t=0_replicate1 condition: no EGF + DMSO	ki#1	ki#1
ki_2_0	GSM1709555	MCF10a PIK3CA H1047R_t=0_replicate2 condition: no EGF + DMSO	ki#2	ki#2
ki_3_0	GSM1709556	MCF10a PIK3CA H1047R_t=0_replicate3 condition: no EGF + DMSO	ki#3	ki#3

- Reference (3) is an opinion article with the last author as the sole author. It is used twice as a self-standing reference, which is confusing, as it suggests there is previous experimental evidence.

We thank the reviewer for pointing this out and agree that it may not be appropriate to cite the article (Guck 2019 Biophysical Reviews, formerly Reference (3), currently Reference (76)) in all instances. The references to this opinion article have now been removed from the introduction:

“The extent to which cells can be deformed by external loads is determined by their mechanical properties, such as cell stiffness. Since the mechanical phenotype of cells has been shown to reflect functional cell changes, it is now well established as a sensitive label-free biophysical marker of cell state in health and disease (1-2).”

“Alternatively, the problem can be reverse-engineered, in that omics datasets for systems with known mechanical phenotype changes are used for prediction of genes involved in the regulation of mechanical phenotype in a mechanomics approach.”

But has been kept in the discussion:

“The mechanical phenotype of cells is recognized as a hallmark of many physiological and pathological processes. Understanding how to control it is a necessary next step that will facilitate exploring the impact of cell mechanics perturbations on cell and tissue function

This reference seems appropriate to us as it expands on the point that our ability to control cell mechanics will enable the exploration of its impact on cell and tissue function, which is central to the discussion of the current manuscript.

-The authors should mention what PC-corr means. Principle component correlation? Pearson's coefficient correlation?

PC-corr is a combination of loadings from the principal component (PC) analysis and Pearson's correlation for each gene pair. We have aimed at conveying this in the “Discriminative network analysis on prediction datasets” result section. We have now added and extra sentence at the first appearance of PC-corr to clarify that for the readers from the start:

“After characterizing the mechanical phenotype of the cell states, we set out to use the accompanying transcriptomic data to elucidate genes associated with the mechanical phenotype changes across the different model systems. To this end, we utilized a method for inferring phenotype-associated functional network modules from omics datasets termed PCCorr (28), that relies on combining loadings obtained from the principal component (PC) analysis and Pearson's correlation (Corr) for every pair of genes. PC-Corr was performed individually on two prediction datasets, and the obtained results were overlayed to derive a conserved network module. Owing to the combination of the Pearson's correlation coefficient and the discriminative information included in the PC loadings, the PC-corr analysis does not only consider gene co-expression — as is the case for classical co-expression network analysis — but also incorporates the relative relevance of each feature for discriminating between two or more conditions; in our case, the conditions representing soft and stiff phenotypes. The overlaying of the results from two different datasets allows for a multi-view analysis (utilizing multiple sets of features) and effectively merges the information from two different biological systems.”

- The formatting of Table 1 is confusing. Horizontal lines should be added to make it clear to the reader which datasets are human and which mouse as well as which accession numbers belong to the carcinomas.

Horizontal lines have now been added to improve the readability of Table 1. We hope that makes the table easier to follow and satisfies the request. We assume that further modifications to the table appearance may occur during publishing process in accordance with the publisher's guidelines.

- In many figures, data points are shown in different shapes without an explanation of what the shapes represent.

We thank the reviewer for this comment and apologize for not adding this information earlier. We have added explanations of the symbols to captions of Figures 2, 3, 5, and 6 in the main text:

“Fig. 2. Mechanical properties of divergent cell states in five biological systems. Schematic overviews of the systems used in our study, alongside with the cell stiffness of individual cell states parametrized by Young's moduli E . (...) Statistical analysis was performed using generalized linear mixed effects model. The symbol shapes represent measurements of cell

lines derived from three different patients (A), matched experimental replicates (C), two different reprogramming series (D), and four different cell isolations (E). Data presented in (A) and (D) were previously published in ref (29) and (30), respectively.”

“Fig. 3. Identification of putative targets involved in cell mechanics regulation. (A) Glioblastoma and iPSC transcriptomes used for the target prediction intersect at 9,452 genes. (B, C) PCA separation along two first principal components of the mechanically distinct cell states in the glioblastoma (B) and iPSC (C) datasets. The analysis was performed using the gene expression data from the intersection presented in (A). The symbol shapes in (B) represent cell lines derived from three different patients. (...)”

“Fig. 5. Perturbing levels of CAV1 affects the mechanical phenotype of intestine carcinoma cells. (...) In (E), (F), (I), and (J), the symbol shapes represent experiment replicates.”

“Fig. 6. Perturbations of CAV1 levels in MCF10A-ER-Src cells result in cell stiffness changes. (...) Statistical analysis was performed using a two-sided Wilcoxon rank sum test. In (B), (D), and (E), the symbol shapes represent experiment replicates.”

As well as to Figures S2, S9, and S11 in the supplementary material (in Figure S2, the symbol explanation was added to the legends in the figure panels as well):

“Fig. S2. Plots of area vs deformation for different cell states in the characterized systems. Panels correspond to the following systems: (A) glioblastoma, (B) carcinoma, (C) non-tumorigenic breast epithelia MCF10A, (D) induced pluripotent stem cells (iPSCs), and (E) developing neurons. 95%- and 50% density contours of data pooled from all measurements of given cell state are indicated by shaded areas and continuous lines, respectively. Datapoints indicate medians of individual measurements. The symbol shapes represent cell lines derived from three different patients (A), two different reprogramming series (D), and four different cell isolations (E), as indicated in the respective panels. (...)”

“Fig. S9. CAV1 knock-out mouse embryonic fibroblasts (CAV1KO) have lower stiffness compared to the wild type cells (WT). (...) (C) Apparent Young’s modulus values estimated for WT and CAV1KO cells using areadeformation data in (B). The symbol shapes represent experimental replicates. (...)”

“Fig. S11. Plots of area vs deformation from RT-DC measurements of cells with perturbed CAV1 levels. Panels correspond to the following experiments: (A and B) CAV1 knock-down in TGBC cells using esiRNA (A) and ONTarget siRNA (B), (C and D) transient CAV1 overexpression in ECC4 cells (C) and TGBC cells (D). Datapoints indicate medians of individual measurement replicates. The isoelasticity lines in the background (gray) indicate regions of same apparent Young’s moduli. The symbol shapes represent experimental replicates.”

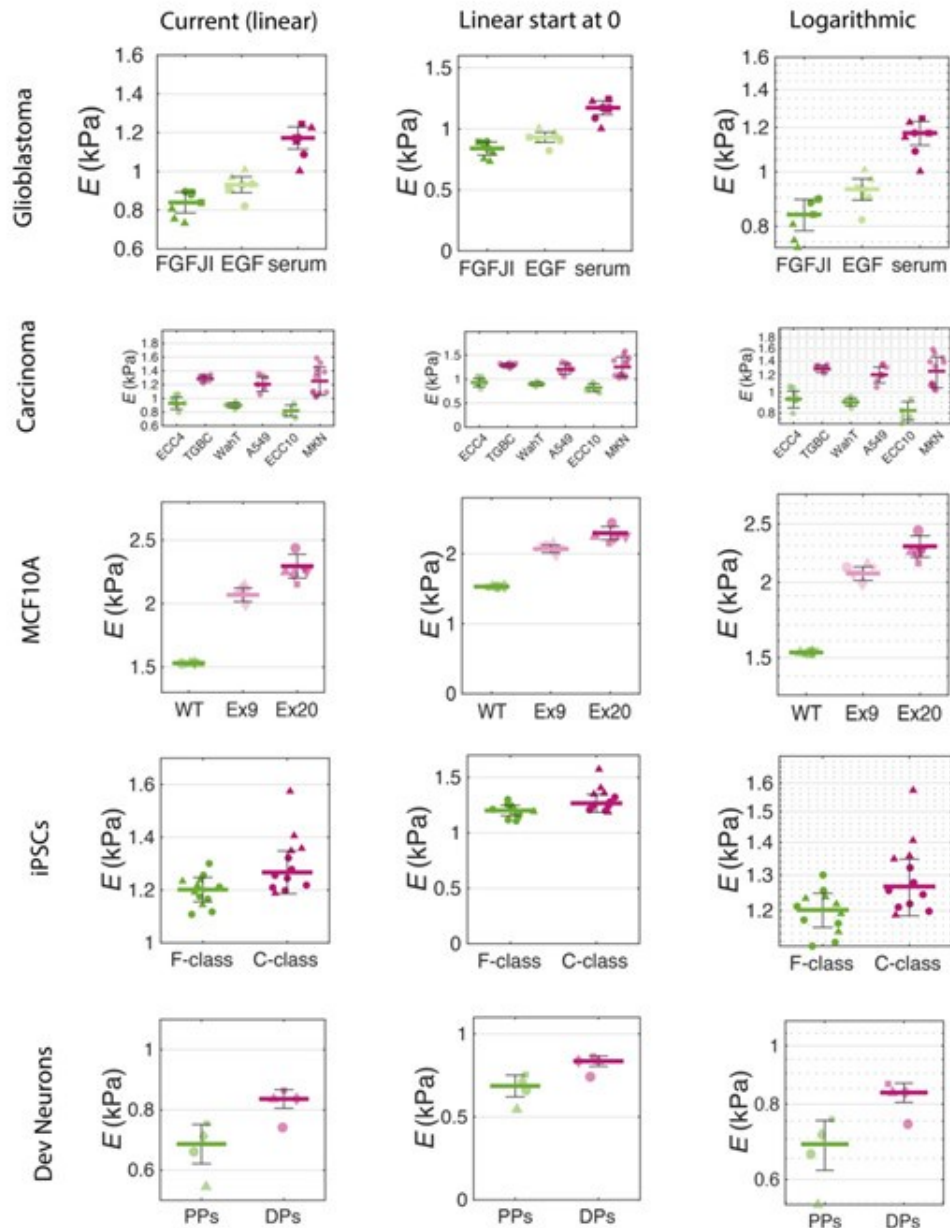
- In Figure 2, the difference in stiffness appears bigger than it actually is because the y-axes are not starting at 0.

While we acknowledge that starting the y-axes at a value other than 0 is generally not ideal, we chose this approach to better display data variability and minimize empty space in the plots.

A similar effect can be achieved with logarithmic scaling, which is a common practice (see Author response image 21 for visualization). We believe our choice of axes cut-off enhances the interpretability of the data without misleading the viewer.

Author response image 21.

Visualization of different axis scaling strategies applied to the five datasets presented in Figure 2 of the manuscript.



Of note, apparent Young's moduli obtained from RT-DC measurements typically span 0.5-3.0 kPa (see Figure 2.3 from Urbanska et al 2021, PhD thesis). Differences between treatments rarely exceed a few hundred pascals. For example, in an siRNA screen of mitotic cell mechanics regulators in *Drosophila* cells (Kc167), the strongest hits (e.g., Rho1, Rok, dia) showed changes in stiffness of 100-150 Pa (see Supplementary Figure 11 from Rosendahl, Plak et al 2018, Nature Methods 15(5): 355-358).

- In Figure 3, I don't personally see the benefit of showing different cut-offs for PC-corr. In the end, the paper focuses on the 5 genes in the pentagram. I think only showing one of the cutoffs and better explaining why those target genes were picked would be sufficient and make it clearer for the reader.

We believe it is beneficial to show the extended networks for a few reasons. First, it demonstrates how the selected targets connect to the broader panel of the genes, and that the selected module is indeed much more interconnected than other nodes. Secondly, the chosen

PC-corr cut-off is somewhat arbitrary and it may be interesting to look through the genes from the extended network as well, as they are likely also important for regulating cell mechanics. This broader view may help readers identify familiar genes and recognizing the connections to relevant signaling networks and processes of interest.

- In Figure 4C, I suggest explaining why the FANTOM5 and not another dataset was used for the visualization here and mentioning whether the other datasets were similar.

In Figure 4C, we have chosen to present data corresponding to FANTOM5, because that was the only carcinoma dataset in which all the cell lines tested mechanically are presented. We have now added this information to the caption of Figure 4. Additionally, the clustergrams corresponding to the remaining carcinoma datasets (CCLE RNASeq, Genetech) are presented in supplementary figures S4-S6.

“The target genes show clear differences in expression levels between the soft and stiff cell states and provide for clustering of the samples corresponding to different cell stiffnesses in both prediction and validation datasets (Fig. 4, Figs. S4-S6).”

Typos

We would like to thank the Reviewer#3 for their detailed comments on the typos and details listed below. This is much appreciated as it improved the quality of our manuscript.

- In the first paragraph of the results section the 'and' should be removed from this sentence: Each dataset encompasses two or more cell states characterized by a distinct mechanical phenotype, and for which transcriptomic data is available.

The sentence has been corrected and now reads:

“Each dataset encompasses two or more cell states characterized by a distinct mechanical phenotype, and for which transcriptomic data is available.”

- In the methods in the MCF10A PIK3CA cell lines part, it says cell liens instead of cell lines.

The sentence has been corrected and now reads:

“The wt cells were additionally supplemented with 10 ng ml⁻¹ EGF (E9644, Sigma-Aldrich), while mutant cell lines were maintained without EGF.”

- In the legend of Figure 6 "accession number: GSE17941, data previously published in ()" the reference is missing.

The reference has been added.

- In the legend of Figure 5 "(E) Verification of CAV1 knock-down in TGBC cells using two knock-down system" 'a' between using and two is missing.

The legend has been corrected (no ‘a’ is missing, but it should say systems (plural)):

- In Figure 5B one horizontal line is missing.

The Figure 5B has been corrected accordingly.

- Terms such as de novo or in silico should be written in cursive.

We thank the Reviewer for this comment; however, we believe that in the style used by eLife, common Latin expressions such as *de novo* or *in vitro* are used in regular font.

| - *In the heading of Table 4 "The results presented in this table can be reproducible using the code and data available under the GitHub link reported in the methods section." It should say reproduced instead of reproducible.*

Yes, indeed. It has been corrected.

| - *The citation of reference 20 contains several author names multiple times.*

Indeed, it has been fixed now:

| - *In Figure S2 there is a vertical line in the zeros of the y axis labels.*

I am not sure if there was some rendering issue, but we did not see a vertical line in the zeros of the y axis label in Figure S2.

| - *The Text in Figure S4 is too small.*

We thank the reviewer for pointing this out. We have now revised Figure S7 (formerly Figure S4) to increase the text size, ensuring better readability. (It has also been updated to include additional fits as requested by Reviewer #2).

| - *In Table 3 "positive hypothesis II markers are discriminative of samples with stiff/soft independent of data source" the words 'mechanical phenotype' are missing.*

The column headings in Table 3 have now been updated accordingly.

| - *In Table S3 explain in the table headline what vi_1 , vi_2 and v are. I assume the loading for PC1, the loading for PC2 and the average of the previous two values. But it should be mentioned somewhere.*

The caption of table S3 has been updated to explain the meaning of vi_1 , vi_2 and v .

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